



HELMUT SCHMIDT  
UNIVERSITÄT

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Universität der Bundeswehr Hamburg

# Source Separation and Echo Detection of Underwater Acoustic Signals of Whales

*Master Thesis*

by

**Ankit Ghosh**

|                  |                                  |
|------------------|----------------------------------|
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## **Abstract**

In this thesis work, a system is proposed for detection of echolocation click of toothed whales and thereby extracting three unique features of the echolocation clicks to identify a toothed whale species. Generalised Weiner filtering based source separation algorithms helped in isolating the clicks from other vocal artefacts - whistle, creeks and other harmonic vocalisations. Separation algorithms are compared using peak-to-noise ratio (PNR), the signal-to-distortion ratio (SDR) and a few simple error calculation method. The separated clicks are the fingerprint of these toothed whales which propelled the motivation to analyse them using methods like Hilbert transform, wavelet analysis and correlation. As detection of echo from a click signal is virtually impossible given that the source location is unknown, these methods helped in the analysis of clicks or rather echoes of the clicks providing determining factors about them. Echo decay time is the first feature which provides information about the saturation time of click pulses, also concluded that each click pulses had superimposed pulses. This intuition about superimposed clicks motivated further to differentiate the click pulse into high energy and low energy part to detect the inlying short-term pulses and provided information about the frequency band in which they exist. Finally, click length is found out through an adaptation of the correlation technique using information about the band-limitation of each pulse.





# Statement

I, Ankit Ghosh, at this moment state that this thesis work has been prepared in entirety by me and with the help of research papers, community and personnel referenced to within this thesis.

Hamburg, June 2018



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# List of Symbols

|                               |                                                                |
|-------------------------------|----------------------------------------------------------------|
| $t$                           | Time.                                                          |
| $\tau$                        | Echo decay time.                                               |
| $f$                           | Frequency.                                                     |
| $N$                           | Window length.                                                 |
| $F_s$                         | Sample frequency.                                              |
| $T_s$                         | Sampling Time.                                                 |
| $x(t)$                        | Input signal.                                                  |
| $x[n]$                        | Discretization of $x(t)$ sampled at $T_s$ .                    |
| $x_d[n]$                      | Denoised $x[n]$ .                                              |
| $x_H(t)$                      | Hilbert transform of $x(t)$ .                                  |
| $x_A(t)$                      | Analytic signal of $x(t)$ .                                    |
| $w(t)$                        | Sliding window for STFT.                                       |
| $X(f,t)$                      | STFT of input signal.                                          |
| $ X(f,t) $                    | Magnitude of STFT.                                             |
| $\angle X(f,t)$               | Phase of STFT.                                                 |
| $\Delta t$                    | Time resolution.                                               |
| $\Delta f$                    | Frequency resolution.                                          |
| $C$                           | Proportionality constant.                                      |
| $M_{j=1\dots J}(f,t)$         | Masking spectrograms.                                          |
| $S_{j=1\dots J}(f,t)$         | Separated magnitude spectrograms.                              |
| $W(f,k)$                      | Basis matrix.                                                  |
| $H(k,t)$                      | Activation matrix.                                             |
| $\beta$                       | Divergence factor.                                             |
| $ [X_{n\beta}]_{F \times T} $ | Normalized magnitude spectrograms over $F \times T$ dimension. |
| $X_P$                         | Percussive initialization signal.                              |
| $X_H$                         | Harmonic initialization signal.                                |
| $N_P$                         | Percussive components.                                         |
| $N_H$                         | Harmonic components.                                           |
| $d_\beta$                     | Divergence cost.                                               |
| $D$                           | Global cost function.                                          |
| $\mathbb{T}_{SP}$             | Temporal sparseness.                                           |
| $\mathbb{F}_{SP}$             | Spectral sparseness.                                           |
| $\mathbb{T}_{SM}$             | Temporal smoothness.                                           |
| $\mathbb{F}_{SM}$             | Spectral smoothness.                                           |
| $S_{P_t}$                     | Percussion enhanced spectrogram.                               |
| $S_{H_t}$                     | Harmonics enhanced spectrogram.                                |

|                          |                                                                                |
|--------------------------|--------------------------------------------------------------------------------|
| $\mathbb{P}(f,t)$        | Power spectral density matrix.                                                 |
| $\mathbb{C}(f)$          | Spatial covariance matrix.                                                     |
| $s_{tar}$                | Ground truth signal.                                                           |
| $e_{interf}$             | Error due to interference.                                                     |
| $e_{artif}$              | Artifacts error.                                                               |
| $V_{t_p}$                | Transverse speed of sound.                                                     |
| $V_t$                    | Speed of sound.                                                                |
| $K$                      | Bulk modulus.                                                                  |
| $B$                      | Shear modulus.                                                                 |
| $\rho$                   | density of the medium.                                                         |
| $C_{ps}$                 | equivalent elasticity in each direction.                                       |
| $S,T,P$                  | Salinity in <i>ppt</i> , Temperature in $^{\circ}C$ , Pressure in <i>bar</i> . |
| $\psi_{s,T}, \psi_{a,b}$ | Mother wavelet.                                                                |
| $s,a$                    | Scaling factor.                                                                |
| $T,b$                    | Shift parameter.                                                               |
| $\gamma_{s,T}, W_x(a,b)$ | Wavelet coefficients.                                                          |
| $cA$                     | Approximation coefficients.                                                    |
| $cD$                     | Detailed coefficients.                                                         |
| $h_0$                    | Low pass filter.                                                               |
| $h_1$                    | High pass filter.                                                              |
| $\lambda_U$              | Universal Threshold.                                                           |
| $P$                      | Sample points to calculate FFT and IFFT.                                       |
| $v$                      | Vector to compute Hilbert transform.                                           |
| $F_{eq}$                 | Equivalent frequencies to different scales.                                    |
| $m_k$                    | $k^{th}$ moment of a function.                                                 |
| $P_x$                    | Power of the wavelet coefficients.                                             |
| $C_f$                    | Center frequency of mother wavelet.                                            |
| $z_c(t)$                 | General chirp.                                                                 |
| $z_l(t)$                 | Linear chirp.                                                                  |
| $z_e(t)$                 | Exponential chirp.                                                             |
| $z_U(t)$                 | Exponential/linear up-chirp.                                                   |
| $z_D(t)$                 | Exponential/linear down chirp.                                                 |
| $z_{Dx}(t)$              | Correlation with up chirp.                                                     |
| $z_{Ux}(t)$              | Correlation with down chirp.                                                   |
| $f_0(t)$                 | Starting frequency.                                                            |
| $f_s(t)$                 | End frequency.                                                                 |
| $f_c(t)$                 | Instantaneous frequency.                                                       |
| $c(t)$                   | Rate of change of frequency.                                                   |

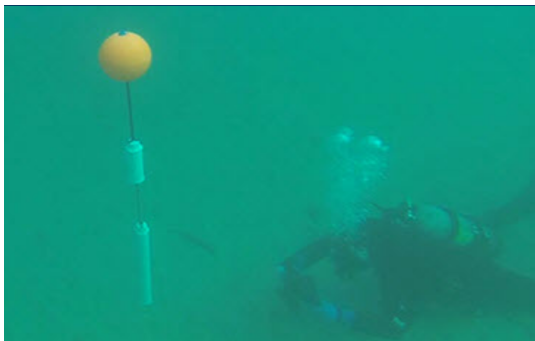
# Chapter 1

## Introduction

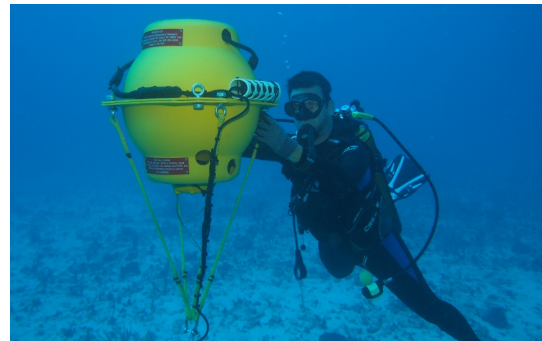
Limitation in vision amid marine surroundings impelled the marine animals relying on sound for activities like navigation, maintaining social interaction, establishing dominance, attracting mates, avoiding predators and most importantly finding food. Understanding different sounds produced by these animals is of particular interest to evaluate the impacts of anthropogenic sound on these acoustically sensitive oceanic mammals.

Passive acoustic monitoring (PAM) is a method practised in the aquatic environment to detect, monitor and localise vocalising marine animals with the help of hydrophones, also known as underwater microphones.

PAM using autonomous acoustic recorders (AAR) assisted in monitoring marine mammals when deployed on the sea-floor for long-term data retrieval. Shallow water compact PAM device like microMARS (micro Marine Autonomous Recording System) as shown in Fig. 1.1(a) used to identify mammal species residing in the delta areas. This device is of a size of a soda can bottle, and the design facilitated in low power consumption and deployed with inflatable objects for anchoring it to the surface of the water. The device has a capacity of 2TB maximum storage with a deployment period of 35 - 140 days continuous recording depending on the recording frequency (250 Hz - 25 kHz).



(a) microMARS<sup>1</sup>

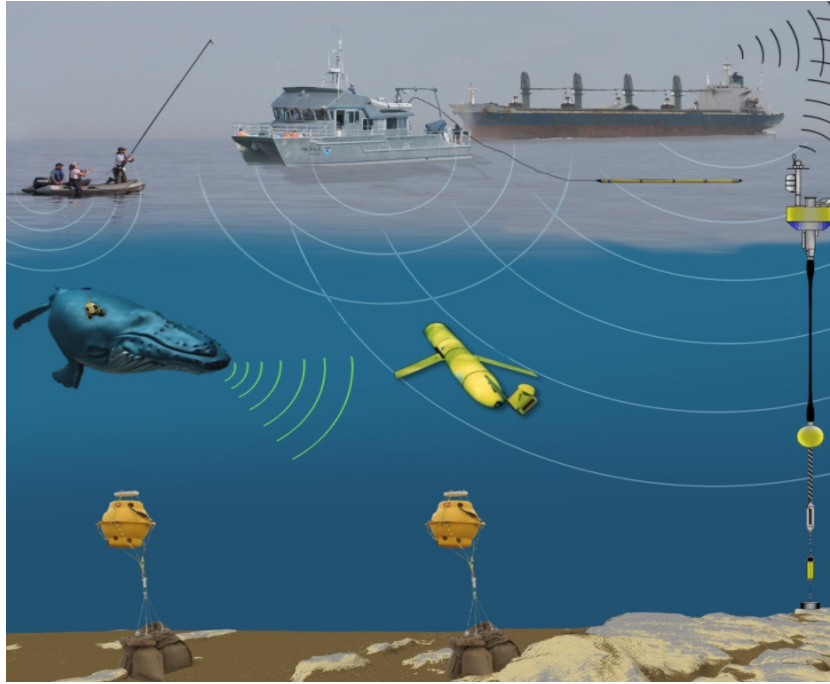


(b) MARU<sup>2</sup>

**Figure 1.1:** *Instruments for Passive Acoustic Monitoring*

<sup>1</sup><http://biowaves.net/services-technologies/hardware/micromars-2/micromars>

<sup>2</sup><https://www.nefsc.noaa.gov/psb/acoustics/psbAcousticsCHAMP.html>



**Figure 1.2:** *Underwater passive acoustic monitoring architecture*<sup>3</sup>

On the other hand, NOAA's (National Oceanic and Atmospheric Administration) CHAMP program saw a successful deployment MARU as shown in Fig. 1.1(b) and soundtrap (picture not shown), two PAM instruments developed by Cornell University and Ocean Instruments respectively, which aid primarily in monitoring humpback whale distribution and analyse the content and structure of whales' calls among others.

## 1.1 Task description

This work considers the marine mammal class Odontoceti or toothed whale because of their widespread nature, well-developed hearing ability and most importantly the capacity to produce a broad range of sound. The identifiable sounds like whistles, creaks and clicks form the primary focus of the thesis recorded with the assistance of PAM.

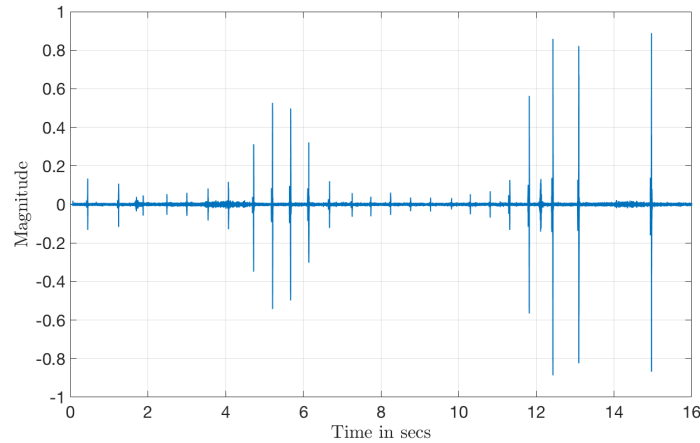
Underwater acoustics consists of the sounds as mentioned earlier among many others; however, this work entailed to separate the sounds produced by cetacean species, and afterwards look for echo content in the separated signal, especially the echolocation clicks. Fig. 1.2 identifies an underwater network planned by NOAA for passively listening to the sounds in shallow water and also at the depth in case of bay area and tidal creeks. Fig. 1.2 demonstrate the use of three devices, namely MARU, soundtrap and electric glider. MARU, as shown in Fig. 1.1(b) are attached at the bottom and generally held at the place by a weight at the bottom. Soundtrap is at the rightmost side of the figure, and the AUV (Autonomous Underwater Vehicle) pictured in the centre of the diagram is an electric glider.

Separation of the vocalisation produced by the cetacean suborder Odontoceti is the first part of

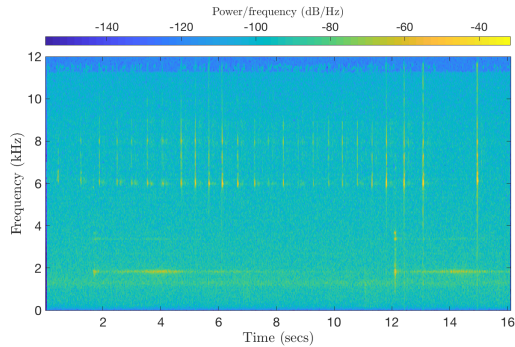
<sup>3</sup><https://www.nefsc.noaa.gov/psb/acoustics/>

the task which involves decimating the captured acoustics into tonal sounds or harmonics and clicks or percussions. The creaks or progressive, either decreasing or increasing, clicks can be classified as a part of whistle or harmonics. For tackling the separation problem, blind source separation techniques proved useful with the basis as matrix factorisation and Wiener filtering. Three particular methods are applied considering the sound to be monaural and are Non-negative Matrix Factorization (NMF), Median Filtering (MF) and Kernel Additive Modeling (KAM). The evaluation considered a newly developed metric based on Peak-Signal-to-Noise Ratio (PSNR), which is used in image processing and to eliminate ambiguity, the name modified to Peak-to-Noise Ratio (PNR). Further evaluations performed are the signal to distortion ratio (SDR), simple errors evaluations like Magnitude squared coherence (MSC), Pearson correlation coefficient (PCC) and Root mean square (RMS) error.

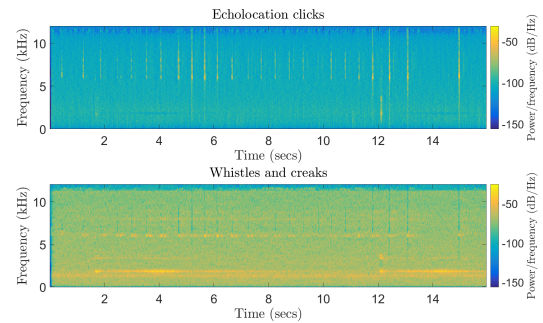
Concluding the separation technique, based on three primary processes, the thesis tries to address the problem of detecting the echo content of the signal. The perennial problems of shallow water communication are discussed as the mammals reside and hunts in those depths of oceans. Creation of synthetic impulse response based on the environment parameter elaborated to deconvolve the separated click signals. Afterwards, Hilbert transform is used to detect the echo decay time of the click signal and subsequently localise the click energies with continuous wavelet transform post denoising of the signal with discrete wavelet transform. After that, due to curiosity synthetic chirps are generated and correlated to identify the click length in the



(a) Time-domain representation of Dall's Porpoise sound



(b) Time-frequency representation of Dall's Porpoise sound



(c) Separation performed on the sound shown in Fig. 1.3(b)

**Figure 1.3:** Recorded sound of Dall's porpoise attenuated by a factor of 20 for better listening.

estimated separated click signal. The proposed system in Fig. 1.4 follows in the subsequent chapters.

## 1.2 Outline of the project thesis

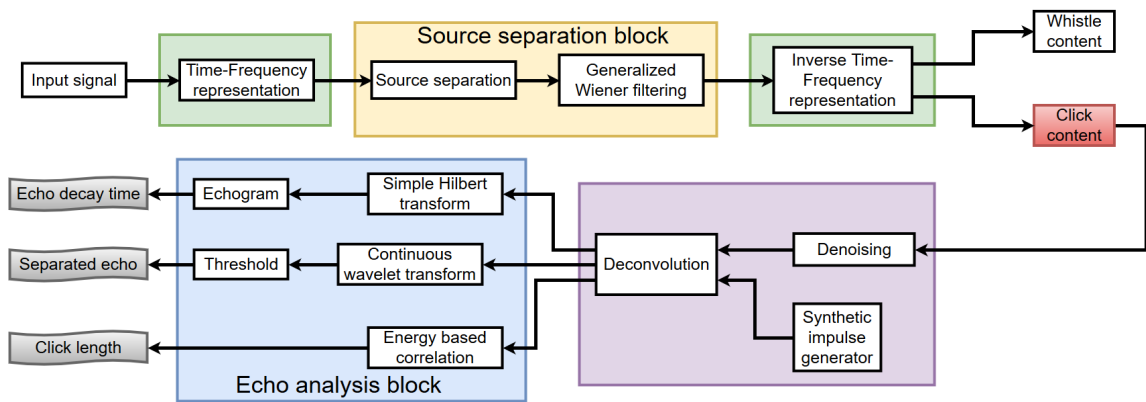
The thesis has been divided into three major parts - introduction to the class of marine animals, source separation on the voice excerpts recorded employing PAM, finally detecting and eliminating the echoes from separated sounds.

Chapter 2 broadly discusses Odontoceti and the sound produced by them. This chapter ascertains the preference of this distinct class of cetaceans for the work. Furthermore, it describes the parameters used for time-frequency representation of the recorded signals.

Chapter 3 introduces the source separation algorithms along with comprehensive explanation given on NMF, MF, KAM. Consequently, their performance based on PNR, SDR and simple errors shown as an evaluation. A complex source separation algorithm proposed based on the evaluation.

Chapter 4 elaborates echo detection. In this chapter shallow water communication problem is explained and generation of the synthetic impulse response discussed. The discrete wavelet transform is applied henceforth to minimise the burbling effect produced by separating the signal. After that proposition of Hilbert transform to obtain the echo decay time of clicks is shown along with continuous wavelet transform for extracting the echoes based on thresholding. It furthermore contains an energy-based correlation technique to find the click lengths of a toothed whale species.

Chapter 5 contains discussion, limitation and future improvements concluding the thesis work.



**Figure 1.4:** The proposed system for extracting click features. The time-frequency representation is elaborated in Chapter 2, source separation block in Chapter 3 and Echo analysis block in Chapter 4. Denoising and deconvolution block is discussed as an introduction to echo analysis.

## Chapter 2

# Underwater Acoustics of Marine Mammals

Marine conservation lacks far behind than land conservation as IUCN (International Union for Conservation of Nature) considers marine mammalian class (Cetaceans) to be data deficient. Passive acoustic monitoring is vital in gaining information about these species, which are a significant indicator of the general health of the ocean and the marine ecosystem. The aquatic animals monitored typically use sounds to perform necessary daily activities, for performing a detailed analysis of these animals.

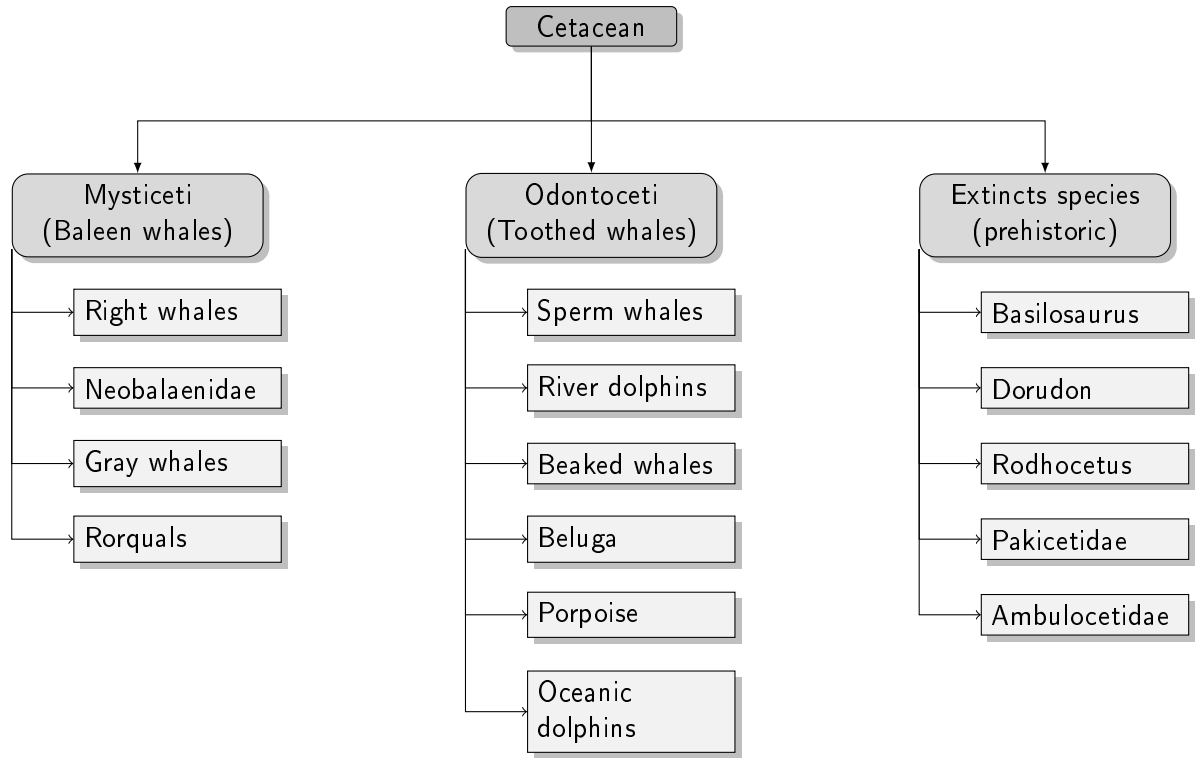
This chapter provides a short introduction on cetaceans followed by the cetacean tree of evolution along with an explanation for considering the specific species of Odontoceti and the utilised sound demographics in the thesis. Afterwards, the elaborated of TF (Time-Frequency) diagrams as preprocessing step since spectrogram information is vital in performing the source separation task.

Chapter 2 focuses on various windowing parameters utilised in work and recorded animal sounds helped in conducting the work.

### 2.1 Cetacean

Cetaceans are considered one of the most intelligent species. They are distinctive and highly specialised mammals. According to the California Academy of Science [2,3], there are eighty-six species of cetaceans living in the present world. Human activities are encroaching their natural habitats endangering the population and thus monitoring the marine animals gain prominence.

Cetaceans grouped into two major division - Mysticeti (baleen whales) and Odontoceti (toothed whales). Of the two of them, Odontoceti is the class which have numerous peg-like teeth making them active predators unlike their counterparts and thus are at the top of marine food chain. In Fig. 2.1, the branching of modern-day cetacean are shown. Three predominant mysticetes are sighted until today in abundance and tabulated. Large varieties of Odontoceti are available, but river dolphins in Yangtze and Ganges-Brahmaputra are on the steep decline which inspires to contribute in the field of passive acoustic monitoring for conservation.



**Figure 2.1:** Family tree of the Cetacean - the modern day species and the extincts

The next section contains elaboration of sounds aided the thesis work prefaced with the criterion on choosing toothed whale vocalisation and discarding baleen whales for this work.

## 2.2 Toothed whale acoustic

Mysticeti and Odontoceti differ in many ways. The most important one is the presence of teeth, as mentioned in Sec.2.1. The other significant difference in nature among the two is that most toothed whales maintain a social structure while baleen whales do not remain in groups for a significant period. Odontoceti or toothed whales' strong social structure enhanced their interactions than from their counterpart, Mysticeti. Moreover, being an active predator, use of echolocation for preying make them interesting subjects for utilising their sounds in the presented work.

The most common sound used by them (Odontoceti / toothed whales) is the biosonar or commonly known as echolocation clicks. These clicks are mostly in ultrasonic range and the arrangement of teeth aid in listening to the echo of this particular kind of sound. In addition to these ultrasonic clicks, another acoustic is captured due to odontocetes' societal structure as found by Olesiuk et al. [4] for killer whales (Orca) and Wells et al. in 1987 [5] for Bahamas dolphins when they were studying dolphin schools few kilometres off the coastline. Other common vocalisations [?] that are recorded by different PAM devices listed as follows:



- Whistles or tonal
- Burst pulse
- Low frequency moans
- Jaw claps
- Creaks
- Groans
- Boings
- Cries
- Trills

Pods(cetacean social structure) communicates through whistles, clicks as described by Møhl et al. in [6] while studying sperm whales, are used for navigating a prey, and burst pulses tend to occur when social interaction takes place, as in during a hunting activity. Jaw claps are considered to be playful gestures among the calves in a pod or between the female and calf. Van der Woude in [7] first recorded Low-frequency moans in the toothed whale (bottlenose dolphins). The other sounds are not prominent but are closely related to each other. Minke whales recorded at North Pacific makes boings [8].

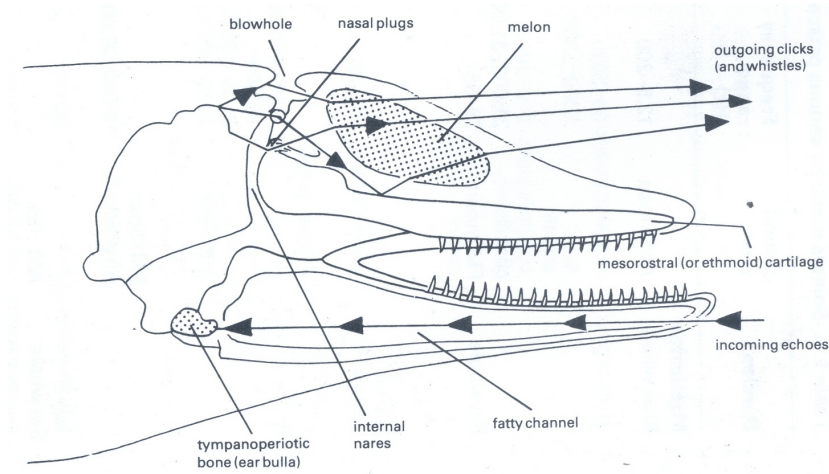
In this thesis work, processing is done to separate echolocation clicks, creaks and whistles. The best argument for considering vocalisations of whales (clicks, whistle and creaks) because of their usage in communication, like, clicks help in ascertaining the shape, size of an object and distance between the entity producing clicks and the reflecting objects. Others' (vocalisations') purpose is yet to be known precisely, and only found documentation is about the perception. The exclusion of baleen whale vocalisation is due to their specific songs, which are quite complicated and researchers are baffled in pointing specific use of the produced melodies by Mysticeti.

The following subsection provides showcasing individual interpretation with proper visualising diagrams on the mentioned sound types.

### 2.2.1 Use of sounds

Light and odour do not travel well underwater, and hence visibility along with olfaction (an act of hearing) are ineffective to communicate in an aquatic environment. However, sound travels at least four times faster underwater than in air, equating to the numerous kind of whale sounds recorded utilising PAM are tabulated in Sec. 2.2, being recorded by different monitoring devices like MARU (Fig. 1.1(b)) and others. Of these underwater symphonies, produced mostly by whales, it is interesting to analyse the structure, and frequency content of the own vocalisations as the sound producing mechanisms of whales are analogous to a certain extent to human counterparts, but efficient airflow in the larynx makes the sounds so convincing. Fig. 2.2 pictured the sound production by Odontoceti with the help of dolphin's cranium structure. The discussion of baleen whale's sound production technique is in Sec. 5.1.1.

The toothed whales use larynx and a specialised air-sac located near their blow-hole [1]. The low-frequency sounds use larynx for their production, but most sounds, especially the production of high-frequency sounds use a much complex mannerism through a specialised organ called *melon*. Air-sacs in addition to specific soft tissues vibrates the air in the nasal passage producing the overall range of vocalisations.



**Figure 2.2:** *Dolphin acoustical cranial structure [1]*

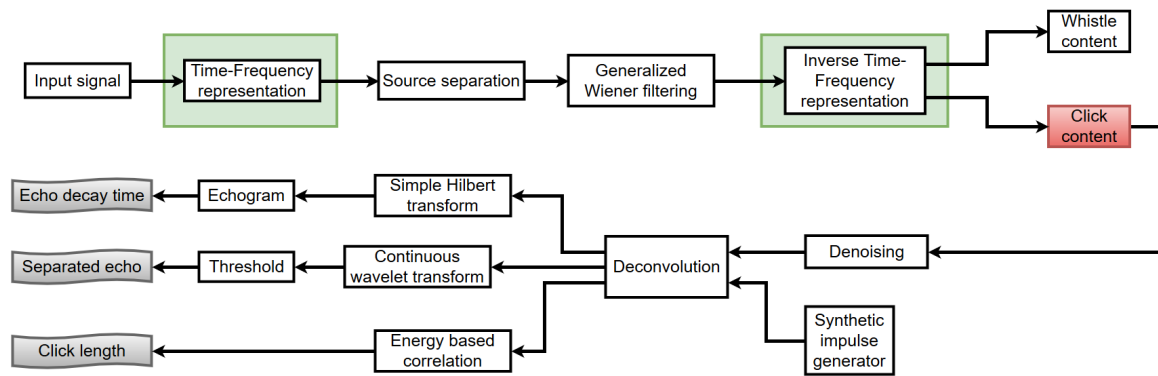
Fig. 2.2 provides intricate details of different parts of the sound producing regions of a dolphin. Evan and Peters [1] describe the production of sound by a dolphin. It is believed that they (dolphins) intake air through the blow-hole, which is then passed along the nasal passage producing short click pulses and afterward focussed by a mass of tissue in the forehead called melon, before emanating either as a single click or as a burst of pulse as required for a particular activity concerned.

The fatty tissue accomplishes the reception of sounds in the lower jaw which is shown by Graf et al. in [9]. The sounds on reception by the teeth are guided through a thin region in the lower jaw like a wave-guide to the middle ear. The space between the lower jaw and teeth acts as an acoustic window. Another suggestion is that the teeth in lower jaw may also act as a receiver which may play a role of an amplifier or attenuator to specific sound reception. The regularly spaces teeth structure help in acting as a passive resonant receiver. Fig. 2.2 illustrates the procedure of producing clicks by a dolphin pictorial as described above. Description of Baleen whale vocalisation structure is in Sec. 5.1.1.

In the following section, a different class of sounds are pictured for a better understanding of their composition in different recording fragments and also for later reference. Fig. 2.3 provides the topic of discussion, the green highlighted part. The purpose of highlighting is to help the reader following the overall process.

## 2.3 Visualization of sound profiles

The sound profiles treated for the thesis are gathered from a wide variety of sound example sources, ranging from National Geographic Wild (Nat-Geo) [10] to Discovery of Sound in the Sea (Dosits) [11] along with different captive Odontoceti recordings in aquariums. The data from open water bodies includes regions like bay areas, creeks, rivers, coastal areas. Moreover, recordings of the datasets collected obtained by hydrophone array produced by various research institutes of which Fig. 1.1 shows two examples. The recorded sound profiles are of varying length and sample frequencies. In addition to this, some files contain sound from more than two species of toothed whales and also a mixture of some constant noise, which may account for the ship noise to which the hydrophone arrays are attached to or some ambient underwater



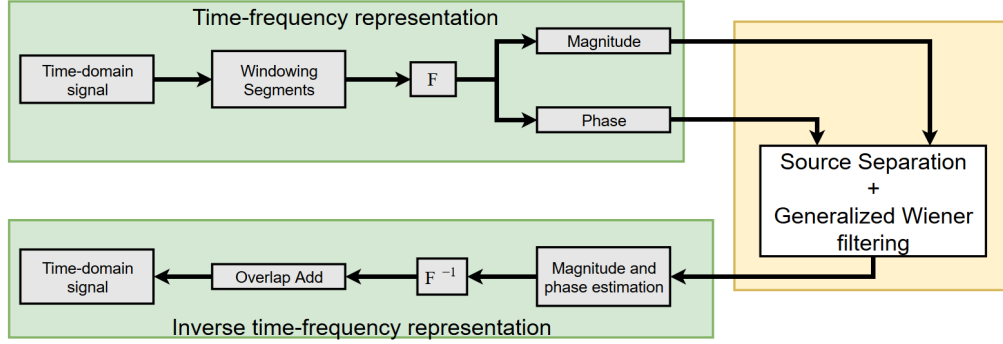
**Figure 2.3:** First step of the overall approach denoted by green blocks. Time-frequency representation and inverse of time-frequency representation is elaborated in this section as an integral part of the overall approach.

noise, that is omnipresent in the environment concerned. To understand the frequencies involved and the variation of characteristics of sound over time, the time domain signal transformed to a time-frequency representation.

The transformations, not only aid in ascertaining the signal constituents but also assist in separating the signals into three different exciting sounds; clicks, creaks and whistles, elaborated in Chapter 3. Sec. 2.3.1 describes the mathematical process of representation of time domain signal into a time-frequency diagram along with an illustration of the constituents. After that, Sec. 2.3.2 conclude the chapter by detailing the sound profiles used along with the people or organisational entities recorded them for research use. This section further motivates the idea of TF representation, the importance of windowing function and visual comparison of resolution change in TF representation with different window length.

Some of the sounds recorded with an acoustic tag attached to the animal, whereas other used hydrophone array towed in their ship/boat. Narwhal whale sounds used in work captured using dipping hydrophones from a small yacht. The other aspect of the sound profiles that worth mentioning is the attenuation factor which applied to the clicks as they are in ultrasonic range and hence are outside of the human hearing range. A factor of 20 attenuates recorded sounds to suit human audible range in some of the cases. The following organisations or peoples mostly recorded the sounds used for the thesis work:

- William A. Watkins - Amazon river dolphin
- www.baiji.org - Baiji (Yangtze river dolphin), finless porpoise
- Johnson M. Woods - Beaked whales
- Peter M. Scheifele - Beluga whale
- Jennifer L. Miksis Olds - Bottlenose dolphin
- Scripps Institution of Oceanography - Dall's porpoise
- Peter Madsen - False killer whales
- Stacy DeRuiter - Harbor porpoise
- Thomas R. Kieckhefer - Orca (Killer whale), common dolphin, white-sided dolphin
- Bernd Würsig - Spinner dolphin



**Figure 2.4:** Optimal signal path for time-frequency processing of a sound input. The signal is transformed into time-frequency representation to work with the magnitude and then the process is reversed to get the time domain signal.

### 2.3.1 Short-time Fourier transform

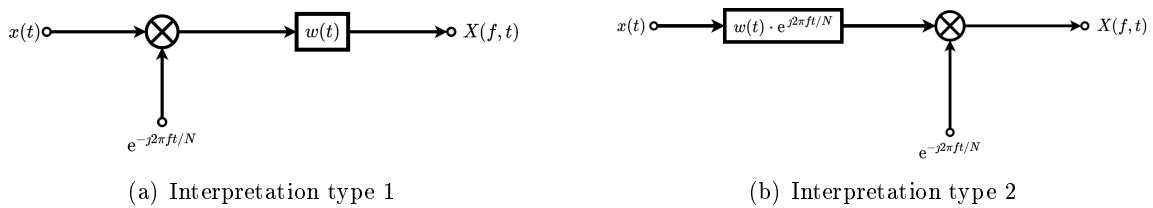
Whale vocalisations contain frequencies that change over time and are non-stationary. In this scenario, fast Fourier transform (FFT) do not provide an optimum result of the changing frequency over time if taken over a long window. Consideration of smaller length of the window aids in contemplating the variation over time as short-term stationary, with sliding approach. Sec. 2.3.1.1 discusses the choice of window length.

The recorded vocalisation can be considered stationary but includes a time-frequency trade-off. During this section, elaboration of the mathematical details of conversion from Fig. 1.3(a) to Fig. 1.3(b) will be appropriated. The STFT analysis follows three broad steps [12] - analysis (sound to representation), transformation (of the representation) and resynthesise (getting back the sound). Fig. 2.4, the steps cited in [12] are shown with inner operations associated in each block. This section elaborates first, and the last step while the middle element, highlighted in yellow, is the subject matter of Sec. 3.

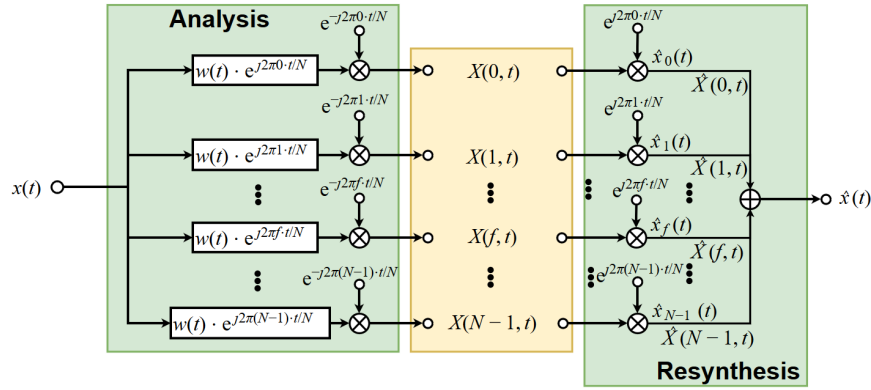
The STFT (Short-term Fourier Transform) of a signal  $x(t)$  is given by [12]:

$$X(f, t) = \int_{-\infty}^{\infty} x(\tau) w(t - \tau) e^{-\frac{j2\pi f\tau}{N}} d\tau, \quad (2.1)$$

where  $X(f, t)$  represents a complex number consisting of the magnitude  $|X(f, t)|$  and phase



**Figure 2.5:** Interpretation of STFT as filtering operation.



**Figure 2.6:** Filter bank description of short-time Fourier transform

$\phi(f, t)$ . It denotes the quasi-stationary element of the time-varying spectrum at frequency bin  $0 < f < N - 1$  and time frame  $t$ . So at each time index  $t$ , the signal  $x(t)$  is multiplied or weighted against the sliding window function  $w(t - \tau)$  which is shown in Fig. 2.8. Thereafter, the integral over  $t$  is performed with a window length of  $N$  which is equivalent to the FFT length considered.

Short-time Fourier transform (STFT) is the process of converting a time-domain signal to its corresponding time-frequency representation. In Fig. 2.5, the two implementations of STFT for  $K - th$  channel is cited. The multiplication of an input signal  $x(t)$  by a sliding window of length  $N$ -samples sampled at  $f_s$  producing a windowed version of the input signal and are further transformed to the spectral domain by Fast Fourier Transform (FFT).

Another interpretation of STFT is that of filtering operation, comparing the sliding windowing function to a filter impulse response. From Fig. 2.5(a), the illustration depicts the idea of convolution between the signal and window function. The physical interpretation of first product is that of modulating the input signal until a certain frequency  $f$ . Rearrangement of the implementation shown in Fig. 2.5(a) will provide Fig. 2.5(b). In this case, the inference is that the input sequence passed through a filter with a linear phase factor, and the output demodulated afterwards. The second interpretation helps in representing the filtering view from both analysis and synthesis side which is shown in Fig. 2.6.

The spectrogram is the way to show the time-varying spectral information in a plot. As STFT is a complex valued function of two variable, it would require 4 dimensions to plot the complete result. However, this method utilises the magnitude of Fourier coefficient computed at each interval, and so the STFT result becomes a real-valued function of two variables and hence helped to cut down one dimension. Here, comes the colour code, which is used by the spectrogram function, to denote magnitude - dark hues for lower / smaller values and brighter hues for higher / large values. Moreover, the logarithm of magnitude is computed to compress the range of values for better representation in the colour scheme. Next, the spectral slices are placed one after the other for better visualisation of a time-varying spectrum.

The horizontal axis of the spectrogram represents time slice, and the vertical axis depicts the frequency segment. If the sample rate of the input signal is known, the highest positive frequency is  $F_s/2$ , and hence the range of the frequency axis/ordinate is  $[0, F_s/2]$  and the abscissa represents the time duration of the input signal.

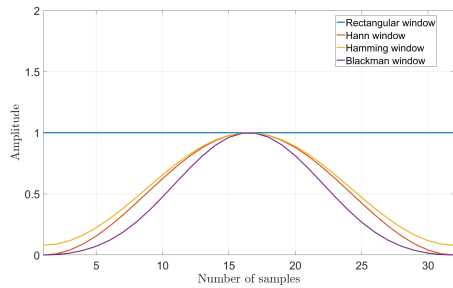
### 2.3.1.1 Windowing function

The discussion in Sec.2.3.1 gave rise to three distinct questions on windowing of the signal. They are as follows:

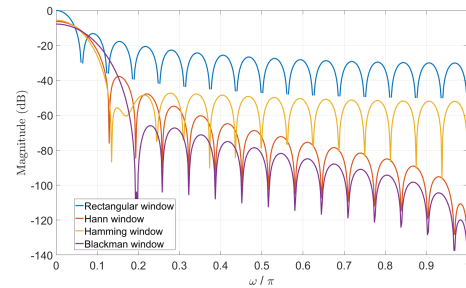
- What could be the shape of the window?
- What could be the length of the window?
- Is there any overlapping when the window slides through the next frame of reference?

The common windows used as windowing function are rectangular, Hann, Hamming and Blackman window [12]. In Fig. 2.7(a), the time-domain representation of the above mentioned windows are represented. The windowing function needs to be tapered at the end assuring in avoidance of discontinuities in the vocalisation segment thereby localising the speech signal in time. Further, considerations must be noted so that the spectral estimate is not affected. The main idea is to choose a window with pronounced main lobe and attenuation of side lobes.

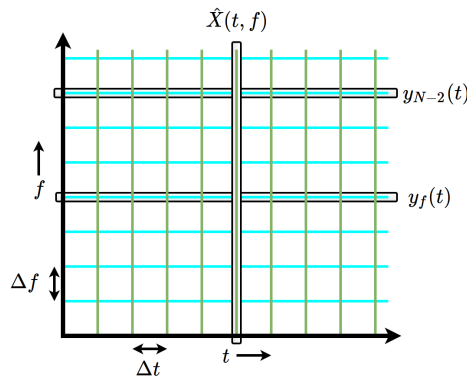
In Fig. 2.7(b), the frequency response of the mentioned windows are plotted. From the Fig. 2.7(b), a trade-off observed between the main lobe and the attenuation of side-lobes. It is seen that width of the main lobe increases with the increased side-lobe attenuation. Blackman window has the strongest side-lobe attenuation, but the width of the main lobe is largest among four mentioned windows. Hence, a certain relation developed from Fig.2.7 that larger main lobe width decrease the resolution, and because of this trade-off, entails the usage of the Hamming window.



(a) Time-domain representation of different windows



(b) Fourier transform of the the time-domain function



(c) Time-frequency blocks of STFT

**Figure 2.7:** Different windowing function

Length of the window needs a deeper look because of the time-frequency uncertainty principle, which essentially defines that there is a resolution trade-off in frequency and time. In this context, the time-frequency tiling is shown in Fig. 2.7(c). If frequency resolution is given by  $\Delta f$  and time resolution by  $\Delta t$ , then the product of them can be written as:

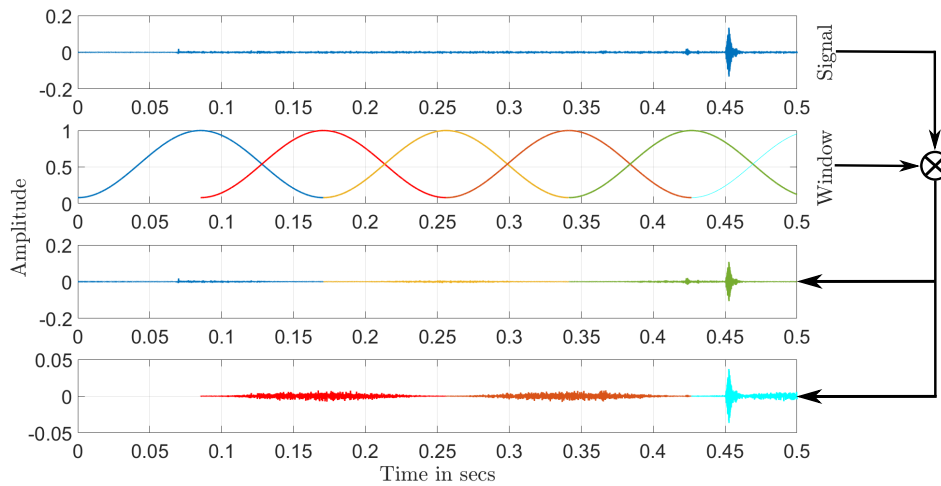
$$\Delta f \cdot \Delta t = C, \quad (2.2)$$

where  $C$  is the proportionality constant.  $\Delta t$  indicates the uncertainty in time and  $\Delta f$  represents uncertainty in frequency. Referring to the Eq. 2.2 and Fig. 2.7(c), if  $\Delta t$  is increased,  $\Delta f$  has to be reduced for justifying the equality. Hence, for a shorter window length, the resolution in time would be higher, however, decreasing resolution in frequency. It is possible to divide these into three [13] broad cases as follows:

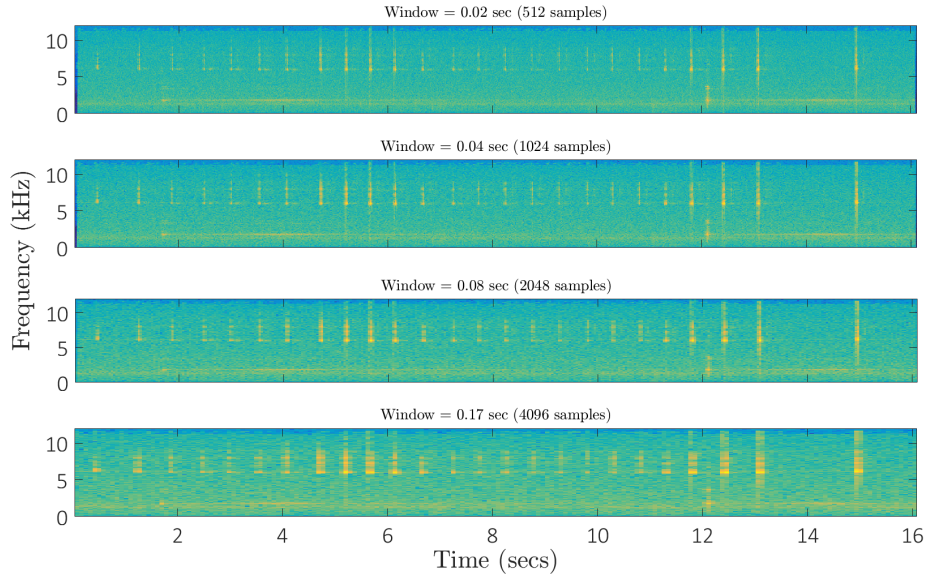
1. The extreme case of Eq. 2.2 is infinite window or  $\Delta t \rightarrow +\infty$ , which essentially translates to  $\Delta f \rightarrow 0$ . This certain case provides the best frequency resolution.
2. The second case is a long time window or  $\Delta t \gg T_s$  (sampling time); hence the uncertainty equation provides shorter frequency window  $\Delta f$ . A longer window provides a better time resolution compared the extreme case, compromising in frequency resolution.
3. The time window is taken to be  $\Delta t \rightarrow T_s$  (sampling time), which ensures a long frequency window. The extended frequency window results in better localisation in time but degrades in frequency resolution.

In general, truncation of a signal corresponds to windowing concerning a rectangular window function. This results in spectral leakage, however, adhere to the signal in consideration. The leakage effect prompts to use a tapering window function, which reduces to zero at the boundary. The use of tapered window gives rise to the second problem, which elaborates in loss of signal elements at the boundary of the window. Overlapping the next windows helped in mitigating with the problem. The next part describes the parametric details used in the project.

In this thesis, the width of the window is taken to be  $0.02\text{ s}$  as input signals have varied sample frequencies (The used recorded dataset contains sample frequency ranging from  $11.02\text{ kHz}$  –  $96\text{ kHz}$ ) with the minimum length of the dataset is  $2.5\text{ s}$ . The consideration of window length



**Figure 2.8:** STFT procedure before FFT computation



**Figure 2.9:** Resolution difference with different windows with Dall's porpoise recording

defined by the nearest power of 2 after computing the number of samples present in a  $0.02\text{ s}$  window for a particular audio recording with a given sample frequency. The result is a variation of window length in between 256 and 2048 samples along with overlap ranging from 50% to 85% for the same reason. The idea implemented is overlap ration is made inversely proportional to the window-length for representing the sparser data in a more meaningful way as an audio recording with denser representation. Fig. 2.8 demonstrates the usage of a window function with a window of length  $0.17\text{ sec}$  (window length = 4096 samples, sampling frequency =  $24\text{ kHz}$ ) with an overlap of 50%. The reason to take such parameters for better visualisation of sparser data.

The time-domain signal in Fig. 2.8 is from Dall's porpoise. In Fig. 2.8, the multiplication of window with the signal is shown in two steps before the application of FFT of each segment. Also, the segments are coloured similarly - the window and products are coloured the same for better understanding. In Fig. 2.9, the STFT result with window length  $0.17\text{ sec}$  (window length = 4096) is shown. The resolution in frequency increased as  $\Delta t$  increased but the resolution in time decreased as a consequence. A decrement of window length to 512 samples or  $0.02\text{ s}$  is found optimum regarding both time and frequency resolution for Dall's porpoise sound. With other recordings in the dataset, a window of  $0.02\text{ s}$  helped in achieving a desired possible trade-off between time and frequency resolution.

### 2.3.2 Implemented sounds

Sound profiles used in the thesis consists of porpoise, dolphins and orcas mainly because of their echolocation. As mentioned in Sec. 2.2.1, the orca profiles also contain other sophisticated inter-community interaction. Moreover, the boat sounds are visible in TF diagram as a constant low-frequency element and seen at the base of corresponding profile diagram. In this section, sound excerpt from an orca pod is depicted explaining the consisting vocalisation of whales.

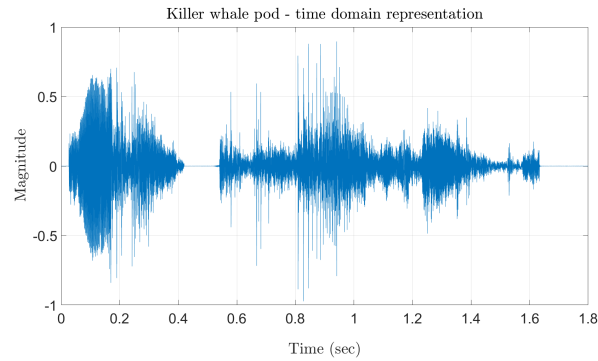
Fig. 2.10(a) show the time domain representation of a killer-whale (orca) pod and Fig. 2.10(b) is the time-frequency representation of time-domain data. Thomas R. Kieckhefer recorded



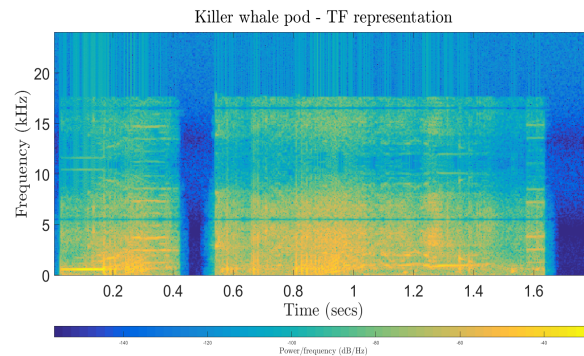
the sound in Johnston Strait, Vancouver at a sampling frequency of  $48\text{ kHz}$ . The dark hues present in the TF diagram or spectrogram represents silence in between communication among the pod while hunting. There are two silence zones seen in this spectrogram intermittent to the communication parts.

The diagram in Fig. 2.10(b) depicts hunting pattern and involves more than one species, killer whales or 'A-pod' and their target<sup>5</sup>. The former being hunter and later, the hunted, and hence whistles, clicks, creaks and low-frequency moans are visible in this particular sound excerpt. The low-frequency moans prominent at the very beginning are from the hunted species or 'A-Pod'. The moaning is seen abundantly at different frequencies in the first half of the communicating region, which is missing in the later communicating region. Few clicks are observed in the first half of the communication and grow considerably in the second communicating region resulting in creaks. This phenomenon indicates that a group of species is involved in hunting.

In Chapter 3, the vocalisations (combination of click and whistles) shown and described in Fig. 2.10(b) are separated to find the echoes integrated with them. The Sec. 2.3 provided brief information about the peoples and organisation helped in the collection of data for the thesis.



(a) Time-domain representation of Killer whale Pod recorded sounds



(b) Time-Frequency representation of Killer whale Pod recorded sounds

**Figure 2.10:** Killer whale pod recordings at Vancouver Bay area while hunting in groups

<sup>5</sup><https://dosits.org/galleries/audio-gallery/marine-mammals/toothed-whales/killer-whale-orca>

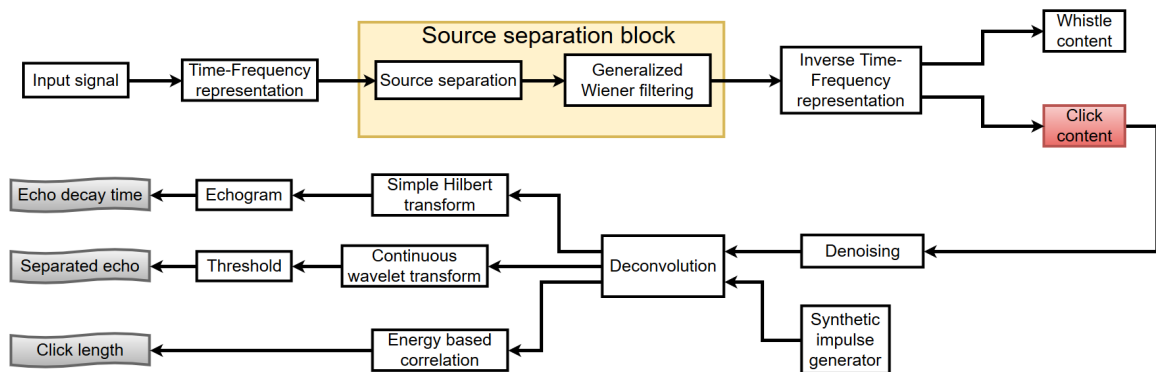


## Chapter 3

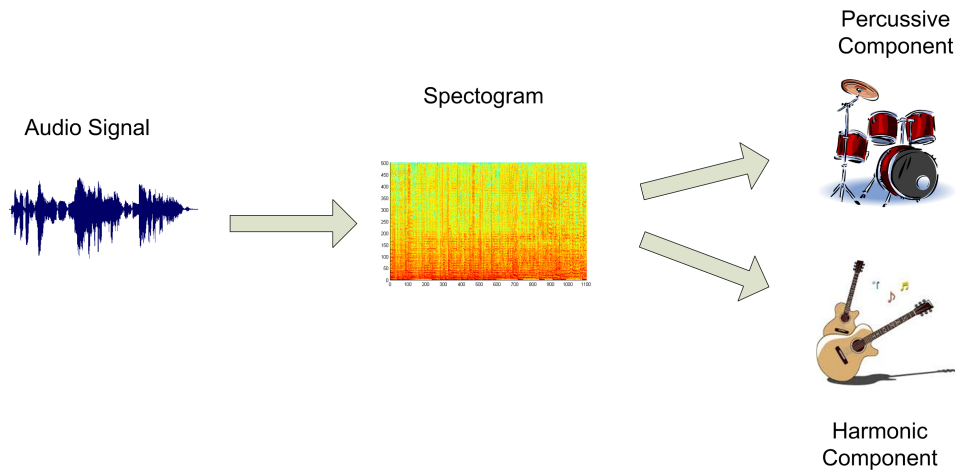
# Source Separation

Cocktail party problem can be used to introduce the basis of source separation which primarily refers to the problem of perceiving speech in noisy social settings. Acoustics from a different point of origin, as an individualistic source, forms a mixture and impinges on a recording system. Conducting a detailed analysis of these sounds helps in finding certain standalone entities of interest before the mixture. Hence, the recorded sound mixture termed as source and extraction of a specific, meaningful part from the source is termed as separation and hence follows the nomenclature which was proposed by Comon and Jutten in [14]. Bee and Micheyl in [15] argued that animal behaviour pattern can be taken under the same problem statement of the cocktail party to learn about their social communication mechanism and finding meaning about the sounds produced by them. This work hence adapts source separation concepts used in music separation in separating the vocal content of marine mammal subclass Odontoceti.

Fig.3.1 reiterates the overall approach to illustrate and highlight the second important aspect of the whole process. It blocks is elaborated in yellow and is emphasised in this chapter giving aiding in separating clicks from the audio recording as marked in red. After that, Fig. ?? shows a skeletal structure separating drum-like sounds (beat pulses) and the harmonics (changing vocalisation). As seen in Fig.2.10(b), the whale vocalisation follows the musical structure of beats (analogous to click structure) and voice (analogous to whistle structure) which motivates to explore the musical source separation technique to whale vocalisations and seen in Fig.3.5.



**Figure 3.1:** Source separation block, highlighted in yellow, described in this chapter as the second step of the thesis



**Figure 3.2:** *Skeletal representation of source separation*<sup>4</sup>

The short vertical pulse in the TF diagram or spectrogram are the clicks or beats, and the long horizontal edges are the whistles or voice or continuous sounds produced during a musical concert. The said structures motivate to implement the standardised source separation concepts utilised in separating musical instruments.

Source separation is a methodology applied to the audio mixtures to estimate the sound produced by each source. The two approaches aid to achieve separation of sounds are as follows:

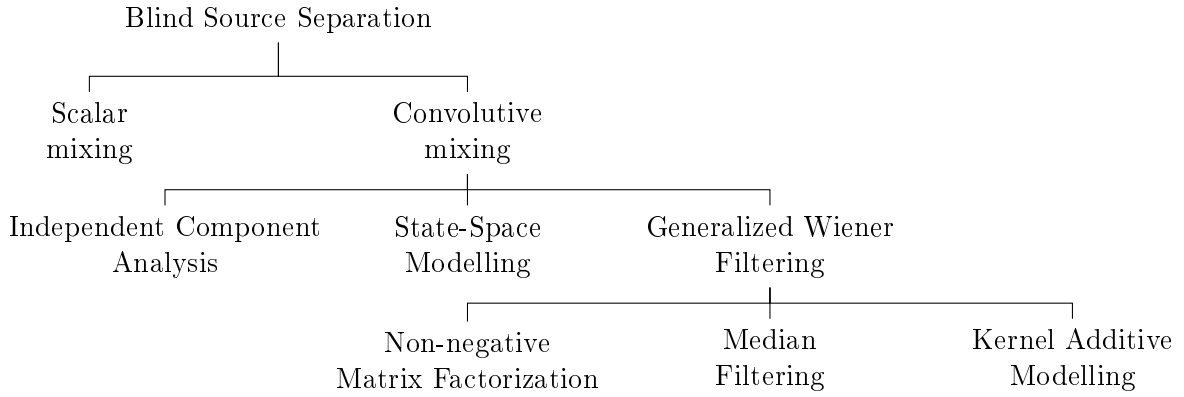
- **Blind Source Separation (BSS)** : The method separate different sources from a mixture, without the aid of information about the mixing process or the source signal [14]
- **Computational Auditory Scene Analysis (CASA)** : This a machine based approach which emulates human auditory system for any scene analysis which was first introduced by Bregman [16]. This method analyses the incoming signal by the use of the heuristic process.

### 3.1 Blind source separation

As described in the introductory section of Chapter 3, the objective of BSS is to estimate sources given only the mixtures and is apt to the thesis work. The little-known details about the captured source audio signals utilised except for the recording frequency of the hydrophones with which the mixtures. There are two classes of BSS algorithm, which depend on the nature of transfer function and are as follows:

- **Scalar mixing** : The transfer function from each source to the sound recording entity is considered to be a simple scalar.
- **Convulsive mixing** : The transfer function is dynamic with a non-trivial impulse response. In normal surroundings and sound occurrence, the mixing is convulsive. The algorithm used in the case of convulsive blind source separation are Independent Component Analysis (ICA), Non-negative Matrix Factorization (NMF) and others.

<sup>4</sup>[http://mir.ilsp.gr/harmonic\\_percussive\\_separation.html](http://mir.ilsp.gr/harmonic_percussive_separation.html)



**Figure 3.3:** *Classification of Blind Source Separation*

The idea of implementation of convolution mixing BSS carried out in many different ways; however, the three specific approach gained importance.

1. **Independent Component Analysis** or ICA [14, 17] assumes that all source are probabilistically independent and identically distributed concerning a non-Gaussian distribution and based on this assumption, the different feature representing non-Gaussian and independence feature maximises leading to the recovery of separated signals. The main shortcoming of this process is that modelling every model as independent and identically distributed is not possible, which is the central assumption of ICA.
2. **State-space modelling**, expressed regarding adaptive filtering problem, where sources are the composition of hidden state and the observation process results in mixtures [18, 19]. The source separation is done through state-interference techniques. This method is not suitable to modelled using linear dynamic model and computation complexity hinder the overall implementation.
3. **Generalized Wiener filtering** is the current paradigm [20–22] for separating sources under Gaussian assumption relying on the basic fundamental characteristic of redundancy in the spectrogram. The term redundancy in generalised Wiener filter method interpreted as the similar structure over equivalent frequency-magnitude and time-magnitude representation of time-frequency representation. The approach aids in the efficient recovery of estimated sources' time-domain waveform by reducing the mixture spectrogram.

Fig.?? elaborates the basic functioning of the source separation technique used for music separation. The audio signal in the figure is a mixture of two class of source, harmonics or continuous for a specific range of time and percussions or the momentary pulses in time. This segmentation is adequately identified in a spectrogram, first analysed by Ono et al. [23], which also helps in visualising different frequency content of the audio signal over time. The horizontal lines in the spectrogram show the harmonic content of the audio source, whereas the vertical lines represent the percussions present in the mixture and afterwards separation algorithm decimates the audio signal into its composition.

### 3.1.1 Generalized Wiener filtering

In this thesis, utilisation of generalised Wiener filtering based convolutive mixing BSS because of their efficiency in modelling dynamic systems reduced computational complexity and ease of

combining two or more algorithm based on performance. The methods developed and modified to fit the need for separation of marine sounds are as follows:

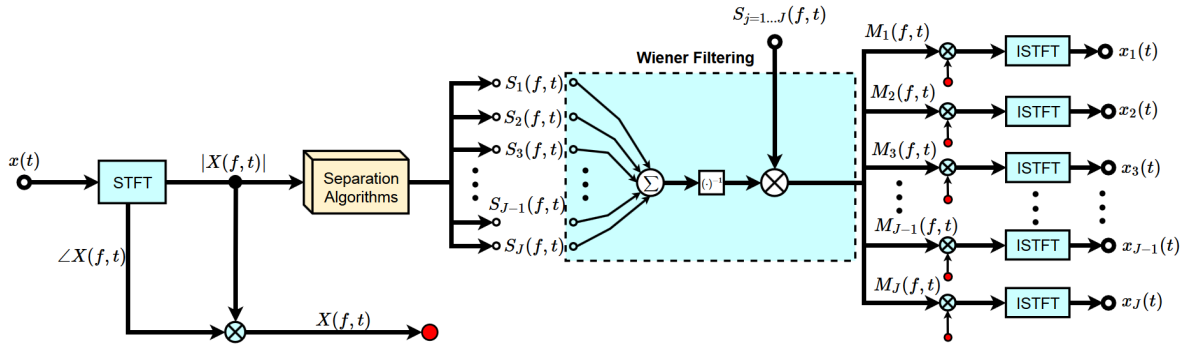
- Non-negative Matrix Factorization (NMF)
- Median Filtering (MF)
- Kernel Additive Modelling (KAM)

Fig. ?? depicts the general architecture of the algorithm applied in the separation process as blocks. The input signal  $x(t)$ , is first applied through the STFT (short-time Fourier transform) block and decimates the signal into phase  $\angle X(f,t)$  and magnitude  $|X(f,t)|$ . The magnitude information is the input to separation algorithms, which outputs the separated magnitude of the sources  $S_{j=1\dots J}(f,t)$ , where  $J$  represents the number of sources to separate from the mixture. Afterwards, the separated magnitude of the sources is filtered to create masks, which alternatively suppress entities related to other sources  $M_{j=1\dots J}(f,t)$ . The mathematical expression of generalised Wiener filtering approach is considered to create masking spectrogram for each source as:

$$M_{j=1\dots J}(f,t) = \frac{S_{j=1\dots J}(f,t)}{\sum_{j=1}^J S_j(f,t)} \quad (3.1)$$

Multiplying the original signal to the masks generated from each source to recreate the estimated source from the mixture in frequency domain representation of the separated sources  $X_{j=1\dots J}$ . Finally, the inverse STFT (ISTFT) provides the time-domain representation of each separated entities  $x_{j=1\dots J}(t)$ .

Next subsections focus into each of the methods in detail with intermediary results of algorithms. In the evaluation, all the methods are evaluated on metrics like peak-signal to noise ratio, signal-to-distortion ratio and others.



**Figure 3.4:** The general source separation architecture

## 3.2 Non-negative matrix factorization (NMF)

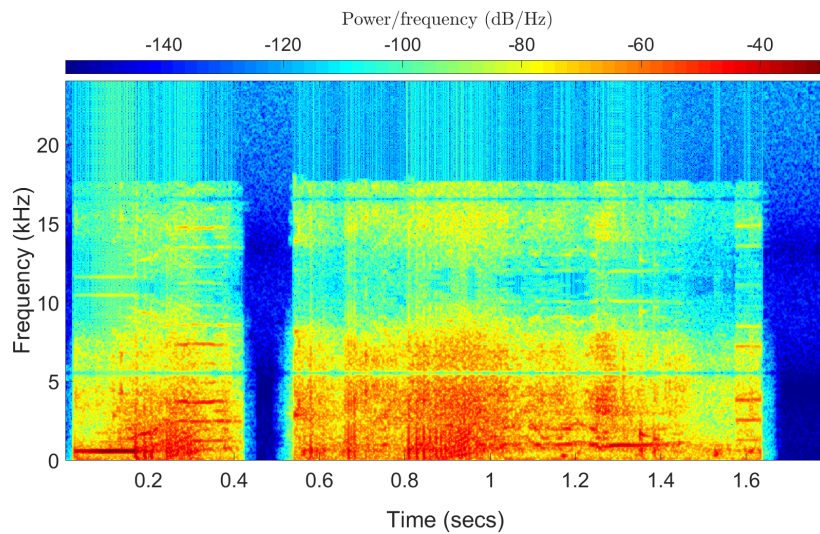
### 3.2.1 Background

As mentioned in the introduction of this chapter, the separation of percussive and harmonic content in a polyphonic monaural mixture is appealing and difficult at the same time. Demonstration of the advantages of using non-negativity as a constraint for matrix factorisation found in [24], [25], show benefits in learning the part representation of the data in consideration. In general consideration, percussion sound pertains to the echolocation clicks, and harmonics are the whistles and creaks.

The last decade witnessed an unprecedented development in the separation of sounds has its inception from matrix factorisation benefits showed by Lee and Seung [26], about segregating a matrix into two or more component matrices. The idea of multiplicative and additive updation of a randomly initialised matrices are also developed in [26]. Several other approaches described in [27–29] which utilised the mathematical ideologies in a meaningful separation of sounds into its harmonic and percussive components using the divergence approach and local minima calculation methodology. Moreover, [30] shows an learning algorithm with better percussion detection leveraging Weiner filtering approach. The disadvantage [20] is an omission of the sparseness and smoothness constraints over time an frequency in spectrogram representation.

In this thesis work, the aim is to detect three main sound types as mentioned in Sec. 2.2, which are whistle, clicks and creaks. The state-of-the-art techniques use outliers based method as pointed out by Ono et al. in [23], which assumes harmonics to be a deviation in temporal slice and percussive onsets are considered to be outliers in a frequency slice. Sec. 3.3 describes this method in detail.

Here, [26], which defines the segregation of a matrix into a product of two non-negative matrices and [20], assisting in the creation of the objective function. Fig. 3.6 represents the algorithm which is considered to develop the separation algorithm combing the methods described in the above papers. The mixture signal is decomposed using an objective function for including three



**Figure 3.5:** Defining smoothness and sparseness with killer whale sound excerpt

specific features as mentioned in [20]:

- Spectro-temporal features
- Anisotropic smoothness
- Anisotropic sparseness

The term *anisotropic smoothness* is defined as the difference in the direction of continuity between the spectrograms of harmonic and percussive sounds. The harmonic spectrogram is smooth/quasi-stationary in the short interval; however, the percussive counterpart is impulsive at the same time interval. When considering frequency interval, the effect is opposite. Percussions are typically smooth, and harmonics are impulsive. *Anisotropic sparseness* is the same as smoothness except that the consideration is in the direction of sparseness. Harmonic sounds are typically sparse in each of the frequency bins over time and percussions being sparse in each time frame over individual frequency bins.

The anisotropy helps in modelling harmonics as sparse in frequency and smooth in time whereas percussions defined as smooth in frequency and sparse in time. Fig. 3.5 is a elongated representation of Fig. 2.10 with a different colour scheme which is elaborated with colour-bar on the plot Fig. 3.5. In Fig. 3.5, if time interval from 0s to 0.2s is considered, the slowly increasing / fairly constant in amplitude artefact defines the harmonic spectrogram, while the short burst represents percussions. The converse is true for 0 Hz to 2.5 kHz.

The algorithm implements an unsupervised learning approach with the smoothness and sparseness criterion helping to discriminate between the harmonic and percussive content. The improvement achieved by merging the preliminary idea of NMF in [26] with the sparseness and smoothness criterion in [20] is of two-fold:

- A threshold is not required as the anisotropies do the discrimination
- It is robust in separating any number of sources, not only flat spectrum

The remainder of the section elaborates the mathematical detail, algorithm for the separation technique, variation of the parameters and results obtained from the method. Additionally, it with tabulate the peak-signal-to-noise ratio (PNR) after separation compared to the ground truth of the audio signal and extends to the signal-to-distortion ratio (SDR).

### 3.2.2 Mathematical foundation of NMF

As defined in [26], NMF is a technique to analyse multivariate data, in which the definition of magnitude spectrogram shown as a product of two non-negative matrices  $W$  and  $H$  as follows:

$$|X(f,t)| \approx W(f,k) \cdot H(k,t), \quad (3.2)$$

where  $k$ -th column of  $W$  is a frequency basis which represents the spectral pattern of the component that is active in the spectrogram and similarly  $k$ -th row of  $H$  defines the temporal gain of activation and represents the time interval over which the  $k$ -th frequency is active.



The magnitude spectrogram  $|X|$  is composed of  $F$  frequency bins,  $T$  frames such that a set of time-frequency units  $X(f,t)$  as described in Eq. 2.1, where  $f = \omega/2\pi$ ,  $f \in F$  and  $t \in T$ . Normalization of the magnitude spectrum is calculated at the very first step of the algorithm to ensure independence from the norm of input signal prior to applying the separated spectrogram to the Weiner filter as stated in Eq. 3.1. The normalization is dependent on the frequency bins, the number of time frames and the constant  $\beta$ . Hence, the normalization is given by:

$$[X_{n\beta}]_{F \times T} = \frac{|X|_{F \times T}}{\frac{\sum_{f=1}^F \sum_{t=1}^T X_{f,t}^\beta}{FT}}, \quad (3.3)$$

where  $F \times T$  represents the matrix size.  $[X_{n\beta}]$  denotes the normalized magnitude spectrogram  $|X|$  with respect to a particular  $\beta$  value. As the separation of click (percussions) and whistle (harmonics) are desired, the magnitude spectrogram is separated into two spectrograms,  $X_P$  or percussive spectrogram and  $X_H$  or harmonic spectrogram.

$$[X_{n\beta}]_{F \times T} \approx [X_H]_{F \times T} + [X_P]_{F \times T} \quad (3.4)$$

After initial decomposition as formulated in Eq. 3.4, each spectrogram is further broken down into its constituent basis and activation matrices as per Eq. 3.2.

$$[X_P]_{F \times T} \approx [W_P]_{F \times N_P} \cdot [H_P]_{N_P \times T} \quad (3.5)$$

$$[X_H]_{F \times T} \approx [W_H]_{F \times N_H} \cdot [H_H]_{N_H \times T}, \quad (3.6)$$

where  $X_H$ ,  $X_P$ ,  $W_P$ ,  $W_H$ ,  $H_P$  and  $H_H$  are non-negative matrices. Both  $X_H$  and  $X_P$  is of size  $F \times T$ . The two approximation are decimated in two matrices  $W_H$ ,  $H_H$  and  $W_P$ ,  $H_P$  as per Eq. 3.2, where  $k$  is either  $N_H$  or  $N_P$  is the components used for each source that is to be separated. More precisely, in this case,  $N_P$  and  $N_H$  denotes the number of percussive and harmonic components respectively used in the factorization process. The choice of the number of components used in each case is guided by the following rule:

- $N_P$  is chosen such that  $F \cdot N_P + N_P \cdot T \ll F \cdot T$
- $N_H$  is chosen similarly as  $F \cdot N_H + N_H \cdot T \ll F \cdot T$

### 3.2.3 Cost function

To find the approximation in factorizing Eq. 3.5 and Eq. 3.6, cost function requires to be defined for quantifying the approximation. It can be performed in two ways:

1. Firstly, measuring the distance between two non-negative matrices, for which Euclidean measure is a viable option. Hence the optimization problem can be defined as:

$$\arg \min_{W, H \geq 0} \{d(\mathbb{S}_{in}, \mathbb{S}_{out})\} \quad (3.7)$$

, where  $\mathbb{S}_{in} = X_{n\beta}$ ,  $\mathbb{S}_{out} = X_P + X_H$ .

2. Secondly, calculation of divergence between the input signal and its decomposed parts. It can be formulated as:

$$\arg \min_{W, H \geq 0} d_\beta(\mathbb{S}_{in} | \mathbb{S}_{out}) \quad (3.8)$$

The functions described in 1 and 2 can be solved by finding global minima. The method used in this work is gradient descent. This advantage for this approach is its simplest implementation however it has few inherent disadvantages, one being sensitive to the step size and other is that the time taken to converge is slow. This work incorporates the divergence-based optimisation technique [31] and includes the anisotropy measurement developed in [20] defining the generalised cost function. The explanation of cost function is in the following section.

### 3.2.3.1 $\beta$ -divergence

Consideration of  $\beta$ -divergence factor is suggested in [31] as a method to minimize the according to Eq. 3.8. However,  $\beta$ -divergence solely cannot determine if a basis function is of percussion or harmonic class and so the divergence function includes the spectro-temporal features and anisotropies. The value of  $\beta$  depends on the divergence method used. In this work, three kind of divergence are exploited - Euclidean distance ( $\beta = 2$ ), Kullback-Leibler ( $\beta = 1$ ) and Itakura-Saito ( $\beta = 0$ ). Equation for  $d_\beta(\mathbb{S}_{in} || \mathbb{S}_{out})$  is defined as follows:

$$d_\beta(\mathbb{S}_{in} | \mathbb{S}_{out}) = \begin{cases} \frac{1}{\beta(\beta-1)} \sum_{f=1}^F \sum_{t=1}^T (\mathbb{S}_{in,f,t}^\beta + (\beta-1)\mathbb{S}_{out,f,t}^\beta - \beta\mathbb{S}_{in,f,t}\mathbb{S}_{out,f,t}^{\beta-1}), & \beta \in (0,1) \cup (1,2] \\ \sum_{f=1}^F \sum_{t=1}^T \mathbb{S}_{in,f,t} \log \frac{\mathbb{S}_{in,f,t}}{\mathbb{S}_{out,f,t}} - \mathbb{S}_{in,f,t} + \mathbb{S}_{out,f,t}, & \beta = 1 \\ \sum_{f=1}^F \sum_{t=1}^T \frac{\mathbb{S}_{in,f,t}}{\mathbb{S}_{out,f,t}} - \log \frac{\mathbb{S}_{in,f,t}}{\mathbb{S}_{out,f,t}} - 1, & \beta = 0 \end{cases} \quad (3.9)$$

### 3.2.3.2 Generalized cost function

Percussive sounds generally described as broadband signals whereas harmonics are the narrowband signals. Spectral smoothness ( $\mathbb{F}_{SM}$ ) and temporal sparseness ( $\mathbb{T}_{SP}$ ) defined in [20] to additionally model percussive spectrograms while temporal smoothness ( $\mathbb{T}_{SM}$ ) and spectral sparseness ( $\mathbb{F}_{SP}$ ) are motivated to model harmonic spectrograms. These factors are weighted by a predefined constant to investigate their contribution further. The formed cost function not only considers the divergence cost due to gradient descent measure based on beta divergence but also the arrangement of data-points, which are either sparse (abrupt along time or frequency) or smooth (continuous along time or frequency) helping in modelling the magnitude spectrogram in detail.

The generalized cost function is then defined as:

$$D = \begin{cases} d_\beta(X_{n\beta}|(X_P + X_H)), & \text{divergence cost} \\ +K_{\mathbb{F}_{SM}} \cdot \mathbb{F}_{SM} + K_{\mathbb{T}_{SP}} \cdot \mathbb{T}_{SP}, & \text{percussive cost} \\ +K_{\mathbb{T}_{SM}} \cdot \mathbb{T}_{SM} + K_{\mathbb{F}_{SP}} \cdot \mathbb{F}_{SP}, & \text{harmonic cost} \end{cases} \quad (3.10)$$

The sparseness objective  $\mathbb{T}_{SP}$  and smoothness objective  $\mathbb{F}_{SM}$  for percussive cost is followed by [20] as:

$$\mathbb{F}_{SM} = \frac{T}{N_P} \sum_{n_p=1}^{N_P} \frac{1}{\sigma_{W_{P_{n_p}}}^2} \sum_{f=2}^F (W_{P_{f-1}, n_p} - W_{P_f, n_p}) \quad (3.11)$$

$$\mathbb{T}_{SP} = \frac{F}{N_P} \sum_{n_p=1}^{N_P} \sum_{\tau=1}^T \left| \frac{H_{P_{n_p}, \tau}}{\sigma_{H_{P, n_p}}} \right| \quad (3.12)$$

From [20], the sparseness objective  $\mathbb{F}_{SP}$  and smoothness objective  $\mathbb{T}_{SM}$  for harmonic cost is modified as:

$$\mathbb{T}_{SM} = \frac{F}{N_H} \sum_{r_h=1}^{N_H} \frac{1}{\sigma_{H_{H_{n_h}}}^2} \sum_{\tau=2}^T (H_{H_{n_h}, \tau-1} - H_{H_{n_h}, \tau}) \quad (3.13)$$

$$\mathbb{F}_{SP} = \frac{T}{N_H} \sum_{n_h=1}^{R_H} \sum_{f=1}^F \left| \frac{W_{H_{f, n_p}}}{\sigma_{W_{H, n_h}}} \right| \quad (3.14)$$

The normalization factor  $\sigma_j$ , where  $j = W_{P_{r_p}}, H_{H_{r_h}}, H_{P_{r_p}}, W_{H_{r_h}}$  are incorporated to make the global objective function defined by Eq. 3.10 independent of the signal norm. Smoothness cost assigns high cost to large changes in frequency bin in case of percussion and when harmonics are considered, high cost is assigned to large changes in time.

The constant weighing of the percussive and harmonic cost, which are given by  $K_{\mathbb{F}_{SM}}, K_{\mathbb{T}_{SP}}, K_{\mathbb{T}_{SM}}$  and  $K_{\mathbb{F}_{SP}}$ , are kept equal while initializing the algorithm. They are hence simplified as:

$$K_{SM} = K_{\mathbb{F}_{SM}} = K_{\mathbb{T}_{SM}} \quad (3.15)$$

$$K_{SP} = K_{\mathbb{F}_{SP}} = K_{\mathbb{T}_{SP}} \quad (3.16)$$

Following [26], the multiplicative update rule is applied in conjunction with [20] as anisotropic components are present in the cost function given in Eq. 3.10. As gradient of the total cost is the weighted sum of the divergence cost, harmonic cost and temporal cost. In general, the gradient can be written as a difference between two non-negative terms:

$$\nabla D(W_{H,P}, H_{H,P}) = \nabla D(W_{H,P}, H_{H,P})^+ - \nabla D(W_{H,P}, H_{H,P})^-, \quad (3.17)$$

where  $D$  is given by Eq. 3.10. If  $\epsilon > 0$  is considered to be the minimum possible distance from a certain  $H, P$ , The term  $\nabla D(W_{H,P}, H_{H,P})^+ = \nabla D(W_{H,P}, H_{H,P}) + \epsilon$  and  $\nabla D(W_{H,P}, H_{H,P})^- = \nabla D(W_{H,P}, H_{H,P}) - \epsilon$ . Hence the multiplicative update rule is given by:

$$Z = Z \odot \frac{\left[\frac{\partial D}{\partial Z}\right]^-}{\left[\frac{\partial D}{\partial Z}\right]^+}, \quad (3.18)$$

where  $\odot$  represents element-wise product and fraction represents element-wise division operator. Substituting  $Z$  with percussive and harmonic basis ( $W_{H,P}$ ) function and activation ( $H_{H,P}$ ) function, the update rule for each of them is derived. The resultant updation rule is as follows:

$$W_P = W_P \odot \frac{\left[\frac{\partial d_\beta}{\partial W_P}\right]^- + K_{SM} \left[\frac{\partial \mathbb{F}_{SM}}{\partial W_P}\right]^-}{\left[\frac{\partial d_\beta}{\partial W_P}\right]^+ + K_{SM} \left[\frac{\partial \mathbb{F}_{SM}}{\partial W_P}\right]^+} \quad (3.19)$$

$$H_P = H_P \odot \frac{\left[\frac{\partial d_\beta}{\partial H_P}\right]^- + K_{SP} \left[\frac{\partial \mathbb{T}_{SP}}{\partial H_P}\right]^-}{\left[\frac{\partial d_\beta}{\partial H_P}\right]^+ + K_{TSP} \left[\frac{\partial \mathbb{T}_{SP}}{\partial H_P}\right]^+} \quad (3.20)$$

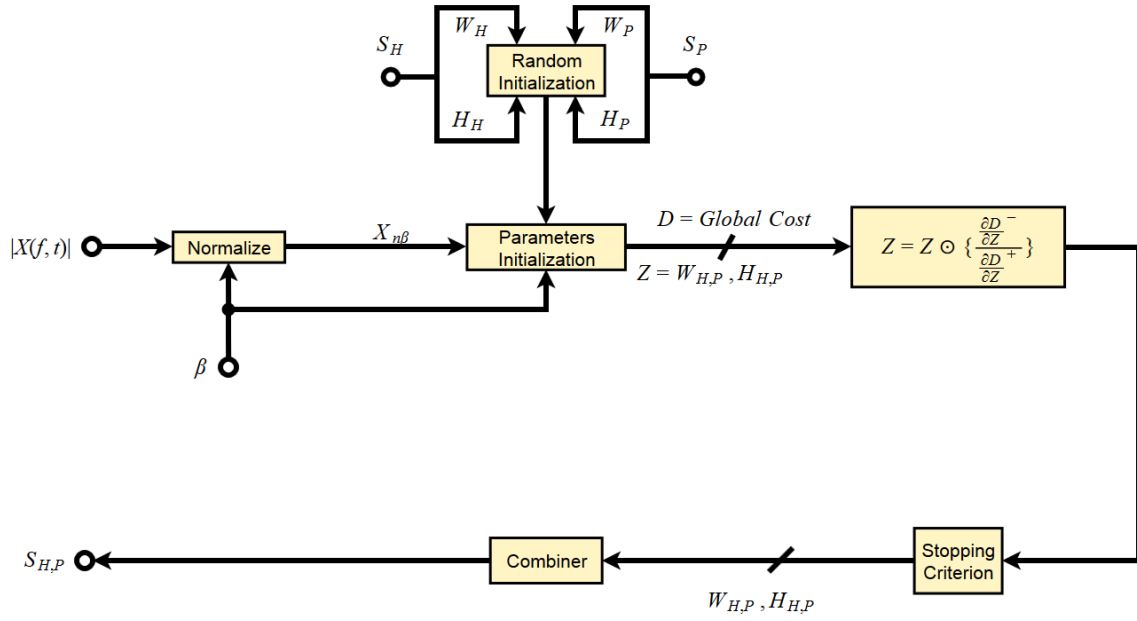
$$W_H = W_H \odot \frac{\left[\frac{\partial d_\beta}{\partial W_H}\right]^- + K_{SP} \left[\frac{\partial \mathbb{F}_{SP}}{\partial W_H}\right]^-}{\left[\frac{\partial d_\beta}{\partial W_H}\right]^+ + K_{SP} \left[\frac{\partial \mathbb{T}_{SP}}{\partial W_H}\right]^+} \quad (3.21)$$

$$H_H = H_H \odot \frac{\left[\frac{\partial d_\beta}{\partial H_H}\right]^- + K_{SM} \left[\frac{\partial \mathbb{T}_{SM}}{\partial H_H}\right]^-}{\left[\frac{\partial d_\beta}{\partial H_H}\right]^+ + K_{SM} \left[\frac{\partial \mathbb{F}_{SM}}{\partial H_H}\right]^+} \quad (3.22)$$

The size of each matrices  $W_H, W_P, H_H, H_P$  as mentioned in Eq. 3.5 and Eq. 3.6. The updation is done iteratively with the number of iteration is varied to identify best-case scenario keeping the complexity of the separation procedure to an optimum. A requirement of minimum 10 iterations for generating a satisfactory result and no significant difference found above 50 iterations.

### 3.2.4 Reconstruction

The recovery of percussive signal (signal with clicks) from percussive spectrogram  $X_P$ , and restoration of harmonic signal (signal with vocals - whistles and creeks) creates the harmonic



**Figure 3.6:** Diagram for implementation of non-negative matrix factorization

spectrogram. They are both computed by the product of their respective basis and activation function. Consequently, the masks are created for each signal (percussive and harmonic) by the help of Eq. 3.1 making the process conservative, i.e. each frequency bin and time bin is scaled to a certain ratio to define the contribution of a respective source to the mixed spectrogram  $X_{n\beta}$ . Thereafter, the general process follows per Fig. ?? giving the separated source as  $x_p$  (clicks) and  $x_h$  (creaks and whistles).

### 3.2.5 Algorithm

The algorithm implemented combines the individual principal of [20, 25, 26, 31] to separate the click & whistle content and Fig. 3.6 proposes separation algorithm for non-negative matrix factorisation which separates source signal into required constituents. As assumed that the varying clicks or creaks, defined in Sec. 2.2, are considered part of whistles, the distinct whistle sounds are treated recursively with the algorithm to separate the creaks and whistle<sup>6</sup>. The first step involves a declaration of  $N_P$ ,  $N_H$ , stopping criterion.

The initialisation of randomly generated matrices,  $W_P$ ,  $H_P$ ,  $H_H$  and  $W_H$ , are fed to the parameter initialisation step along with the  $\beta$ -divergence value as per requirement. In this experiment, setting the value of  $\beta$  at 1.15 provided an optimum separation result. The stopping criterion is the number of iteration required for convergence of the initialised matrices to the normalised version of an input signal. The updation block receives the global cost function and the initialised matrices as its input, which updates each of the initialised matrices until the stopping criterion is valid. Afterwards, the updated matrices are combined to produce the separated click and whistle magnitude spectrogram for further processing.

In the second iteration involves the separation of whistles and creaks. The separated whistles are fed to the parameter initialisation separation and using the same procedure as mentioned

<sup>6</sup><http://www.sciencemag.org/news/2014/08/whales-and-dolphins-squeal-delight>

| Parameter name      | Value for click & whistle extraction | Value for creak & whistle extraction |
|---------------------|--------------------------------------|--------------------------------------|
| $\beta$             | 1.15                                 | 1.15                                 |
| $N_P, N_H$          | 50, 75                               | 50, 35                               |
| Number of iteration | 50                                   | 50                                   |
| Window length       | 0.02 s                               | 0.04 s                               |

**Table 3.1:** Parameters used for non-negative matrix factorization algorithm

keeping  $R_P, R_H$  fixed but different  $\beta$  value which decomposes magnitude spectrogram of whistle into creaks and residual whistle. The algorithm implemented is shown in Algorithm. 1.

### 3.2.6 Separation result

Fig.3.7 pictured the separated whistle and clicks from two recorded species and the parameters shown in Table.3.1. Beta value is taken to be 1.15 with  $N_p = 50$  and  $N_h = 75$  for all the cases as best possible signal-to-distortion ratio and peak-to-noise ratio are obtained with these values. Generation of two separated sources obtained by implementing the Algorithm. 1 plugging mentioned variable values. Recursive implementation of the same algorithm with a slight decrement of the parameter  $N_h = 35$  provides the separation of creaks from the separated clicks. The change in window size is as slowly changing clicks are defined to be creaks. Fig. 3.8. The reason to chose 2.5 s from the recording is for better visualisation of in-lying components. For separation the blind source into whistles and clicks, the recordings are represented in TF diagram for further processing with the values of window length set to 0.02 s with an overlap of 50% to 85% as described in Sec 2.3.1.1. However, during recursively applying the algorithm for creak extraction the window size is increased to 0.04 s for better frequency resolution following the Eq. 2.2 keeping overlap region unchanged. The reason is that creak is defined in this thesis as slowly evolving clicks and hence is concluded as sharply changing over time unlike whistles, which are almost smooth over time.

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#### Algorithm 1: Non-negative matrix factorization

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**Result:** Producing source separated magnitude spectrogram for click and whistle

**Initialization:**

- $|X(f,t)|$
- $N_P, N_H, \beta$ , Stopping criterion
- $F$  = frequency bins,  $T$  = time frames of  $|X(f,t)|$

**Intermediate step:**

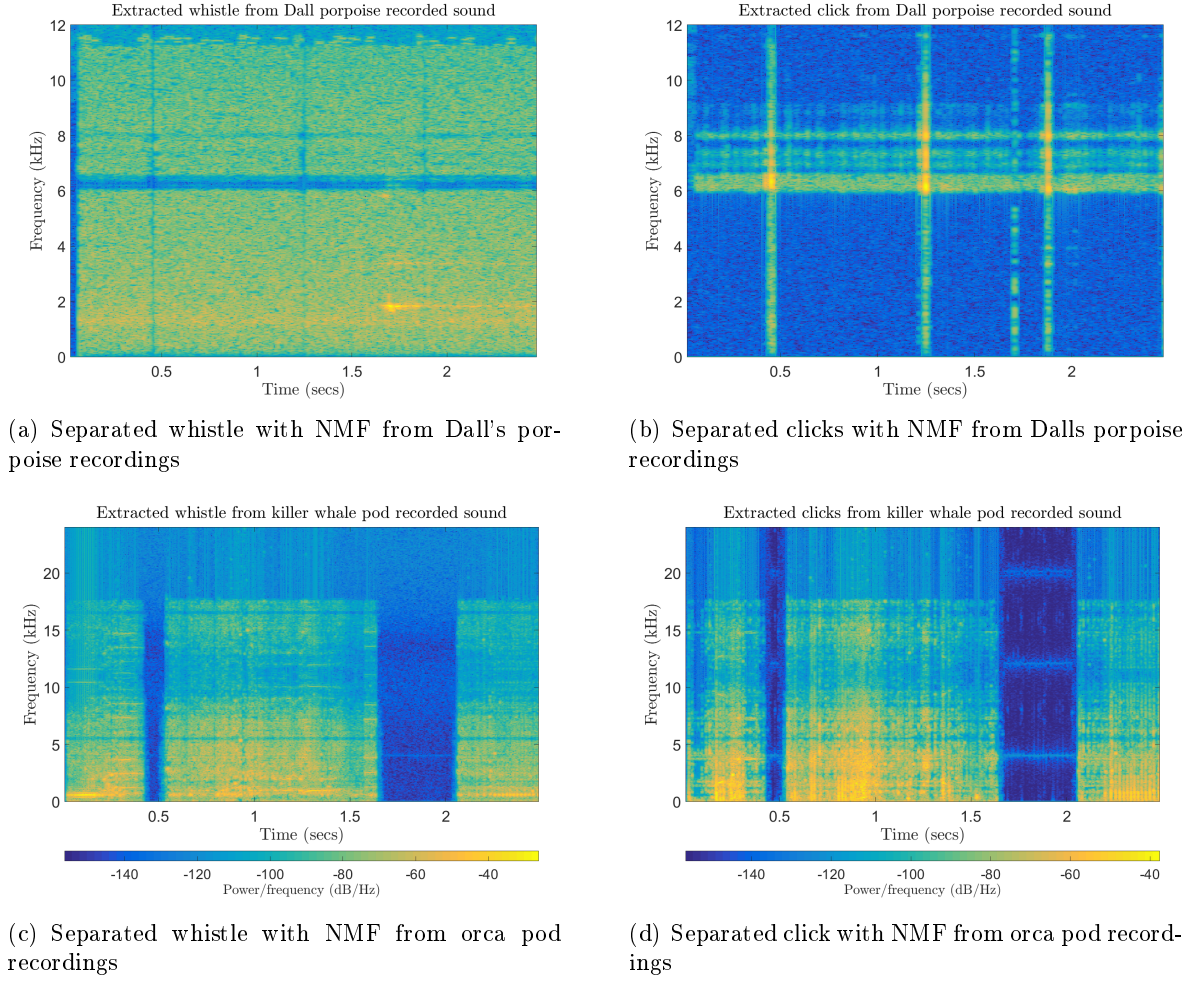
- Random initialization of matrices  $W_P, H_P, H_H, W_H$
- normalize  $|X|$  according to Eq. 3.3

**while**  $\neg$  no. of iteration **do**

- Update  $W_P$  according to Eq. 3.19
- Update  $W_H$  according to Eq. 3.21
- Update  $H_P$  according to Eq. 3.20
- Update  $H_H$  according to Eq. 3.22

**end**

---



**Figure 3.7:** Source separation into two sources with non-negative matrix factorization

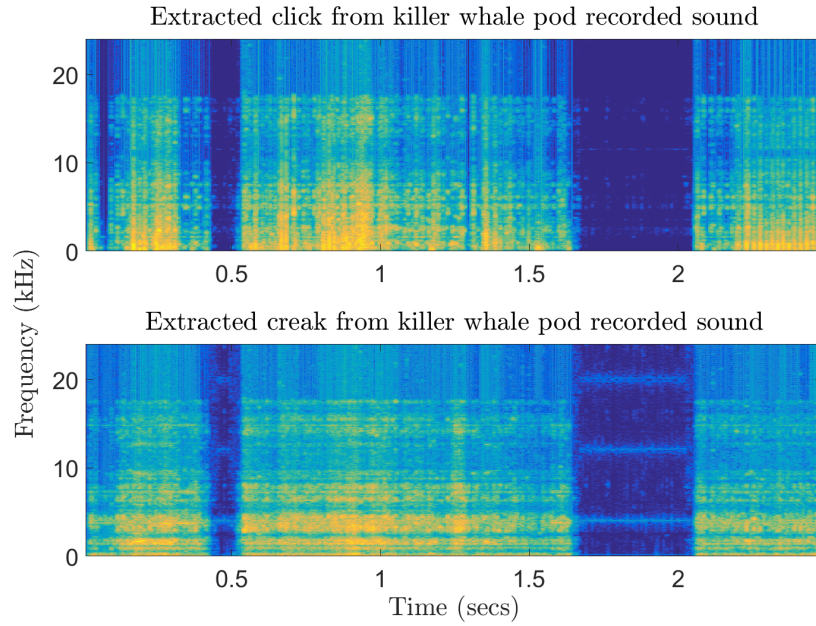
Fig.3.7 shows four sub-plot with click and whistle extractions with the parameters and algorithm mentioned. The separation between the clicks and creaks are performed by further iterating the clicks. Two critical visualisations performed in this particular context. Firstly, observing noise in between two separated high energy clicks which are termed as burbling artefacts in this thesis and provides the basis for signal-to-distortion ratio and signal-to-artefact ratio for evaluating the effectiveness of the separation algorithm. Separation of whistles from the audio observed complementing the clicks, but noise floor seemed higher when compared with the click extraction.

In Fig. 3.8, only constant sounds are visualised in separated creaks, which concludes that these recordings do not have creak content in them. However, the first visualisation suggests further cleaning of clicks with a second iteration of the complete algorithm. Thus iterative treatment of the separation algorithm can reduce the burbling effect helping in a cleaner representation of clicks culminating in the introduction of denoising block before deconvolution.

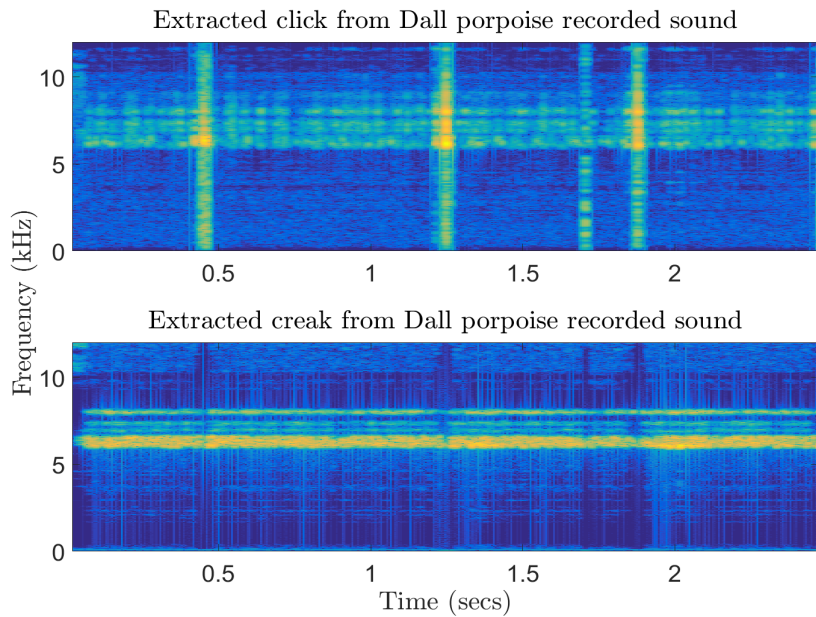
The results from non-negative matrix have efficiently separated the characteristics of sound from a whole recording. The significance of the achieved result is due to an efficient merger of the cost function. However, NMF suffers high computational complexity because of its higher iteration requirement for proper convergence. Secondly, the doubling of window length for

better frequency resolution is somewhat arbitrary and worked for the collected dataset, and need more tests with other possible found datasets. Thirdly, evaluating the value of other parameters with SDR for a couple of dataset and average of the values are taken for the algorithm as variance found with the tested ten recordings of different sampled frequency data are found to be within one percent of the mean.

The computational complexity of this method propelled in looking for a faster algorithm and hence median filtering is tested intending to attaining similar results as NMF.



(a) Creaks separation from clicks of orca clicks



(b) Creaks separation from clicks of Dall porpoise clicks

**Figure 3.8:** No creaks found in Dall's porpoise or in killer whale, as separated by NMF, however static sources are removed



### 3.3 Median filtering (MF)

#### 3.3.1 Background

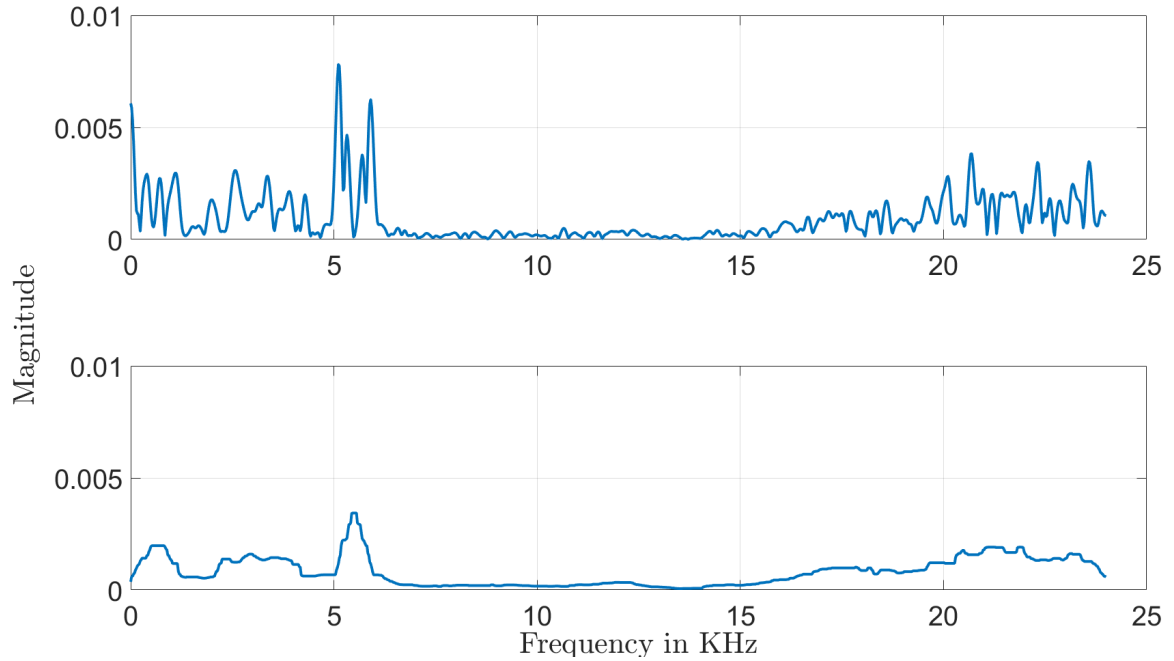
Non-negative matrix matrix factorization is computationally complex and hence time consuming. The main reason is that the global cost function incorporates the percussive and harmonic costs as depicted by Eq. 3.10 on top of divergence cost. In addition to that, the magnitude spectrogram is divided into percussive and harmonic spectrogram, refer Eq. 3.4, and further division to their respective activation and basis matrices which is defined by Eq. 3.5 & 3.6.

Median Filtering is a known used approach in image processing for speckle noise and salt & pepper noise removal from images [32]. It operates by replacing the given sample point of a signal by the median of the window around that given sample. Mathematically, for a given input vector  $v_x(n)$  and its corresponding median filter output  $v_y(n)$  is given by:

$$v_x(n) = \text{median}\{v_y(n-l : n+l)\}, \quad l = w/2 \quad , \text{ when } w \text{ is even} \quad (3.23)$$

$$v_x(n) = \text{median}\{v_y(n-l : n+l)\}, \quad l = w - 1/2 \quad , \text{ when } w \text{ is odd}, \quad (3.24)$$

Where  $n$  denotes the number of samples in the input vector,  $w$  is the window length upon which the median filtering is taking place. The original value is replaced by the middle of the values present within filtering window if the window length is odd. If the filter window length is even, then mean of the middle two values are considered as the median of that window, and the calculated mean replaces the original value.



**Figure 3.9:** Median filtering performed over first time frame. **Top:** Unfiltered magnitude vs frequency plot for first time frame. **Bottom:** Median filtered output for first time frame enhancing the percussive/click component.

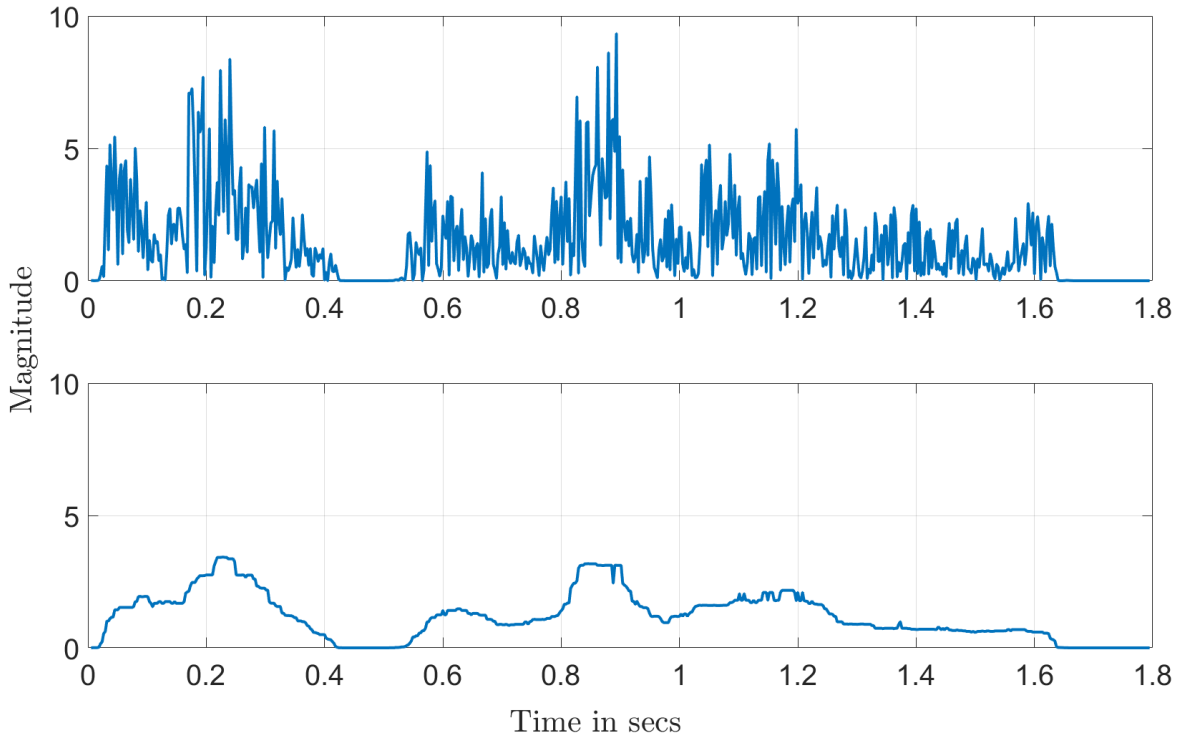
To apply median filtering in spectro-temporal space, over time frames and frequency bins, let's denote each element of the input magnitude spectrogram as  $|X(f,t)|$ . So,  $t^{th}$  time frame can be denoted as  $|X|_t$  and  $f^{th}$  frequency bin can be termed as  $|X|_f$ . Median filtering is performed along each time frames to generate a percussion enhanced spectrogram, as :

$$\mathbb{S}_{P_t} = \text{median} \{|X|_t, l_{perc}\}, \quad (3.25)$$

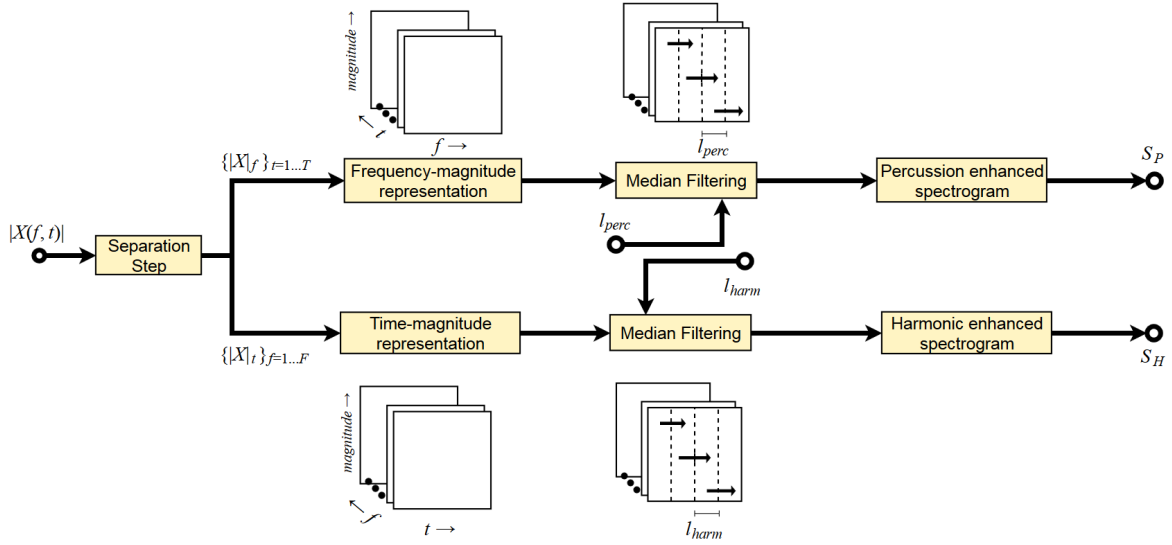
Where  $l_{perc}$  denotes the filter length for creating individual percussion enhanced frames  $\mathbb{S}_{P_t}$ , which are compounded to form percussion enhanced spectrogram  $S_P$ . Similarly, to create the harmonic enhanced spectrogram median filter is performed in a complementary method, i.e., along with the frequency bins as follows:

$$\mathbb{S}_{H_f} = \text{median} \{|X|_f, l_{harm}\}, \quad (3.26)$$

Where  $l_{harm}$  denotes the filter length for creating individual harmonic enhanced frames  $\mathbb{S}_{H_f}$ , which are further combined to form harmonic enhanced spectrogram  $S_H$ . These are further processed to generate masks utilising Weiner filtering technique as described by Eq. 3.1 and then, the masks generated are multiplied with the phase of the original signal to recreate the real-time separated signals. In Fig. 3.9, creation of percussion enhancement for the first time frame is shown. The figure on top shows the magnitude vs frequency plot, where spikes represent the harmonics and are unwanted elements in case of clicks or percussions.



**Figure 3.10:** Median filtering performed over first frequency bin. **Top:** Unfiltered magnitude vs time plot for first frequency bin. **Bottom:** Median filtered output for first frequency bin enhancing the harmonic/whistle component.



**Figure 3.11:** Work-flow of median filtering separation technique

Median filtering with a window length of 25 is done to subjugate the influence of whistles or harmonics, and the filtered result seen in the bottom of Fig. 3.9. The similar experiment performed over the first frequency bin, and the harmonic enhanced bin shown in Fig. 3.10. The visible fluctuations in the magnitude vs time plot in the top of Fig. 3.10 are the outliers for harmonics which are discarded by the median filter of the same length and can be observed by the bottom plot of Fig. 3.10.

### 3.3.2 Order of complexity

The implementation of median filtering is motivated due to its incredibly reduced complexity when compared to non-negative matrix factorisation. Two operation sequentially takes place, to find the median of a given window length, first being sorting of values in the given window and finally is to find the middle element. In the context when the window length is even, mean of the two middle elements are calculated.

Quicksort is exploited to sort data within the median filter window as mentioned by Sedgewick in [33, 34]. The algorithm arranges samples in ascending order of their magnitude with an operational complexity of  $\mathcal{O}(n \log n)$  calculated for an average case performance, and with a recursive operation, the complexity reduces to  $\mathcal{O}(\log n)$ . Calculating the middle of filter window have a complexity of  $\mathcal{O}(n/2)$  and mean calculation takes place when the window length is even, and the complexity is  $\mathcal{O}(1)$ . Hence the total complexity of the whole operation is  $\mathcal{O}(n \log n) + \mathcal{O}(n/2) + \mathcal{O}(1)$  or equivalently  $\mathcal{O}(n \log n)$  per iteration in comparison to the  $\mathcal{O}(FT)$  for non-negative matrix factorization.

### 3.3.3 Implementation of median filtering

Median filtering is fairly simple to implement among the three studied algorithm - NMF, MF and KAM. Fig. 3.11 elaborates the details of processing step. The input magnitude spectrogram is separated into frequency-magnitude and amplitude-magnitude representation to apply

median filtering to both of them. Progressing to the next stage of median filtering, each of the representation of spectrograms requires a certain length of the median filter and a stopping criterion to create percussion and harmonic enhanced spectrograms.

Generation of a percussion-enhanced spectrogram requires a median filtering window length of  $l_{perc}$  through each time-frame of the frequency-magnitude representation. On the contrary, creating the harmonic enhanced spectrogram required  $l_{harm}$  as the window length and calculated for each frequency bin to smooth-en out the offshoots produced by percussions. The median filter window is the lone parameter that is varied to summarise different results obtained because of the window.

Typically, both the windows are kept short to maximise the smoothing of enhanced spectrogram thereby increasing time consumed for the whole separation procedure. In this work, typically the window length is kept at 10 samples which are then varied till 50 samples to observe the degradation. When window length is relatively small, the fluctuation represented by the magnitude of the outlier remains in the filtered output result whereas a higher window length accounts to loss of information in click/percussion & whistle/harmonic enhancement. Table. 3.2 provides the set of optimal value used for median filtering approach.

$S_{H,P}$  denoted the enhanced spectrograms, where  $H$  represents the harmonic enhanced spectrogram, and  $P$  denote percussion enhanced spectrogram. They are the input for further post-processing generating the separated signal.

---

**Algorithm 2:** Method to proceed for median filtering

---

**Result:** Separated magnitude spectrograms by median filtering

---

initialization:  $|X|$

$W_{len}$  = length the window used for STFT

$l_{perc}$  = median filter window for percussion enhanced spectrogram

$l_{harm}$  = median filter window for harmonic enhanced spectrogram

**for**  $f = 1$  **to**  $F$  **do**

| Update using Eq. 3.25

|  $f = f + 1$

**end**

**for**  $t = 1$  **to**  $T$  **do**

| Update using Eq. 3.26

|  $t = t + 1$

**end**

---

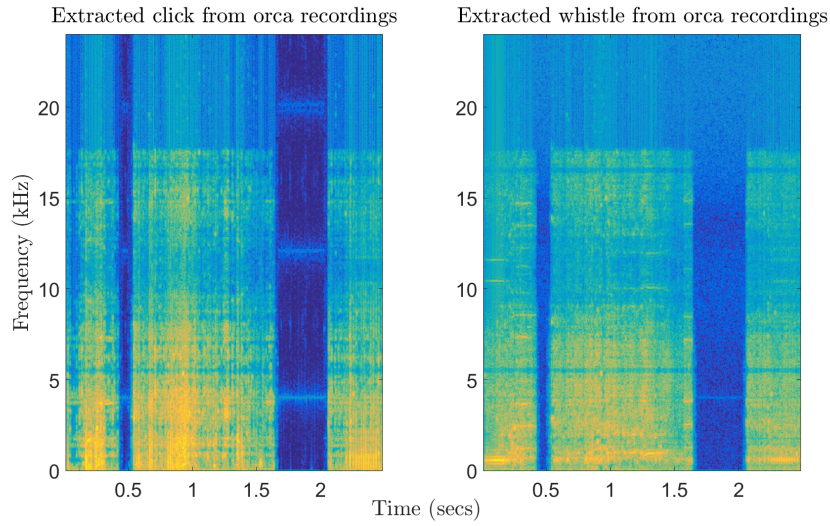
### 3.3.4 Separation results

Median filtering can be used to detect the sharp pulses like drum and analogously the echolocation clicks in a swift manner which when put into perspective simplifies to the fact that a 2.5 s recording excerpt separated in 0.18 s in average and if any single kind of separation trained a few times MF does the same task in about 0.15 s. The time taken for separation by the other two methods are at least the same time as the recording excerpt taken into consideration and with NMF, if the parameter values of  $N_P$  and  $N_H$  are below 20, operation time rise to twice the audio file in the process.

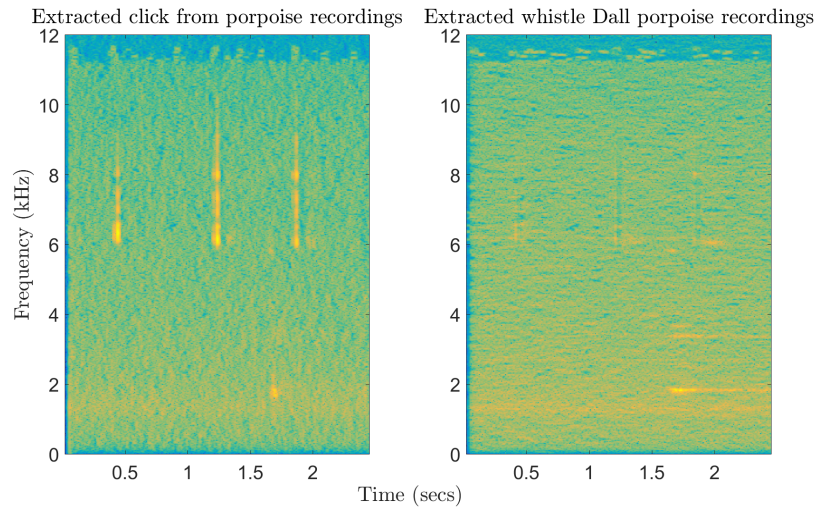
| Parameter name | Optimal value |
|----------------|---------------|
| $l_{harm}$     | 15 samples    |
| $l_{perc}$     | 12 samples    |

**Table 3.2:** *Parameters used for median filtering algorithm*

Fig. 3.12 shows the separation of the same recordings with median filtering approach. The harmonic window is taken to be 15 *samples* and the percussive window set to 12 *samples* for best separation result based on the evaluation criteria of PNR (peak-to-noise ratio). Other evaluation includes SDR (signal-to-distortion ratio) and other simple error calculation as RMS error, Pearson correlation coefficient and magnitude squared coherence between the ground truth and the separated results and found in Sec. 3.5. The median filtering window is acceptable until the



(a) Click and whistle extraction from orca recordings



(b) Click and whistle extraction from Dall porpoise recordings

**Figure 3.12:** *Source separation into two sources with median filtering*

harmonic window of length 50 *samples* and the percussive window of length 35 *samples*.

The main advantage of median filtering being computationally fast and moreover easy to implement. The parameters required to perform the separation also restricts to only 2, which simplify the whole process. However, disadvantage looms over this process regarding burbling artefacts. These particular artefacts are more visible than that of NMF in case of click detection, which remains the primary goal of this step. A state-of-the-art separation technique is used, known as Kernel Additive Modelling (KAM), which incorporates NMF to model the sources and afterwards economic modelling is performed using median filtering to reduce such artifacts. The next section elaborates this separation algorithm.

### 3.4 Kernel additive modelling (KAM)

In both of the previous cases, the complete mixture is used as input of the separation procedure. In case of NMF, a global cost function developed for modelling the entire spectrogram. Whereas for the median filtering technique, the approach based on frame or bin operation.

Kernel additive modelling [22] motivates the separation problem using random source initialization thereby modelling the sources, which is the central theme for NMF, and applying median filtering simultaneously for minimizing the local cost function similarly to [35], thus combining the rigorous work on Generalized Additive Models (GAM) [36], showcasing each source depends only on a specific coordinate. The work of Friedman et al. in [37] further points out that a particular projection of the input source considered under the terminology Projection Pursuit Regression (PPR). The Kernel additive modelling revolves around the concept of Kernel back-fitting as coined in [22] and is broadly follows three crucial steps - separation of the source or input signal, modelling constants of the sources locally and finally implementing the convergence criterion.

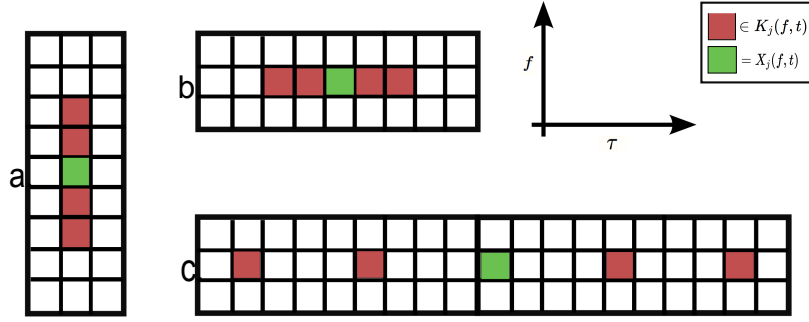
The significant difference between the methods described in Sec.3.2, 3.3 and KAM are that the total audio input considered for separation rather than magnitude only approach.

#### 3.4.1 Mathematical background

##### 3.4.1.1 Source modelling

In the case of underwater acoustic sounds, the sounds captured by hydrophone arrays, and hence they can be thought to be premixed before analysis and as each of the sounds are considered monophonic or heard from one position at any instantaneous time. When hydrophone arrays are involved, the recorded sounds may be described as multichannel, otherwise in a limited case when only single hydrophone involved in recording. The source/input signal spectrogram can be represented by the sum of  $N$  unknown signal's spectrogram as  $\tilde{X}_{1...N}$ , where the separation of  $x$  done as:

$$x(t) = \sum_{j=1}^J x_j(t), \quad (3.27)$$



**Figure 3.13:** Examples of proximity  $k$ -nearest neighbour proximity kernels for modelling cetacean sounds. a. nearest neighbours for percussions, b. nearest neighbours for harmonic elements, c. elements which are repetitive or periodic.

where  $X_{1...J}(f, t)$  are the spectrogram of  $x_{1...J}(t)$  and  $X(f, t)$  is the spectrogram of  $x(t)$ . The spectrograms are all of the size  $F \times T \times I$ .  $F$  represents the frequency bins,  $T$  provide the time frames present, and  $I$  denotes if the input signal is monophonic or stereophonic, which is defined by the value of 1 or 2 respectively. For the thesis work,  $I = 1$  is considered throughout because of the recording technique involved, which incorporates single hydrophones capturing sounds an instantaneous sound. From [38] each time-frequency bin of STFT  $X_j(f, t)$ , under local stationary assumptions, are independent and normally distributed. Moreover, in [39], each bin of the STFT spectrogram is said to be locally Gaussian modelled for a multivariate centred around that particular bin and is defined as:

$$\forall(f, t), X_j(f, t) \sim \mathcal{N}_c(0, \mathbb{P}_j(f, t) \cdot \mathbb{C}_j(f)) \quad (3.28)$$

$$\forall(f, t), X(f, t) \sim \mathcal{N}_c(0, \sum_{j=1}^J \mathbb{P}_j(f, t) \cdot \mathbb{C}_j(f)), \quad (3.29)$$

where  $\mathbb{P}_j(f, t) \geq 0$  is the power spectral density of source  $j$  at time-frequency bin  $(f, t)$  which is a non-negative scalar and accounts for the power content at that bin.  $\mathbb{C}_j(f)$  [38] represents the spatial covariance matrix which encodes the covariance between adjacent sources of  $X_j$  at  $f$ . Where Eq. 3.28 gives the distribution around a particular TF-bin, Eq. 3.29 For a sound recorder, it is assumed that the data represent a heteroscedastic behaviour, i.e. the variability of a variable is unequal across the range of values of a second variable that predicts it. The variability motivates the use of a function  $\Psi_j(f, t)$  which gathers the latent explanatory parameters to model the entire dataset which formulates to:

$$\Psi_j(f, t) = \{\mathbb{P}_j(f, t), \mathbb{C}_j(f)\} \quad (3.30)$$

The power spectral density term can also be defined as latent explanatory function as it models the slowly moving artefacts over time frames and frequency bins separately. Eq. 3.30 as developed in [22] defines the source cost function in terms of negative log-likelihood function

as noise is negotiated in a flexible way:

$$\mathcal{L}_j = -\log_p(X_j|\Psi_j) \quad (3.31)$$

The auto-covariance matrix  $\mathbb{C}'(f,t)$  can be calculated as the product of STFT of a certain source  $X_j(f,t)$  with its complex conjugate. After that, estimating the spatial covariance matrix as the summation of auto-covariance matrix normalised to the trace of an auto-covariance matrix and the number of time frames. Afterwards, calculating power spectral density matrix as the sum over the overall spectrogram normalised to the number of sources needed to be separated. Hence, the estimation of latent explanatory parameters of  $\Psi_j$  defined in Eq. 3.30 are  $\hat{\mathbb{P}}_j$  and  $\hat{\mathbb{C}}_j$  which aids in MMSE (minimum mean-squared error) estimation of  $\hat{X}_j(f,t)$  of through generalized spatial Weiner filtering model [40]:

$$\mathbb{C}'_j(f,t) = X_j(f,t) \cdot X_j(f,t)^* \quad (3.32)$$

$$\hat{\mathbb{C}}_j(f) = \frac{1}{T} \sum_t \frac{\mathbb{C}'_j(f,t)}{\text{trace}(\mathbb{C}'_j(f,t))} \quad (3.33)$$

$$\hat{\mathbb{P}}(f,t) = \frac{X(f,t) \cdot X(f,t)^*}{J \cdot I} \quad (3.34)$$

$$\hat{X}_j(f,t) = \frac{\hat{\mathbb{P}}_j(f,t) \cdot \hat{\mathbb{C}}_j(f)}{\sum_{j'=1}^J \hat{\mathbb{P}}_{j'}(f,t) \cdot \hat{\mathbb{C}}_{j'}(f)} \cdot X(f,t), \quad (3.35)$$

where  $\hat{X}_j(f,t)$  represents STFT estimate based on the latent explanatory parameters of the same. This can be multiplied with the input STFTs corresponding time-frequency bin and consequent inverse STFT will provide reconstructed separate sources. Eq. 3.28 gives the source cost function which is minimized by the Eq. 3.35. This paved the way to model each of separated source as locally constant and is described in subsection 3.4.1.2.

### 3.4.1.2 Local parametric model

In the second step of KAM, each power density functions  $\hat{\mathbb{P}}_j$  are to be modelled as locally constant incorporating k-NN (nearest-neighbours) proximity kernels [41]. The requirement of k-NN proximity kernels for estimating the influence of adjacent elements on the element under consideration. This method is applied in both horizontal and vertical direction as shown in Fig. 3.13 to determine the relation of a particular element within a given window with its adjacent window element. Hence, the local approximation of gradient descent function and thereby deferring of computation or backfitting until classification of the outliers. This method gathers the self-similar elements along time and frequency with a pseudo-repetitive period.

For each TF-bin  $(f,t)$ , the set of  $k$ -nearest-neighbours  $K_j f,t$  whose power spectral density closely follows  $\mathbb{P}_j(f,t)$  of source  $X_j(f,t)$  is formulated [22] as:

$$\forall (f',t') \in K_j(f,t), \quad \mathbb{P}_j(f',t') \approx \mathbb{P}_j(f,t), \quad (3.36)$$

where  $(f',t')$  denotes neighbour TF-bins of  $(f,t)$  corresponds to a pre-defined periodicity and



is shown in Fig. 3.13.  $\mathbb{P}_j(f, t)$  can be modelled in terms of NMF as:

$$\mathbb{P}_j(f, t) = \sum_{k=1}^K W_j(f, k) H_j(k, t) \quad (3.37)$$

The Eq. 3.36 provides proximity kernel models which assists in defining the model cost function  $\mathcal{L}_j$  as in Eq. 3.31. If each separated source  $X_j$  is computed using one observation, as the input signal is monophonic in nature, contamination from the adjacent sources is a common occurrence. In order to minimize the error due to the interference from other sources, which is motivated by Dempster et al's Expectation-Minimization approach [42], the following cost function has to be minimized at each TF-bin which can be written as:

$$\hat{\mathbb{P}}_j(f, t) = \arg \min_{\mathbb{P}_j(f, t)} \sum_{\forall (f', t') \in K_j(f, t)} |X_j(f, t) - \mathbb{P}_j(f, t)| \quad (3.38)$$

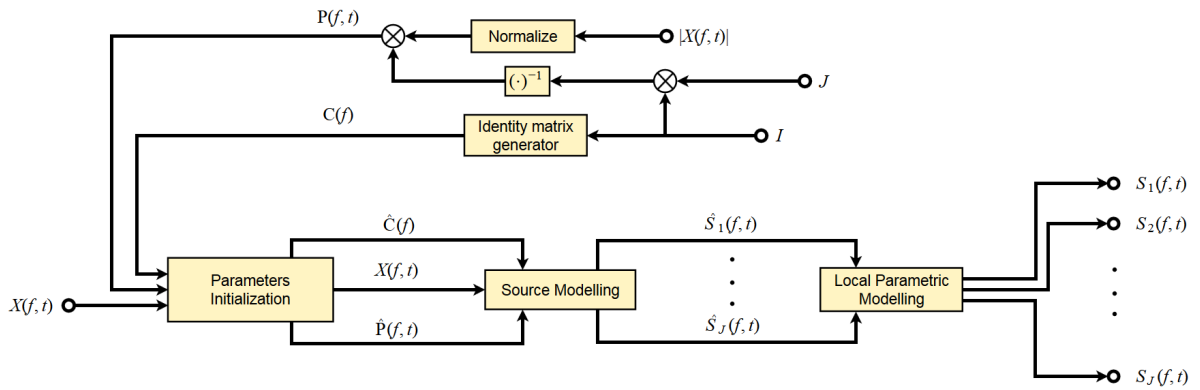
which is equivalent to the following as:

$$\hat{\mathbb{P}}_j(f, t) = \text{median}(X_j(f, t) \mid (f, t) \in K_j(f, t)) \quad (3.39)$$

Algorithm.3 coded the separation algorithm showcasing the arrangements of concepts for separating the of harmonic and percussive spectrograms. Finally, the stopping criterion is defined either by a convergence of new and old estimates or by iteration number.

### 3.4.2 Implementation of kernel additive modelling

The KAM formulation revolves around the parametric estimation of the input signal, which is source estimation and local model estimation, minimised further via source cost estimation



**Figure 3.14:** Prototype for kernel additive modelling

| Parameter name                        | Value for click & whistle extraction | Value for creak & whistle extraction |
|---------------------------------------|--------------------------------------|--------------------------------------|
| Number of sources to separate ( $J$ ) | 2                                    | 3                                    |
| Type of sound ( $I$ )                 | 1 (monophonic)                       | 1 (monophonic)                       |
| Number of iteration                   | 5                                    | 5                                    |
| Window length                         | 0.02 s                               | 0.02 s                               |
| Harmonic stability                    | 0.05 s                               | 0.05 s                               |
| Percussive stability                  | 100 Hz                               | 100 Hz                               |

**Table 3.3:** Parameters used for kernel additive modelling algorithm

as per Eq. 3.35 and Eq. 3.39. For algorithmically structuring of the formulation is generated as defined in [22, 42, 43]. Firstly the input mixture spectrogram, the neighbourhood and number of iteration are given as a stopping criterion to the input of separation function. Afterwards the latent explanatory function  $\hat{\mathbb{P}}_n$  and the covariance matrix function  $\hat{\mathbb{C}}_n$ . After that, each separated source, which is harmonic (whistle, creeks) and percussive (clicks) in this case, are to be modelled using the Eq. 3.35. Later, the latent explanatory function or the power spectral density in the case of audio separation are modelled locally for each of the source  $\hat{X}_j(f, t)$ , which fed to the inverse STFT block for recreating time-domain separated signal.

Fig. 3.14 provides a detailed description of the block diagram for the separation according to kernel additive modelling with variables incorporated in each block. The normalisation of input spectrogram is the first step followed by the initialisation and recursive learning process to minimise the cost functions with the output being leaned kernels and estimated separated spectrograms.

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**Algorithm 3:** The kernel backfitting formula

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**Result:** Source PSDs  $\hat{\mathbb{P}}_n$  and covariance matrices  $\hat{\mathbb{C}}_n$  for Weiner filtering

Inputs:

- Mixture STFT  $X(f, t)$
- Neighbourhoods  $K_j(f, t)$
- Number of iterations

Initialization:

- Update power spectral density according to Eq. 3.35
- $\hat{\mathbb{C}}_j(f) \leftarrow I \times I$  identity matrix
- normalisation of  $|X|$

**for**  $j = 1$  **to**  $J$  **do**

1. Calculate the auto-covariance matrix as Eq. 3.33
2. Update the estimate of spatial covariance matrix as Eq. 3.34
3. Calculation of the estimation of each source using Eq. 3.35
4. Minimise the error at each time-frequency bin with Eq. 3.39

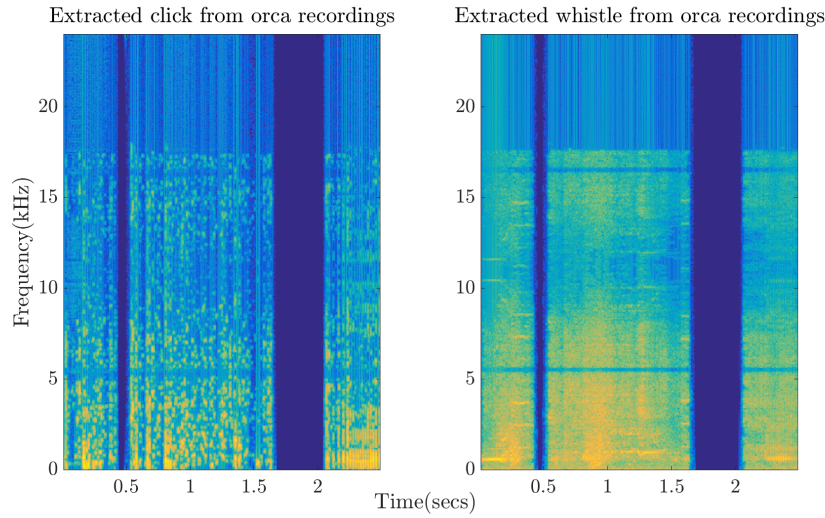
**end**

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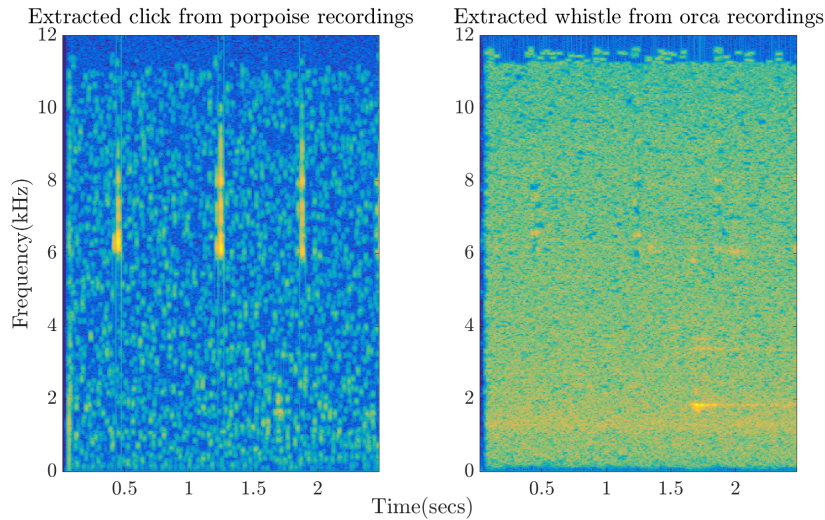
### 3.4.3 Separation results

The separated results obtained from KAM had two particular visible effects because of the implementation of two previously explained algorithm. The burbling artefacts in the separated clicks found to be considerably less when observed compared to NMF and MF based separation. Visual improvement found in Fig. 3.15 due to the incorporation of the pseudo-periodicity in KAM before source and local modelling. The pseudo-periodicity helped in tracking the echolocation clicks, which are almost at a regular interval. However, because of the source modelling in addition to local modelling of each window elements provide a better whistle extraction which is visible and audible.

The inclusion of pseudo-periodicity has another adverse effect as the creaks became a part of whistles as they do not comply with the pseudo-periodic nature of the clicks. In this regard, a declaration about the number of sources to be separated made during initiation. KAM processes



(a) Click and whistle extraction from orca recordings



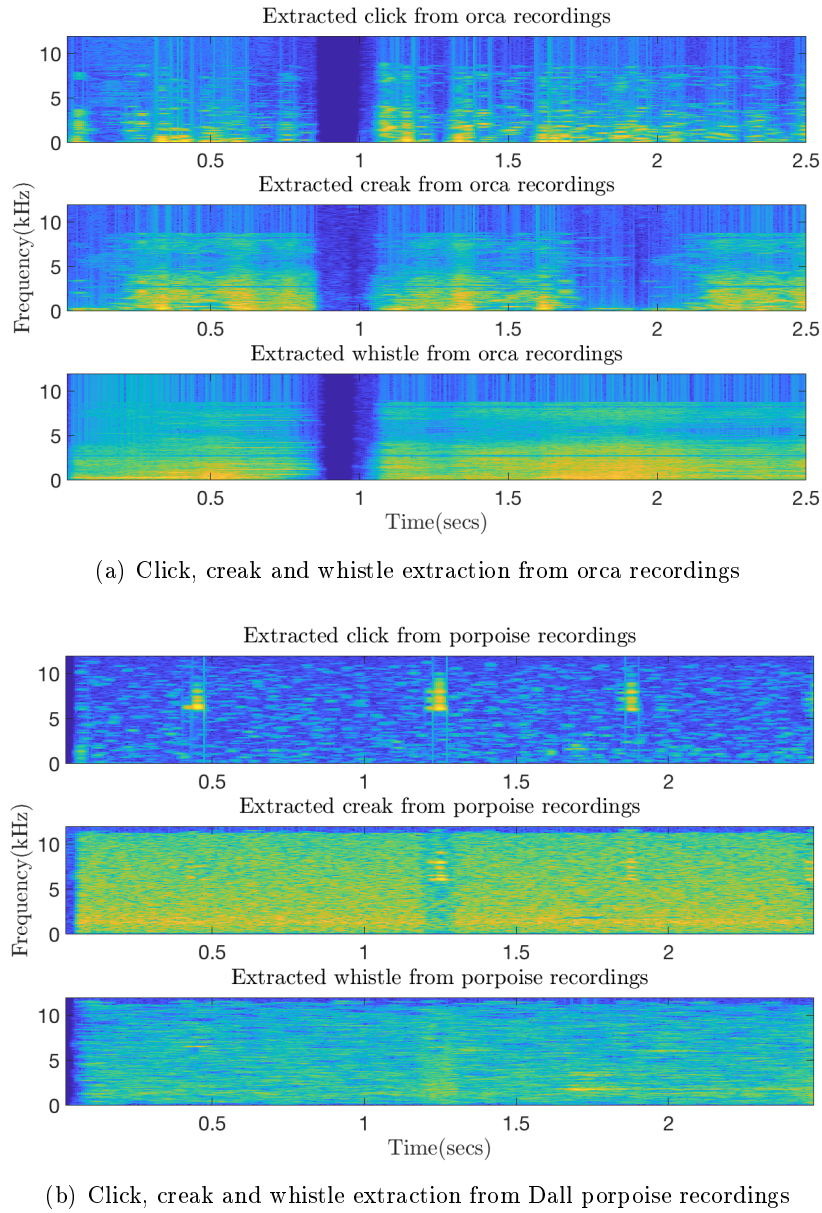
(b) Click and whistle extraction from Dall porpoise recordings

**Figure 3.15:** Source separation into two sources with kernel additive modelling

the harmonics or whistles based on their changing gradient and hence discriminates against them by slopes before separating them.

The absence of creak is again showed an improved and clean extraction of click as found in Fig. 3.16. However, when extracting three sources, whistle visualisation becomes blurry as the parameters given for such separation contains single parametric entry only for a particular band. The conclusion obtained from these results indicates the superiority of NMF because of its flexibility to change parameters for separating more than one sources.

Next section elaborates on the different evaluation process created (PNR, PCC, MSC, RMS) and adapted (SDR, SAR) to justify the use of a particular class of algorithm.



**Figure 3.16:** Source separation into three sources with KAM method. A few creaks are found by the method but deteriorating the clicks.

### 3.5 Evaluation

The visualisation results obtained by the different algorithm gave only half of the details about their efficiency even when listened to it. With these two particular characteristics, each of three algorithms can be declared to perform necessary separation but to ensure the best possible algorithm among the three, evaluation tools are generated and adapted to get certain measures for defining the best suitable algorithm.

Evaluation for blind source separation being scarce defined broadly in two categories. Firstly, without ground truth based approach as in the case of PNR (peak-to-noise ratio) and secondly, ground truth based approaches like SDR (signal-to-distortion ratio) and different simple error calculations like the RMS (root mean square) error, PCC (Pearson correlation coefficient) and MSC (magnitude squared coherence).

This section first talks about peak to noise ratio or non-ground truth-based approach and then introduce the ground truths before remaining evaluation concepts.

#### 3.5.1 Peak-to-noise ratio (PNR)

Peak to noise ratio is an evaluation technique which is developed to understand the clarity of separated click signals. The method can be seen as a signal-to-crest ratio between two consecutive peaks and then averaged. The separated clicks by the three implemented algorithm visually merged with burbling artefacts, and this measure (PNR) helps in revealing the algorithm helps in a cleaner representation of the separated clicks.

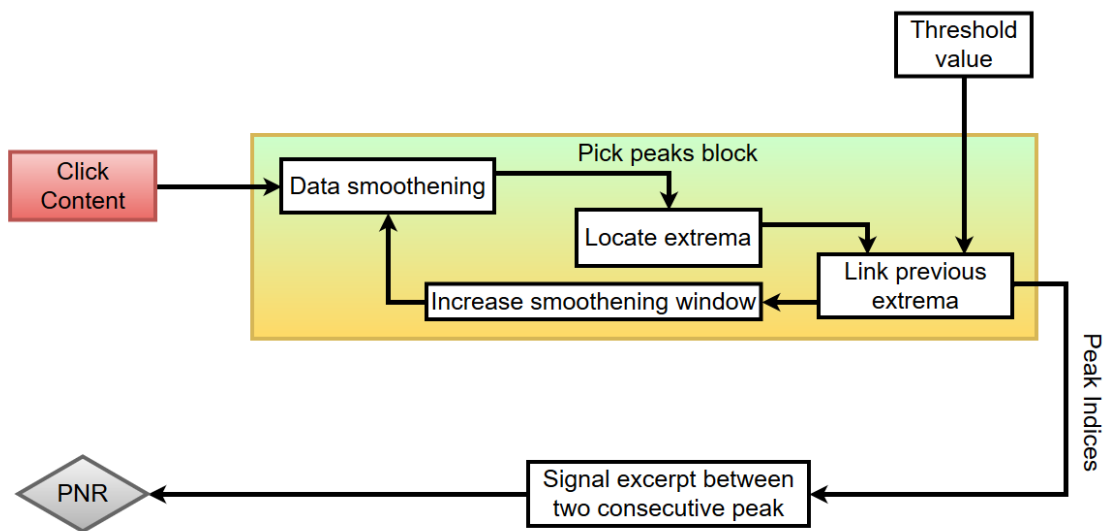


Figure 3.17: Global error calculation block diagram

### 3.5.1.1 Algorithm

The algorithm follows Fig. 3.17. The important block in this evaluation step is the pick-peaks block which helps in ascertaining indices of the peak from separated clicks with a given threshold. The threshold is taken to be a value in between 0 and 1 as the separated click content is already normalised. The data is smoothed over the entire signal length with a window length of 0.01 s and increased by the same amount in the successive iteration until the length of the input signal. Thereafter, extrema in that window are found, and in the last step of the pick-peaks algorithm, previous extrema recorded linked to the extrema of the current increased window. When traversing the entire length of the input, the threshold is applied to get each peak of separated click content.

### 3.5.1.2 Qualitative measure

PNR or peak to noise ratio is a the higher, the better form of qualitative measure. Fig. 3.18 demonstrates that NMF scores higher than other two methods implemented, but the confidence interval in case of PNR increases with the sampling frequency. If referred to Fig. 3.7, the artefacts between two clicks are visible in case of porpoise which adds to the noise, and when the clicks are closely spaced as in orca clicks, or in case of rapid production of clicks, these artefacts increased rapidly and hence reduced the PNR value for NMF at higher sampling rate.

Median filtering conforms to the same pattern as NMF with increasing sample frequency but only concerning median value rather than the confidence interval. When considering KAM, the results remain quite similar when only whistle and clicks are separated. To conclude the best possible generation of separated click contents, NMF can be used rather than the other

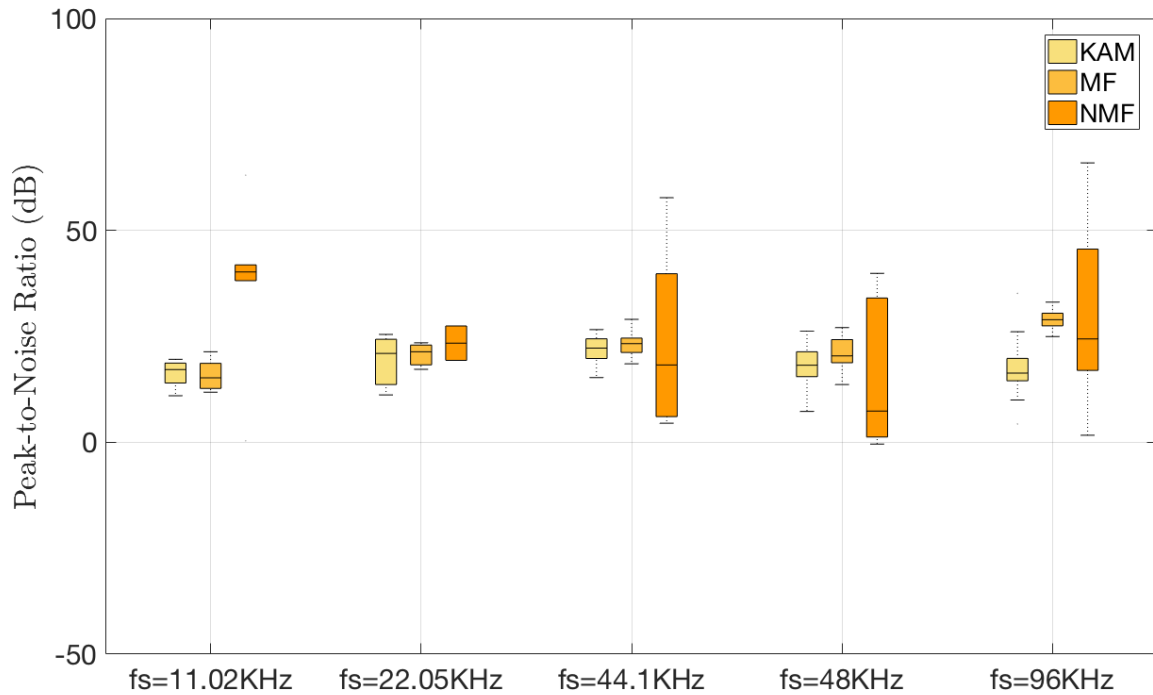


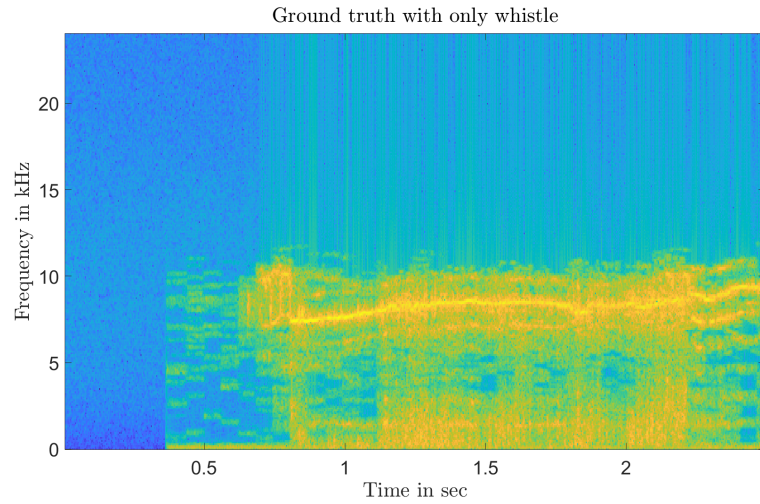
Figure 3.18: Peak-signal to noise ratio



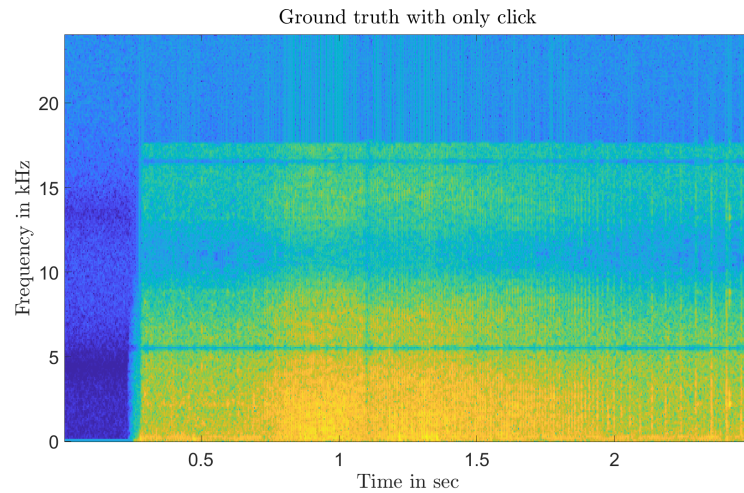
two algorithms, as decimating the audio recordings with higher sample frequency can result in better separation and with lesser complexity than that of KAM and more reliable than MF separated results.

### 3.5.2 Ground truth based evaluation

This section considers four evaluation aspects based on predefined audio recordings, obtained from animals remaining in captives. Hence, two audio in the dataset contained either only whistles or clicks in the entire recording with the sampling frequency of  $48\text{ kHz}$ . All other signals with similar sample frequencies are superimposed and after that separated with a different algorithm. The comparison between the separated whistles and the reference ground truth signal depicted in Fig. 3.19(a) and comparing the clicks against the ground truth click signal pictured in Fig. 3.19(b). The first part of the section evaluates the algorithms based on a signal



(a) Time-frequency representation of captive Baiji recorded whistles

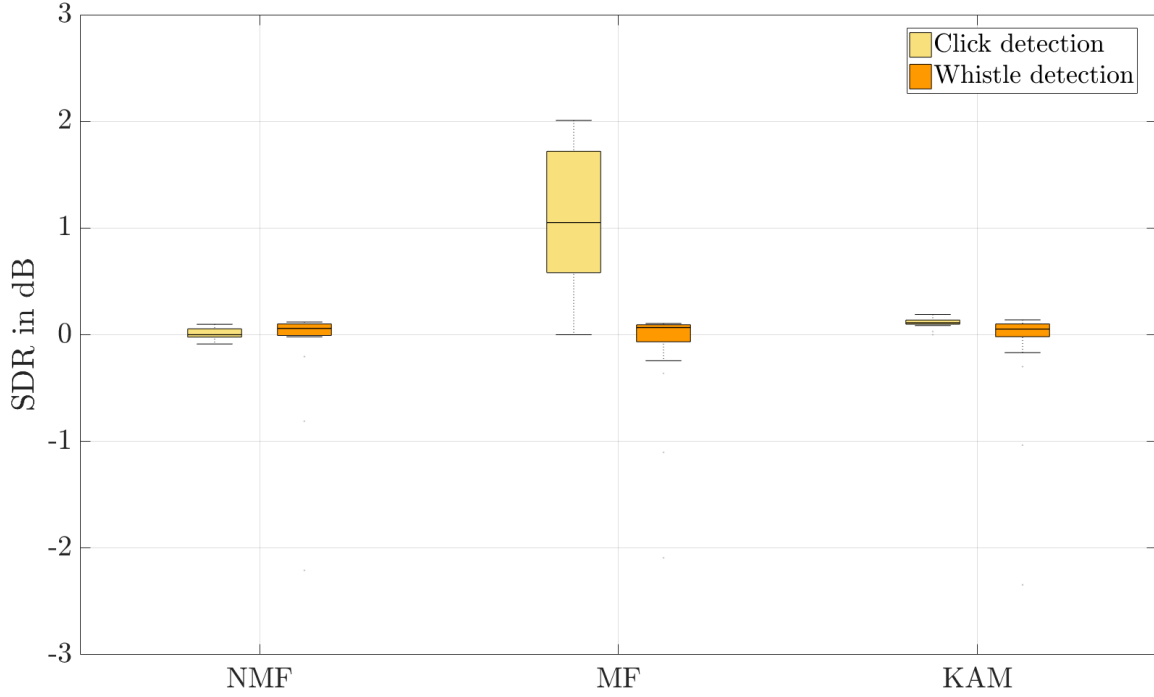


(b) Time-frequency representation of captive Amazon river dolphins recorded clicks

**Figure 3.19:** *Ground truths for each class of sounds*







**Figure 3.21:** *Signal-to-distortion ratio*

Fig. 3.21 shows the boxplot for SDR. The qualitative measure for SDR is the closer it is to zero, better is the separation process. NMF performed the best in separating the clicks whereas, for whistle detection, MF performs best. The negative values observed in the boxplot is because the datasets merged with the ground truth have whistles and clicks combined which results in having more whistle or click components than the ground truth signal. The other reason is a higher degree of production of burbling artefacts. The adverse effect in NMF is minimal in case of whistle detection because of its calculative global cost function, whereas click detection does not show any adverse results in case of KAM due to the implementation of the pseudo-periodic algorithm through a peak-picking process.

To conclude this evaluation, the proposal of a network consisting of heterogeneous separation technique (merging of two separation technique into one) put forward for separating clicks by NMF and the whistles through MF improves the overall SDR. This practice helps in faster separation overall with a higher degree of accuracy. The boxplot showing stronger evidence in doing overall separation through NMF, but the proposed mixing of the separation helps in reducing the time required by three-quarter of the time taken by NMF alone.

In next section, the simple errors are discussed further to make strong arguments in favour of the heterogeneous separation algorithm.

### 3.5.2.2 Simple error calculation in the separation of the blind signals

A scarcity of standard evaluation for blind source separation, three traditional error calculation methods are introduced to observe the efficiency of implemented techniques based on the ground truth based approach. The basic idea to implement these simple error calculation techniques

is to verify the similarity between the ground truth signal to the separated signal. For ground truth based evaluations, the initial steps followed according to the SDR evaluation method.

The evaluation consists of RMS error [48] calculation between the ground truth and separated signal. Moreover, Pearson correlation coefficient [49] and magnitude squared coherence [50] are calculated to make a measure of similarity between the ground truth and the separated signal.

The RMS error is calculated by the following formula:

$$cRMS = \sqrt{\frac{1}{N} \sum_t \left( (\hat{x}(t) - x(t))^2 \right)}, \quad (3.41)$$

where  $N$  is the number of samples,  $\hat{x}(t)$  is the estimated separated click ( $c$ ) or whistle ( $w$ ) content and  $e = w, c$  and  $x(t)$  is the ground truth input signal considered for the evaluation. Pearson correlation coefficient is given by the following formula:

$$cCORR = \frac{C_{\hat{x}x}}{\sqrt{C_{\hat{x}\hat{x}}C_{xx}}} \quad (3.42)$$

$$C_{\hat{x}x} = \frac{1}{N} \sum_t (\hat{x}(t) - \bar{\hat{x}})(x(t) - \bar{x}) \quad (3.43)$$

$$C_{\hat{x}\hat{x}} = \frac{1}{N} \sum_t (\hat{x}(t) - \bar{\hat{x}})^2 \quad (3.44)$$

$$C_{xx} = \frac{1}{N} \sum_t (x(t) - \bar{x})^2 \quad (3.45)$$

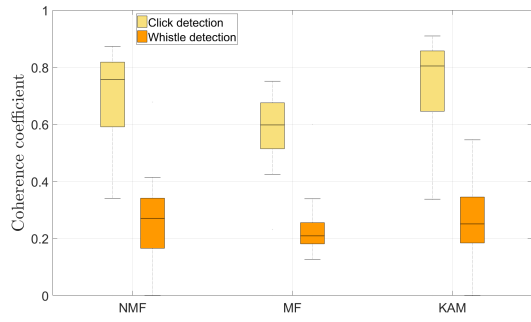
where  $C_{\hat{x}x}$  represents the cross correlation of the ground truth and the separated result,  $C_{\hat{x}\hat{x}}$  and  $C_{xx}$  are the auto-correlation of separated result and the ground truth input. Finally, the magnitude squared covariance is calculated using the `mscohere` function inbuilt in Matlab and then normalised the result to sample length as:

$$MSC = mscohere(\hat{x}_e(t), x_{std}(t)) \quad (3.46)$$

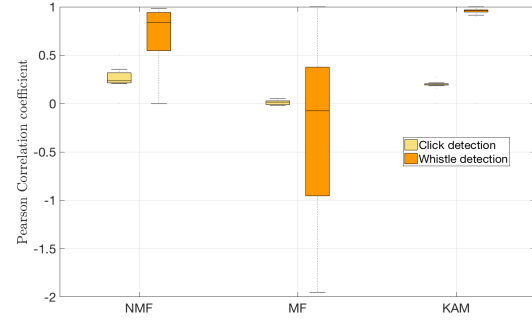
$$cMSC = \frac{1}{N} \sum_t MSC \quad (3.47)$$

The qualitative measure provided by RMS error follows the rule that suggests closer the value is towards zero, better is the result. Coherence and correlation evaluation follow an opposite principle which stated as - closer is the value towards unity better is the separation. Fig. 3.22(a), 3.22(b) shows the coherence and correlation results obtained. Click detection is almost similar by both NMF and KAM when compared with coherence results whereas whistle detection is comparable for all the algorithm concerned. Correlation results suggest NMF as a better algorithm for click detection and KAM as a better performer for whistle detection among the three.

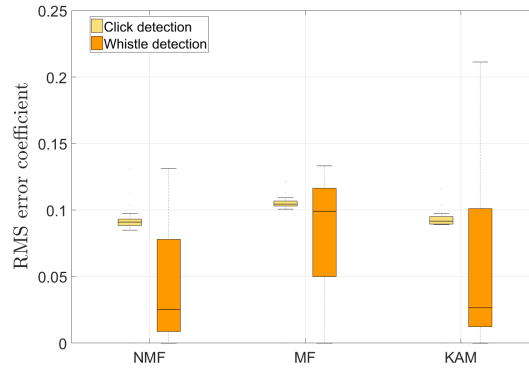
RMS error evaluation performance is shown in Fig. 3.22(c). In this particular evaluation, NMF is edging out KAM on click detection based on the qualitative measure of RMS error evaluation. If the highest value considered as the reference, MF or NMF can be regarded as a better whistle detector, but that changes when the median value adjudged, and KAM joins NMF as the better whistle detector.



(a) Magnitude squared coherence error between the ground truth and the separated signal



(b) Correlation between the ground truth and the separated signal



(c) Root Mean Square error between original and separated signal

**Figure 3.22:** Simple error calculation

### 3.6 Further improvements

Visualizing the evaluations for three separation method studied, implemented and optimised for separation of underwater acoustic mammal vocalisation, reduction of separation time and the overall complexity of the operation deemed to be of primary concern. Median filtering has the lowest complexity as well as the better computational time, however, lacks behind in every evaluation performed. Three methods of obtainable heterogeneity in developing mixed algorithms explained in detailed in next paragraph.

In the different evaluation process, NMF always remained an option to go for click detection efficiently and hence can be concluded as the useful click detection algorithm. For whistle detection, KAM performs better in different algorithm than MF, however, is comparable to NMF in some cases. So, to reduce programming complexity, NMF can be used for separating the sounds into particular sets of whistles and clicks. However, maintaining the accuracy of the result at highest possible order KAM should be implemented for whistle separation along with NMF for click separation which entailed too much higher computational complexity. The third combination can be NMF for click detection and MF for a whistle when the time is a constraint and well as computational power available for separation purposes.

In next chapter, the separated clicks or echolocation are analysed to detect three unique characteristics of this particular vocalisation.



## Chapter 4

# Echo Detection

Echo detection requires specific positioning of the source and the receiver. This is vague in case of whale sound recording and hence the chapter will document fundamental characteristics of the shallow water communication to generate a synthetic response taking into the fundamental parameters like temperature, salinity and pressure. Prior to the generation of synthetic impulse response, denoising is explained and later half of the chapter concentrated into analysing the recorded whale clicks. The third subsection is hence termed as echo analysis because it is not known how the hydrophone is place in an already unknown marine environment.

### 4.1 Shallow water communication

Mammals, in general, require air for their respiratory procedure and Odontoceti (toothed whales) are no different than that. Most of the whales' habitats lie in shallow water. Daily activities, specially coordinated hunting and socialisation, are found to occur in this region which helps in exploiting the concepts of shallow underwater acoustic communication methodologies. The term shallow water in underwater acoustics refers to the coastal shoreline until a depth from which the sea-floor slope increases and is roughly considered to be of 200 m <sup>6</sup>.

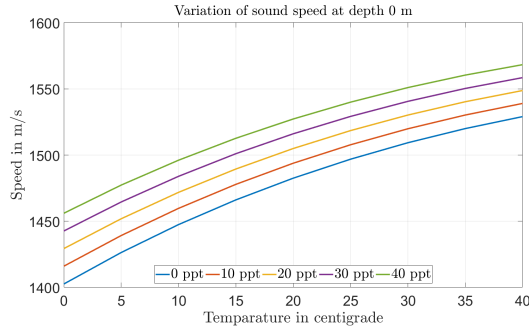
In shallow underwater channels, communication is severely affected by the effect of multi-path propagation which gives rise to time-delayed attenuated versions of the line-of-sight signal. Clicks, used by whales for localisation, understanding the shapes and size of nearby objects, undergoes the same phenomenon which motivates to separate the line-of-sight entity from the echo. The sound is the way of communication for toothed whale, interests in having an insight in the speed of sound in those shallow water areas. The speed of sound varies based on the elasticity of medium [51] and is related as

$$V_{t_p} = \left( \frac{K + \frac{4}{3}G}{\rho} \right)^{\frac{1}{2}} \quad (4.1)$$

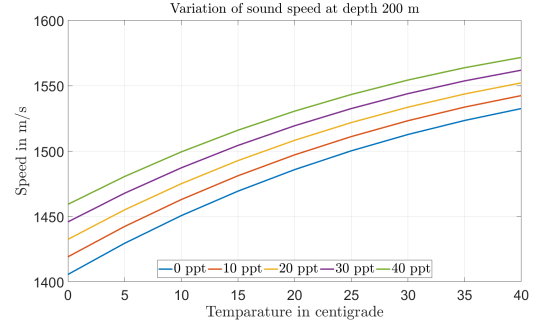
$$V_{t_s} = \left( \frac{G}{\rho} \right)^{\frac{1}{2}} \quad (4.2)$$

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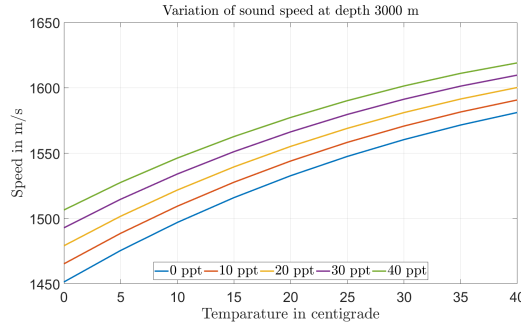
<sup>6</sup><https://dosits.org/science/advanced-topics/shallow-water-propagation/>



(a) Sound speed variation within constrained parametric condition at 50 m



(b) Sound speed variation within constrained parametric condition at 200 m



(c) Sound speed variation within constrained parametric condition at 3000 m

**Figure 4.1:** Variation of sound speed for a selected range of parameters

$$\Leftrightarrow V_t = \left( \frac{C_{ps}}{\rho} \right)^{\frac{1}{2}}, \quad (4.3)$$

where  $V_{t_p}$  and  $V_{t_s}$  are longitudinal and transverse waves consecutively and  $V_t$  is the equivalent sound speed.  $K$  and  $G$  are the bulk and shear modulus which comprises the elastic behaviour of water.  $C_{ps}$  is the equivalent elastic constant with subscript  $ps$  denoting the directionality of elastic constants concerning type and direction of wave travel.  $\rho$  denotes the density and is considered constant for sea water. This being the reason for sound to travel much faster in water ( $\sim 1500 \text{ m/s}$ ) than in air ( $\sim 343 \text{ m/s}$ ) at  $20^\circ \text{C}$ . Additionally, salinity, pressure (which is dependent on depth) and temperature affect the speed of sound profoundly. The speed of sound shown in Fig. 4.1 describes the variation based on three parameters - temperature, salinity and depth. The UNESCO equation [52] aided in determining the variation of the speed of sound in shallow water which is given by:

$$V_t(S, T, P) = C_w(T, P) + A(T, P)S + B(T, P)S^{\frac{3}{2}} + D(T, P)S^2, \quad (4.4)$$

where  $S$  is salinity in *ppt*,  $T$  is temperature in  $^\circ \text{C}$  and pressure  $P$  is in *bar*.  $C_w$ ,  $A$ ,  $B$  and  $D$  are the coefficients.

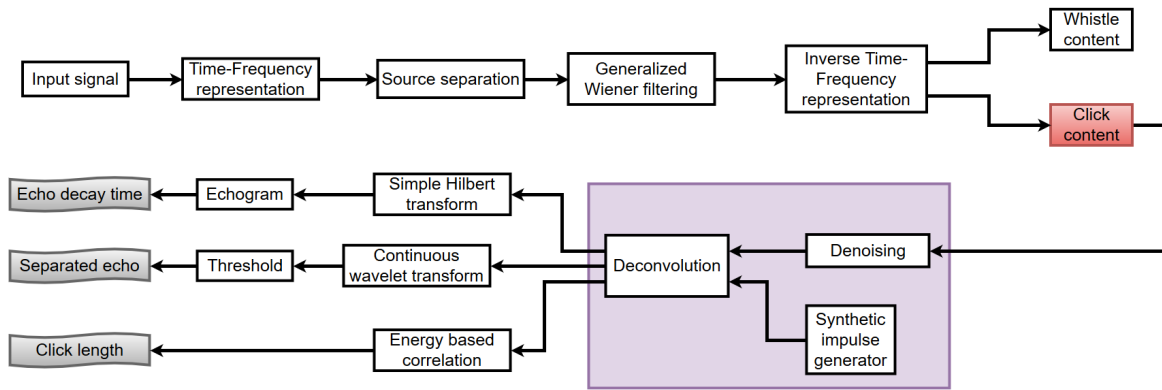
The toothed whale vocalisations recordings mentioned in Sec. 2.3 collected from bay areas, rivers and delta regions. Moreover, the sounds captured are mostly of their hunting patterns except for the captive Odontoceti species' recordings, where different observable communicating patterns found to occur. The primary motive to reintroduce the environment is to narrow down the parameter set that affects the sound speed.

| Name of Parameter   | Range of values                       |
|---------------------|---------------------------------------|
| Temperature         | $0^{\circ}C - 40^{\circ}C$            |
| Optimal temperature | $\approx 12^{\circ}C$ (coastal areas) |
| Depth               | $\approx 0m - 3000m$                  |
| Optimal depth       | $50m$                                 |
| Salinity            | $0ppt - 40ppt$                        |
| Optimal salinity    | $16ppt$ (coastal areas)               |

**Table 4.1:** Parameters used for estimating sound speed variations

The recordings of sounds mentioned in Sec. 2.3 took place in diverse areas, ranging from coastal areas in Canada and USA<sup>7</sup> to Maldives<sup>8</sup>. The temperature in these areas ranges from  $0^{\circ}C - 30^{\circ}C$ , giving information on the temperature range. Secondly, the hunting pattern of toothed whale involves coordinated attacks on seals<sup>9</sup>, fish schools<sup>10</sup> and others which commonly extends from  $\approx 0m - 200m$ . The notable exception being sperm whales which dive nearly  $3000m$  for catching squid<sup>11</sup>. These data gave an impression of the depth, and hence the similar pressure. The third important factor is the salinity of the water. At bay area and delta regions, the salinity varies from  $0ppt - 40ppt$ . Fig. 4.1, pictured the increasing nature of the sound speed within the specified parametric limit. Fig. 4.1(a) and Fig. 4.1(b) shows two extreme depth points of shallow water in which toothed whales' surface hunting activity occurs. Fig. 4.1(c), shows the sound speed at  $3000m$  for sperm whales.

The parameters in Table.4.1 aided in the creation of synthetic impulse response of marine recording environment. Parameters used are reflection coefficients, attenuation factor and cross-sectional area of an underwater environment and collected from various real-time experiments to estimate the transfer function of the water channel.



**Figure 4.2:** Click processing prior to finding unique features of an echolocation click.

The first section worked around shallow water communication to develop the violet pre-processing block for echo analysis. Next two sections develop the idea contained in the said

<sup>7</sup><https://www.nodc.noaa.gov/dsdt/cwtg/cpac.html>

<sup>8</sup><https://www.seatemperature.org/asia/maldives/>

<sup>9</sup><http://www.newsweek.com/scientists-explain-video-orca-punting-seal-80-feet-air-388554>

<sup>10</sup><https://whyevolutionistrue.wordpress.com/2013/03/09/dolphins-trap-fish-with-nets-made-of-mud/>

<sup>11</sup><http://www.bbc.com/earth/story/20150417-deep-sea-encounter-with-giant-whale>

coloured box and also talk about the limitation of such approach before entering into the echo analysis concepts. Denoising block explains in the first iteration along with citing the reason for using this block late in the proposed architecture. After that, the idea of synthetic impulse response generator is explained briefly with all the constraints taken into consideration. Afterwards, implementation of fast deconvolution process documented explaining the primary motive to obtain better clarity about the clicks.

## 4.2 Denoising

### 4.2.1 Motivation

In the real world, contamination of signals with noise and no data is clean barring a few cases where noise is negligible, i.e. signals with high signal-to-noise ratio. Underwater acoustic signals are no exception and the hydrophones used to record the sounds picks up the contaminated mammal vocalisation. Removal of noise would give a clearer estimate of the signal which helps in better detection of the echo. Noise is the unwanted elements added to the original signal with different amplitude, the removal of known as denoising. In this thesis, denoising based on different wavelets are studied to have an overview of which particular mother wavelet is best suitable for denoising the signals.

### 4.2.2 Background: Discrete wavelet transform

The discrete Wavelet transforms primarily used in denoising and compression signals. In the thesis, as denoising is done through discrete wavelet because DWT can represent naturally occurring signals with few coefficients and hence form a sparser representation of the signal. The discrete wavelet transform is comparable with discrete multi-rate filter-banks, and Mallat gives a conceptual representation of the same, [53, 54] and is pictured in Fig. 4.3, showing decomposition and reconstruction of the signal.

The fundamental scaling in DWT, set to 2, with general scaling effected by raising the base scale to a power of integer as  $2^j$  ( $j = 1, \dots, \infty$ ) and the translation is an integer multiplication to the scale of the wavelet as  $2^j k$  ( $k = 1, \dots, \infty$ ). This process is termed as dyadic scaling and shifting helping in eliminating redundancy of coefficients. Elimination of redundancy helps in restricting the number of coefficients to the number of samples in the analysing signals. DWT decomposes the signal into its high-pass, and low-pass components and the low-pass components are then divided recursively into its high pass and low pass components. The output of the low-pass components termed as the approximation coefficients, and that of high-pass termed as detail coefficients.

The input signal filtered in each step into its high-pass and low-pass sub-bands as shown in Fig. 4.3 discarding half of the samples as per the Nyquist criterion. In a second stage, recursively filtering low-pass sub-band into its high-pass and low-pass narrower sub-bands like  $cD_2$ ,  $cA_2$  and so on. In each step, sparser higher DWT coefficients represent signals and noise as smaller DWT coefficients. The advantage of doing so helps in analysing the signal in narrower sub-bands with different resolution and helping in denoising signals with suitable threshold criterion to filter out the noise coefficients.

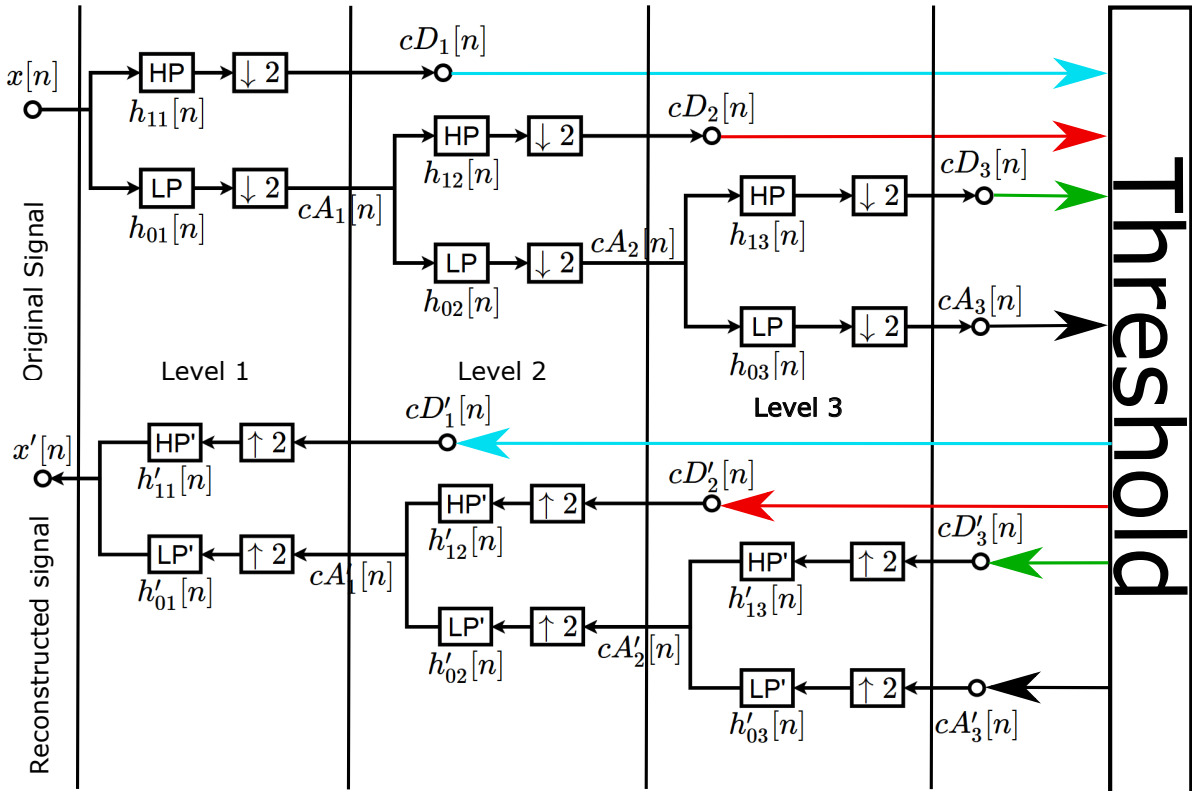


### 4.2.3 Mathematical formulation: Discrete wavelet transform

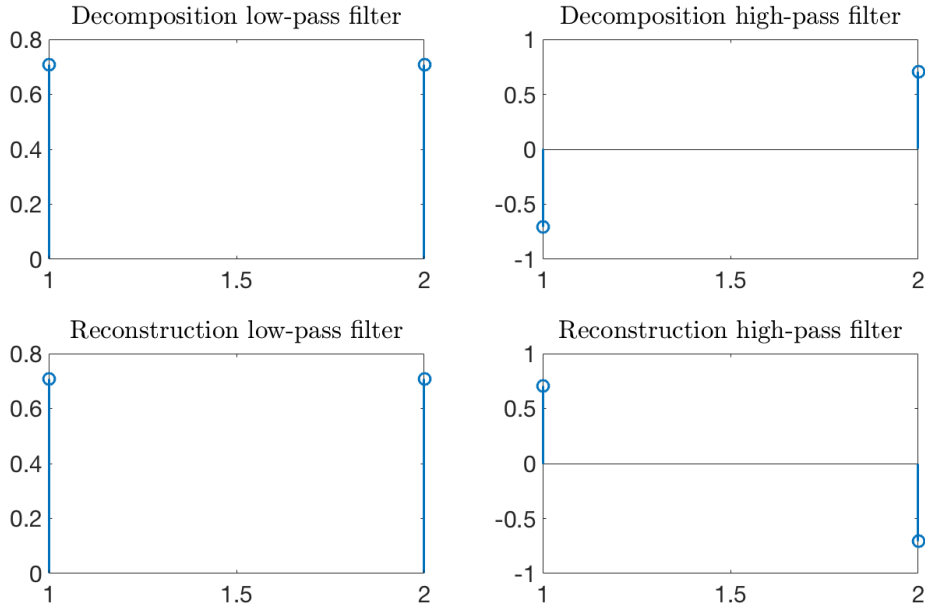
The discrete wavelet transform (DWT) is an implementation of the wavelet transform using a discrete set of the wavelet scales and translations which decomposes the signal into a mutually orthogonal set of wavelets.

The DWT of a signal  $x[n]$ , which is formed from  $x(t)$  by sampling at sampling time  $T_s = \frac{1}{F_s}$  with  $F_s$  being the sampling frequency of the signal, is calculated by passing it through a series of filters. First, the samples are passed through a low pass filter with transfer function  $h_{0j=1,2,3}$  resulting in a convolution of the two signals. The signal is also decomposed simultaneously using a high-pass filter  $h_{1j=1,2,3}$ . The outputs giving detail coefficients (from the high-pass filter) and approximation coefficients (from the low-pass). It is important that the two filters are related to each other, and they are known as a quadrature mirror filter.

However, since removing half the frequencies from the signal, half the samples can be discarded according to Nyquist's rule. The filter output of the low-pass filter  $h_{0j=1,2,3}$  in the diagram above is then sub-sampled by 2 and further processed by passing it again through a new low-pass filter  $F_{j=1,2,3}$  and a high-pass filter  $h_{1j=1,2,3}$  with half the cut-off frequency of the previous one. This decomposition has halved the time resolution since only half of each filter output characterises the signal. However, each output has half the frequency band of the input, so the frequency resolution has doubled.



**Figure 4.3:** Mallat's algorithm demonstration for a three level decomposition and re-construction of a signal. HP represents the high-pass and LP is low pass with detail coefficients are denoted by  $cD_j (j = 1, 2, 3)$  and  $cA_j (j = 1, 2, 3)$  represents the approximate coefficients.



**Figure 4.4:** Haar / db1 wavelet filters for decomposition and reconstruction of signals

The filter-bank implementation of wavelets can be interpreted as computing the wavelet coefficients of a discrete set of child wavelets for a given mother wavelet  $\psi(t)$ . In the case of DWT, the mother wavelet  $(\psi_{s,T}[n])$  is shifted and scaled by powers of 2 which can be formulated as:

$$\psi_{s,T}[n] = \frac{1}{\sqrt{s}} \psi\left(\frac{T-n}{s}\right), \quad (4.5)$$

where  $s$  is the scale parameter and  $T$  is the shift parameter. Wavelet coefficient  $\gamma$  of a signal  $x[n]$  is defined as the projection of  $x[n]$  onto the wavelet. This can be defined by the following equation:

$$\gamma_{s,T}[n] = \sum_{n=-\infty}^{\infty} x[n] \cdot \frac{1}{\sqrt{s}} \psi\left(\frac{T-n}{s}\right) \quad (4.6)$$

In Eq. 4.6,  $\gamma_{s,T}[n]$  can be viewed as a convolution of  $x[n]$  with a dilated, reflected, and normalized version of the mother wavelet,  $h[n] = \frac{1}{\sqrt{s}} \psi\left(\frac{-n}{s}\right)$ , sampled at the points  $s = 2^j$ . The translation parameter  $T = 2^j \cdot k$  which provides the resultant transform as:

$$\gamma_{j,k}[n] = \sum_{n=-\infty}^{\infty} x[n] \cdot \frac{1}{\sqrt{2^j}} \psi\left(\frac{k \cdot 2^j - n}{2^j}\right) \quad (4.7)$$

Fixing  $j$  at a particular scale,  $\gamma_{jk}$  is a function of  $k$  only. The relationship between coefficients of quadrature mirror filters is as follows:

$$h_{0j}[n] = (-1)^n \cdot h_{1j}[1-n] \quad (4.8)$$

| Parameter Name          | Value                                                     |
|-------------------------|-----------------------------------------------------------|
| Wavelet Families        | Debauchies, Symlets, Coiflets, Fejer-Korovkin, Orthogonal |
| Level of decompositions | 6                                                         |
| Thresholding technique  | rigorous SURE                                             |

**Table 4.2:** Parameters used for denoising using DWT

As an example, consider the discrete Haar wavelet, at  $n = 2$ , whose mother wavelet function is given as  $\psi[n] = [1, -1]$ . Then the dilated, reflected, and normalized version of this wavelet is  $h_{1j}[n] = \frac{1}{\sqrt{2}}[-1, 1]$ , which represents high-pass decomposition filter for the discrete Haar wavelet transform.

Each step of Mallat decomposing is formulated as :

$$cA_j[n] = \sum_{k=-\infty}^{\infty} cA_{j-1}[k]h_{0j-1}[2k-n] \quad (4.9)$$

$$cD_j[n] = \sum_{k=-\infty}^{\infty} cD_{j-1}[k]h_{1j-1}[2k-n] \quad (4.10)$$

In Eq. 4.9 and 4.10,  $j$  denotes each step of decomposition where  $cA_0(n) \approx x(n)$ . Each successive decomposition step utilizes  $cA_j(n)$  as the input signal to that step producing  $cA_{j+1}(n)$  and  $cD_{j+1}(n)$ . The maximum decomposition step possible is  $\log_2 N$ , where  $N$  is the number of samples present in the input signal. Post thresholding of the coefficients to minimise the noise components, Eq. 4.9 and 4.10 are used but the filter are transposed.

#### 4.2.4 Threshold: Discrete wavelet transform

The wavelet coefficient is affected by linear shrinkage associated with a resolution level of a coefficient. Donoho et al. [55–57] proposed non-linear thresholding which can be implemented in either as hard threshold or as a soft threshold.

In hard and soft thresholding, values of the DWT coefficient lower than the threshold, set to 0. The difference lies in the coefficient values higher than the threshold. In case of hard thresholding, coefficients higher than the threshold value is left unchanged and in soft thresholding, the values higher than the threshold, are subtracted from the threshold. This thresholding is given by:

$$\text{Hard Threshold} : \begin{cases} x_d[n] = x[n], & |x[n]| > \lambda \\ x_d[n] = 0, & |x[n]| < \lambda \end{cases} \quad (4.11)$$

$$\text{Soft Threshold} : \begin{cases} x_d[n] = x[n] - \lambda, & |x[n]| > \lambda \\ x_d[n] = 0, & |x[n]| < \lambda \end{cases}, \quad (4.12)$$

where  $\lambda$  is the threshold,  $x_d[n]$  is the denoised version of  $x[n]$ . The universal threshold value

$(\lambda_U)$  [55] which does not depends on noise estimates is given by:

$$\sigma_t = \frac{\text{median}(cD_1[n])}{0.6745} \quad (4.13)$$

$$\lambda_U = \sigma_t \cdot \sqrt{2 \log N} \quad (4.14)$$

The other thresholding techniques proposed are rigorous SURE thresholding technique developed from Stein's unbiased risk estimate (SURE), minimax thresholding [57].

#### 4.2.5 Discrete Wavelet Families

The implementation of the discrete wavelet transform helps in denoising separated source to detect echoes in a noise cleaned estimate of the separated signal. For this purpose, different wavelet functions [58] like Debauchies, symlet, coiflets, Fejer-Korovkin, bi-orthogonal and reverse bi-orthogonal are tested to obtain the different SNR improvement of a separated click signal. The idea of cleaning the signal after separation is because some pseudo-periodic noise can be left post-separation. Each separation procedure tracks the energy change either over time or frequency axis, which in turn filters out noise as they are outliers for each of whistle (or harmonics, which are smooth in time) and clicks (or percussions, which are smooth in frequency) and hence discouraging to include a denoising block before the separation of sources. The primary reason being that the sounds aided in this work are visibly clean with clearly audible clicks and whistles.

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#### Algorithm 4: Rigorous SURE thresholding

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**Result:** Producing the threshold criterion for each level for soft thresholding

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**Initialization:**

- DWT coefficients  $C_{DWT}$
- Length of coefficients at each level stored in an array  $L_{DWT}$
- Level of decomposition  $lev_{DWT}$
- Threshold rescaling factor  $sf_{DWT}$

**Intermediate step:**

- $Lrev_{DWT}$  = storing all the values of  $L_{DWT}$  in reverse order except the last value
- $L'_{DWT}$  = storing all values of  $L_{DWT}$  except the first one
- normalize the coefficients w.r.t.  $sf_{DWT}$

**for**  $i \leftarrow 1$  **to**  $level$  **by** 1 **do**

```

Level(i)coeff = (CDWT[L'DWT(i) : LrevDWT(i)])2
N1 = length(CDWT) - 2 : -2 : -length(CDWT); N2 = length(CDWT) - 1 : -1 : 0
CS1 = cumulativesum(Level(i)coeff)
thresi = value(position(min(N1 + CS1 + N2.*Level(i)coeff)))
```

**end**

---

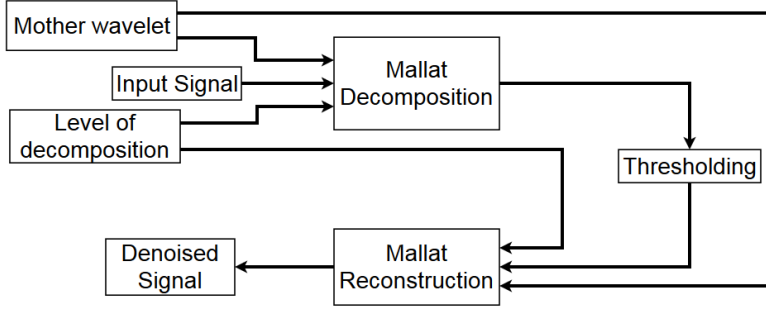


Figure 4.5: Implementation of Discrete wavelet transform to denoise a signal

### 4.2.6 Implementation

The implementation of the discrete wavelet transform is as per the block diagram in Fig. 4.5. The choice of the input signal and the mother wavelet describes the initialisation step. The second step involves the creation of filter matrices to decompose the signal into detailed and approximation coefficients for which setting the number of decomposition level performed before the start of the operation. The usage of *Rigorous SURE* thresholding function with soft thresholding criterion aided in thresholding step. Afterwards, the reconstruction of denoised signal follows Mallot's reconstruction algorithm.

The rigorous SURE [59] thresholding works through Algorithm. 4 which provides a threshold for each level of decomposition.

In order to see the performance of denoising process, the absolute value of the input signal is divided by the absolute of difference of the mean of an input signal to the denoised version of the input signal is calculated. Thereafter logarithm of the result is considered to generate the performance of denoising. The mathematical formula is given below:

$$SNR_{gained} = 10 \log_{10} \left( \frac{\sum_{n=0}^N x^2[n]}{\sum_{n=0}^N (\bar{x}[n] - x_d[n])^2} \right) \quad (4.15)$$

### 4.2.7 Results

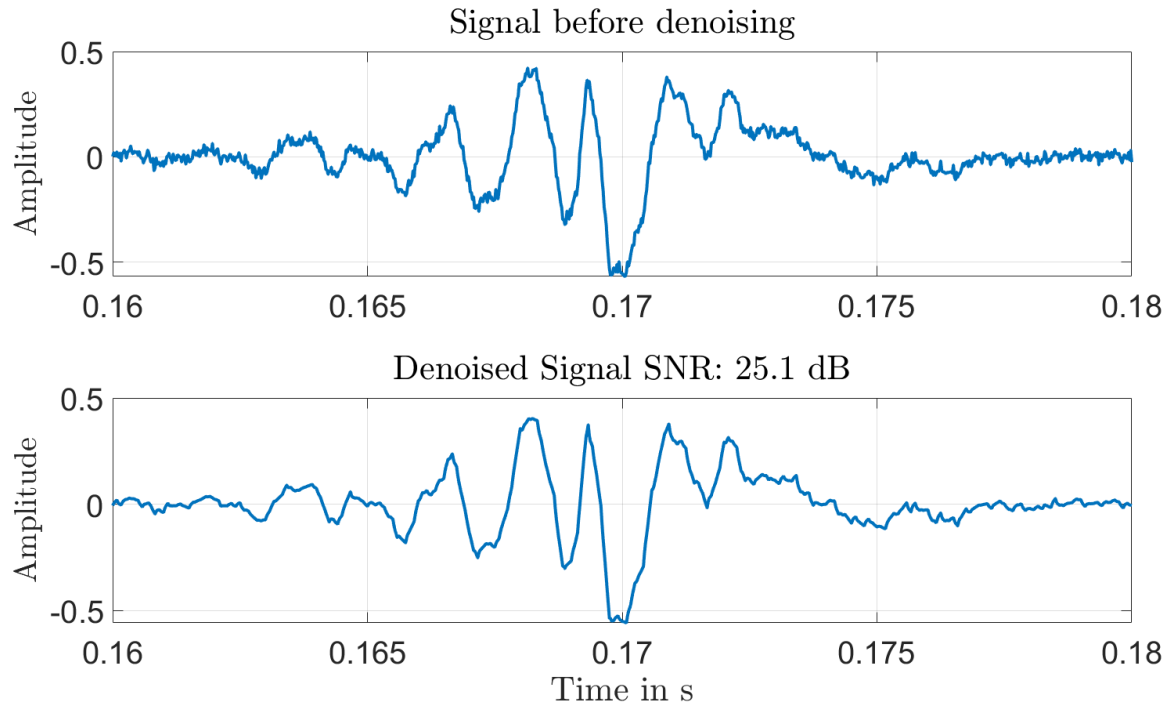
The clicks extracted from the killer whale hunting recording is denoised with all the wavelet family to check the most suitable mother wavelet that can be used in denoising the separated clicks. Higher the value of  $SNR_{gained}$  computed by Eq. 4.15, better is the denoising capabilities. A single click extracted from the audio file of the separated click of killer whale's hunting characteristics. The extracted part is then denoised with different mother wavelets as mentioned in Sec. 4.2.5 following the overall algorithm as pictured in Fig. 4.5 along with the thresholding technique as in Algorithm. 4. Fig. 4.6 pictured an example, where the input signal is denoised using the wavelet *db21*. *db21* is from Debauchies' wavelet family, which is found to provide an optimal gain in SNR with a 5 level decomposition and reconstruction of the signal. The term optimal signifies less computationally intensive process in wavelet transform which in turn depends on the wavelet filter length which is proportional to the computational time.

| Type of wavelet      | Maximum $SNR_{gained}$ in dB | Minimum $SNR_{gained}$ in dB |
|----------------------|------------------------------|------------------------------|
| Biorthogonal         | 27.5 ( <i>bior3.9</i> )      | 14.2 ( <i>bior1.1</i> )      |
| Reverse Biorthogonal | 25.3 ( <i>rbio5.5</i> )      | 14.2 ( <i>rbio1.1</i> )      |
| Debauchies           | 25.1 ( <i>db21</i> )         | 14.2 ( <i>db1</i> )          |
| Symlet               | 25.1 ( <i>sym21</i> )        | 14.2 ( <i>sym1</i> )         |
| Coiflets             | 24.8 ( <i>coif5</i> )        | 20 ( <i>coif1</i> )          |
| Fejer-Korovkin       | 24.5 ( <i>fk22</i> )         | 16 ( <i>fk4</i> )            |

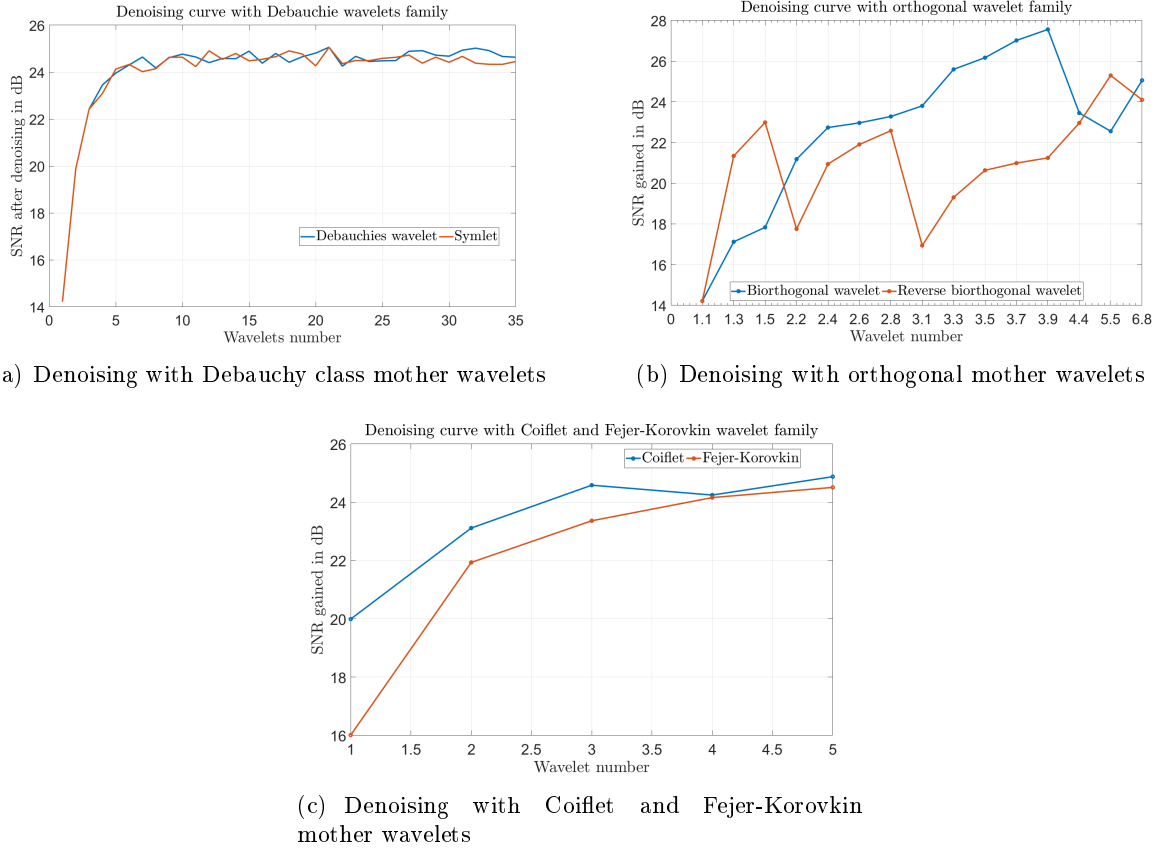
**Table 4.3:** Maximum and minimum  $SNR_{gained}$  by applying different wavelets

Table.4.3 tabulates the gain in SNR obtained by different wavelets which and subsequently pictured in Fig. 4.7. The discarded Debauchies' and symlet wavelet from 36–45 because of their higher computational time requirement to a marginal improvement regarding gained SNR. In Fig. 4.7(c), for Fejer-Korovkin wavelet, the x-axis is different than Coiflets, but they are plotted together because both classes of wavelet have five type of mother wavelet classified based on the filter length.

The next section discusses about the possible method to generate a synthetic impulse response and practical suggestion but not implemented as the middle block between the denoising block and echo analysis block, as limited success is achieved.



**Figure 4.6:** Denoising of killer whale excerpt click signal with *db21* wavelet having a filter length of 42 and 5 level decomposition-reconstruction process. The denoised signal SNR defines as the SNR gained from the original signal to the output of DWT algorithm implemented.



**Figure 4.7:** Denoising Killer whale (*Orca*) separated clicks with different mother wavelets

### 4.3 Synthetic impulse response

The idea of synthetic impulse response generator is catered as the recorded sound is affected by reflections, attenuation, absorption and hence the click structure of a whale vocalisation is distorted. The synthetically prepared impulse response can help in deconvoluting the denoised signal and clicks having further clarity. The process of creating synthetic impulse response requires a wide set of parameters, foremost is the positional details of the source and receiver. Next, is the environmental condition underwater as it regulates the sound speed affecting vocalisation of the whales. Thirdly, the angle at which receiver receives the whale vocalisation.

The remainder of the section discusses about a possible range of parameters and the boundary conditions which could be implemented with a machine learning approach to successfully estimate impulse responses blindly for different marine environment.

As this section is added as introspection to make the recording independent of parameters affecting the whale vocalisations, the idea and initial implementations are detailed in this section but deemed as out of scope to the thesis work.

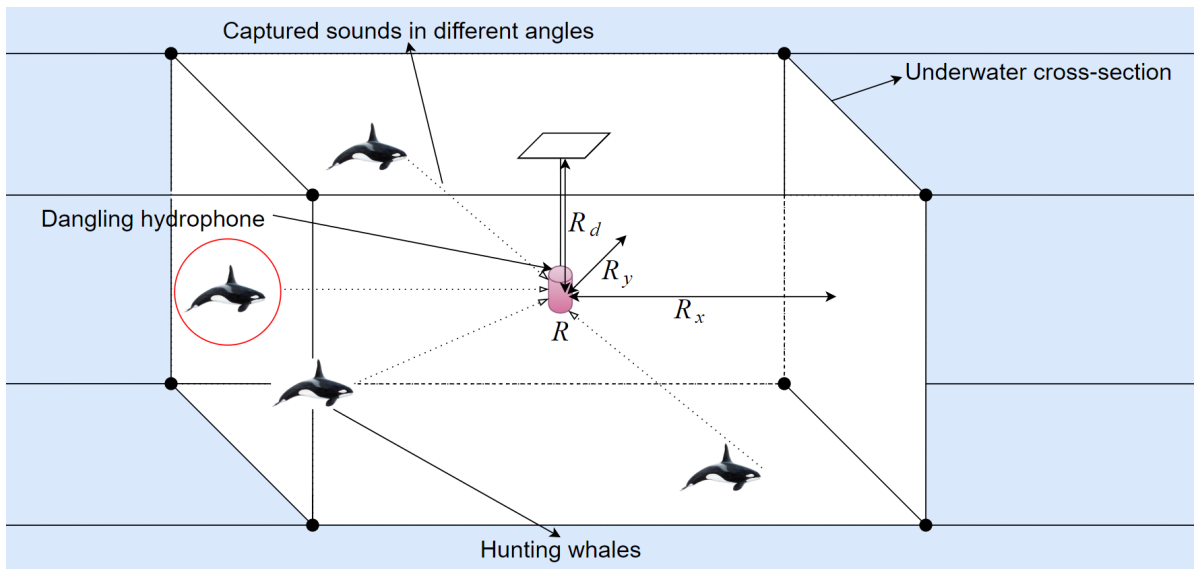
### 4.3.1 Model underwater cross-section

To estimate the channel impulse response, the position of hydrophone needs fixation and to achieve such scenario, a cross-sectional area in a shallow water is considered. The consideration of a cross-section helps in defining position of the receiver ( $R$ ); dangling at a depth of  $R_d$ . Taking the cross-section as a reference, lets consider that the hydrophone is situated at a distance of  $R_x$  and  $R_y$  from the wall of the cross-sectional area. Sources are constantly moving in this scenario and can be at any angle with the horizontal of the receiver. To simplify the scenario, one whale is taken at each of the time and the position of a particular whale is also considered analogous to the hydrophone. Fig. 4.8 shows a model to elaborate the assumptions in a cross-section. Fixing the position of both source and the receiver, viewpoint shifts to parameters like reflection coefficients from the boundary of cross-sectional wall, absorption coefficients and the speed of sound for a certain marine environment. Fig. 4.8 demonstrates the idea of creating an underwater cross-section to create synthetic impulse response.

The absorption coefficient depends on factors like frequency, temperature, depth, salinity and acidity [60,61]. The mathematical formulation to compute the absorption coefficient is given by:

$$\alpha = C_b + C_m + C_{pw}, \quad (4.16)$$

where  $C_b$  determines the boric acid contribution,  $C_m$  gives contribution due to magnesium sulphate and  $C_{pw}$  provides the pure water contribution with each of the contributing factor demonstrated in [60] having their dependencies on salinity and temperature; described by an unit of  $dB/km$ . Reflection coefficients are the next set of parameters. These coefficients depends on the boundary condition of the cross-section. The cross-section is considered from the water surface as the huntings and social structure have their predominance in surface level; defining one constraint. The primary scattering area is the water surface, with a scattering coefficient of 0.88 maximum. The other objects are the toothed whales in pursuit of their target and lastly the target itself. Other subjects are treated negligible given the primary



**Figure 4.8:** A model showing the necessary consideration to create a synthetic impulse res.



ones. These scattering objects degenerate the echoes and enhancing these echoes is possible by signal processing techniques like Hilbert transform and continuous wavelet transform. The separated click signals in consideration can be defined as a broadband pulse modulated at a centre frequency which is limited in time and frequency.

Finally, the dimension of the cross-section needs defining. This demonstrates the final limitation as this dimension needs to be instantaneous because of the whales in motion. The methodology is defined in two ways during researching on the topic as:

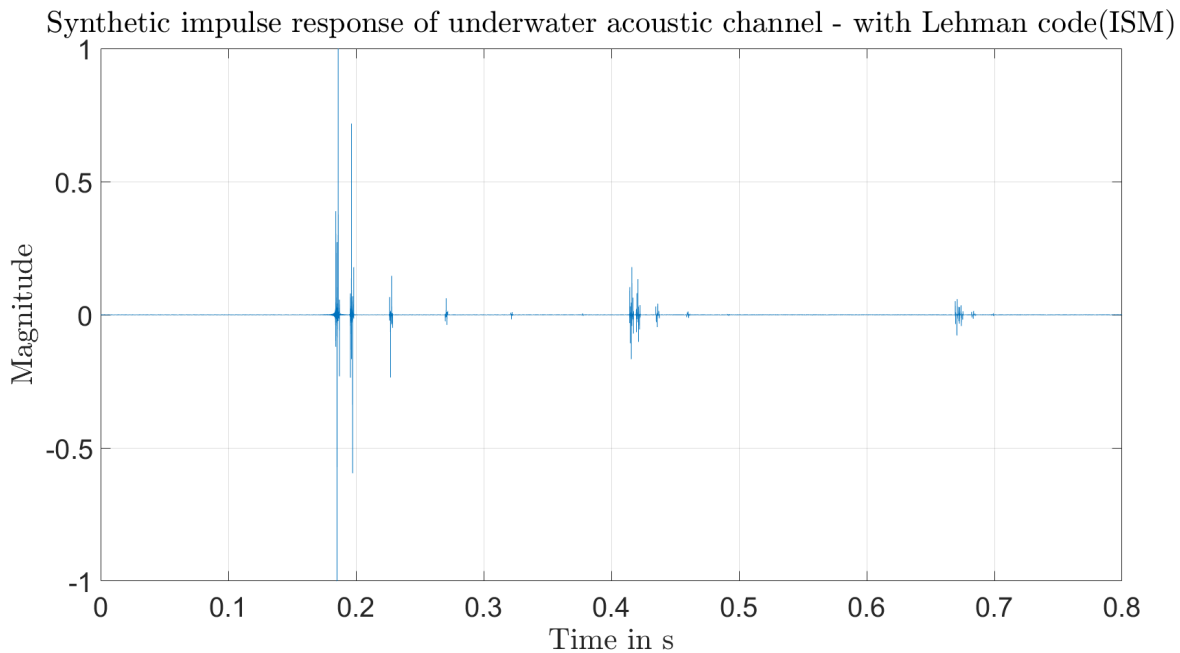
1. The hydrophone is kept at the centre of the cross-section with vocalising whale is considered at any of the six surface of the cross-section.
2. The other way suggest both the entities; vocalising whale and hydrophone; are on the surface of cross-section with the hydrophone remaining at the centre of its residing surface.

Each of the virtual positioning has its own advantage. When considering the first way, the diagrammatically opposite position is occupied by the target and hence reflection coefficient can be fixed to a value closer to 1. Whereas, in the second method, the target of whales can be neglected.

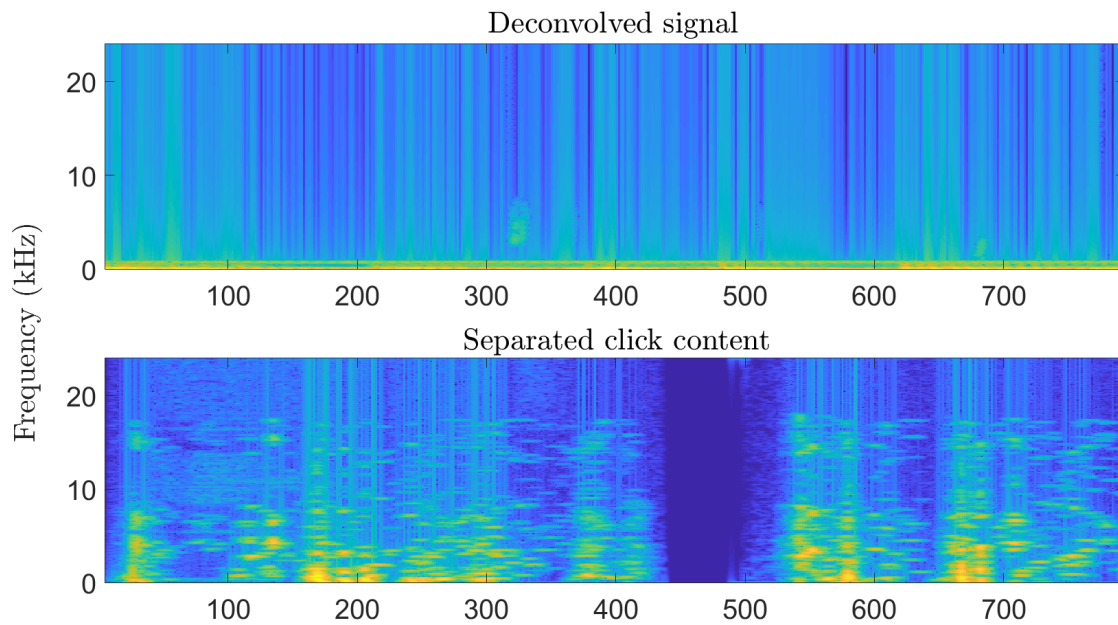
#### 4.3.2 Test implementation and initial results

The implementation of explained model is done in the following way. The method describes synthetic impulse response due to a single whale adapting the work of McGovern [62] and Lehman-Johannson [63]. The first method of calculation adapts the basic nature of creation of a *Room Impulse Response*. The first definable characteristics is the dimension of the cross-section, which is defined as cuboid of dimension  $200m \times 200m \times 100m$  and consequently the hydrophone is placed either at  $100m \times 100m \times 50m$  according to first method and whale is placed at  $0m \times 0m \times 50m$  as shown in Fig.4.8 and according to the second method, the hydrophone position is shifted to the position  $200m \times 200m \times 50m$ , i.e. at the other end of the cube. The sound speed is taken to be  $1520 m/s$  and the reverberation time is taken as  $800ms$  with the reverberation type as *T60*. The absorption coefficient, being dependent on frequency, is calculated for any particular recording.

The impulse response is generated with Lehmann-Johannson's image source model [63] with the position of the hydrophone set at the the boundary of the cross-section with respect to a particular whale and shown in Fig4.9. Fig.4.9(b) provides the deconvolved signal and two probable clicks are seen, one after  $300ms$  mark and the other one is before  $700ms$ . This concludes with possible localisation performed by machine learning algorithm could help in determining the actual click and also determining number of whales and their orientation with respect to the hydrophone position. Due to lack of data from different ocean environment, the denoised clicks are processed further bypassing the deconvolution step and this module is considered as the limiting block of the whole process. A definite machine learning algorithm with varying parameters may generate close to ideal response for any certain marine environment.



(a) Synthetic impulse response generated with documented parameters



(b) deconvolved signal with the synthetic impulse response

**Figure 4.9:** *Denoised click pre-processing for echo analysis*

## 4.4 Echo analysis

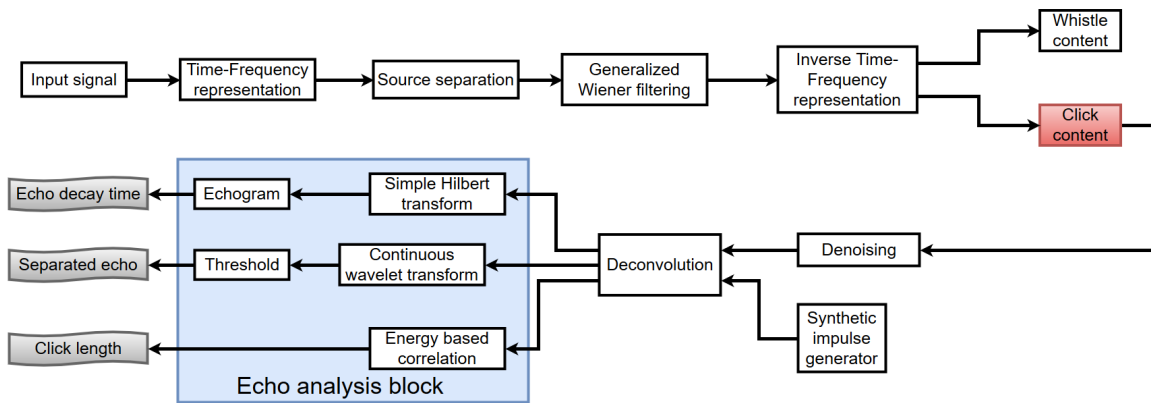
This section elaborate three unique findings from the separated clicks for the dataset used in the thesis. The blue highlighted section in Fig. 4.10 is focussed. Simple Hilbert transform is described first to find the echo decay time followed by the concentrating the energies in a click pulse through CWT and after that performing energy based correlation for denoting the click

length considering clicks as fingerprints for toothed whales [64, 65]. This interests in finding echoes and separating them from the main clicks. In addition to that, the echoes further help in estimating the number of objects the clicks had encountered within a certain range.

The idea of non-destructive testing (NDT) [66] is implemented in this cause to detect overlapping echoes produced due to the multi-path propagation of shallow water communication. NDT exploits the idea of finding cracks within a construction site or knowing the internal composites of any complex material through sound pulses. The primary methods followed in the detection of cracks or abnormality are Hilbert transform and continuous wavelet transform. These sound pulses bounce off the material from the source, fixed at a certain point, and consequently received by a fixed placed receiver. If there lies any possible crack, the receiver receives the bounced sound at a later point of time with a lower amplitude. The stored reference echoes of a particular material correlated with the received echo to determine features of the crack. This work tries to utilise similar concept in abstractly determining the echoes as the sounds aided in this work are the only source of information. Hilbert transform approach is the primary focus of discussion before the continuous wavelet transform as the developed structure tries to extract two major features of a click:

1. Echo decay time
2. Separating high energy clicks from the overlapped echoes of lower energy

Hilbert analysis helps in extracting the features mentioned in Point. 1 based on the assumption of a decaying waveform. The insights from these results motivated the use of CWT in ascertaining the number of clicks in one single click and elaborates the characteristics mentioned in Point. 2. Hence, Sec. 4.4.1 uses a simple adoption from that of Hilbert-Huang approach [66] for providing the initial speculations obtained via echogram analysis. Later, in Sec. 4.4.2, the idea presented in the introduction of Sec. 4.2 is further developed and implemented in visualising the echoes.



**Figure 4.10:** Three methods generating the unique features of an echolocation clicks.

#### 4.4.1 Hilbert transform

Echo signal, in general, can be defined as the signal which arrives at a time later than that of the line-of-sight (original) signal. In case of underwater captured sounds, each overlapping click signals formed by the echoes from clicks produced before and each click is proved [67] to contain more than one clicks which are the primary motivating factor in focusing on the Hilbert analysis of a signal. Hilbert transform helps in computing the analytic signal providing the same frequencies as the input signal but shifted by  $90^\circ$ . The following section describes the mathematical foundation of the proposed model in the following section:

##### 4.4.1.1 Mathematical foundation

Assuming that a signal decays over time, let's assume that  $x(t)$  is a time domain signal, with amplitude  $A$ :

$$x(t) = Ae^{\frac{-t}{\tau}} \cos(\theta(t)), \quad (4.17)$$

where  $e^{\frac{-t}{\tau}}$  defines the decay with  $\tau$  as the decay time constant. The argument of  $\cos$  is not defined as whale vocalization is considered to be either linear or exponential chirp. The Hilbert transform of Eq. 4.17 is given by:

$$x_H(t) = \mathcal{H}\{x(t)\} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \quad (4.18)$$

$$\Leftrightarrow x_H(t) = -Ae^{\frac{-t}{\tau}} \sin(\theta(t)) \cdot \frac{d}{dt}(\theta(t)) \quad (4.19)$$

The term analytic is tailed to the Hilbert transformed signal as it is the analytic function of a continuous complex variable. The result contains both real and imaginary components as

$$x_A(t) = x_{A_r}(t) + jx_{A_i}(t) \quad (4.20)$$

$$\Leftrightarrow x_A(t) = x(t) + j\mathcal{H}\{x(t)\}, \quad (4.21)$$

where  $x_{A_r}(t) = \text{Re}\{x_A(t)\}$  and  $x_{A_i}(t) = \text{Im}\{x_A(t)\}$  are real valued functions and it can be shown that  $x_{A_r}(t) = x(t)$  and  $x_{A_i}(t) = \mathcal{H}\{x(t)\}$  [68],  $\mathcal{H}\{\cdot\}$  describes the Hilbert transform. The magnitude of analytic signal which can be given by the polar magnitude vector  $|x_A(t)|$  as

$$r = |x_A(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x(t)\}^2}, \quad (4.22)$$

which helps in creating envelope of the signal. Natural logarithm of Eq. 4.22 can be given by following considering  $\theta(t) = C$ ,  $C$  being constant. The reason to take the argument as constant due to the fact that each click have a certain bandwidth and center frequency. This assumption

provides the value  $\frac{d}{dt}(\theta(t)) = 0$ :

$$\ln |x_A(t)| = \ln(A^2 \cdot e^{-\frac{t}{\tau}} (\sin^2(\theta) + \cos^2(\theta)))^{\frac{1}{2}} \quad (4.23)$$

$$\Leftrightarrow \ln |x_A(t)| = -\frac{t}{\tau} + \ln A \quad (4.24)$$

$$\Leftrightarrow \ln r = -\frac{t}{\tau} + \ln A \quad (4.25)$$

which considers the fact of  $\sin^2(\theta) + \cos^2(\theta) = 1$ . From Eq. 4.25, the value of  $\tau$  will provide the echo decay time of the decaying signal

#### 4.4.1.2 Simple Hilbert transform - Implementation and algorithm

The proposed method follows Fig. 4.11. The first step calculates the Hilbert transform of the input signal which provides the imaginary part of the analytic signal as described in Eq. 4.21. The second step involves in finding the magnitude of the analytic signal as in Eq. 4.22. The penultimate step calculates the logarithm of resultant magnitude to calculate the echo decay time. The representation of the finding forms an echogram [69], which is a standard across NDT technology. The standard method has a three-dimensional plot with depth on the y-axis, different colours represent time in x-axis and magnitudes. As only an estimation of depth in the underwater scenario, The echogram presented in the thesis is the only logarithm of magnitude vs time of the recorded signal.

The first step involves a description of the Hilbert transform, and other steps follow the equations as mentioned in the paragraph before. Fast Hilbert transform [70] is performed with the aid of fast Fourier transform which has a fundamental basis of transform pair as  $x_H(t) \Leftrightarrow \mathbf{X}_H(f)$ . The process involves following steps:

1.  $P$ -point FFT of the input signal is calculated in the first step of  $x(t)$  with  $P$  sample points sampled at  $T_s$
2. A vector  $v(i)$  is created as follows:

$$v(i) = \begin{cases} 1, & i = 1, (\frac{P}{2}) + 1 \\ 2, & i = 2, 3, \dots, (\frac{P}{2}) \\ 0, & i = (\frac{P}{2}) + 2, \dots, P \end{cases} \quad (4.26)$$

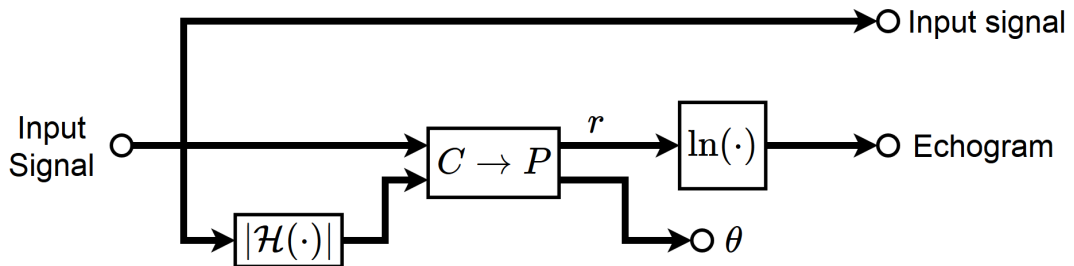


Figure 4.11: Simple Hilbert transform analysis to find Echo decay time

3. Element wise product of  $v$  and  $\mathbf{X}$  is calculated
4.  $P$ -point Inverse FFT of the sequence is calculated thereafter to get the Hilbert transform of the input signal

Teloni et al. [71] proved the consistency of click structure of sperm whale clicks which consists a three pulse formation and on the recorded signal in the front, back or at an angular position ahead of the species which is in motion continuously. Click structure of orca pods and other species having dense clicks conforms to the result presented in [71] but toothed whales like the porpoise, common dolphins contains two pulse structure as can be seen in Fig. 1.3(a).

#### 4.4.1.3 Echo Decay time: Results

A test with four overlapping sine wave is tested to find justification of the performed algorithm and consequently shown in Fig. 4.12. The implementation is done by producing a sine waves with sampling frequency  $8\text{ kHz}$  and carrier frequency  $0.8\text{ kHz}$ . Later three delayed and attenuated version of the created signal is superimposed on the original signal to create the overall signal. Subsequently, the whole signal is treated with the algorithm mentioned in Algo. 5. The result showed three steep jumps with the last being most prominent signifying all the sine wave are saturated. Second observation is visualising three jumps whereas four signals are present in the test signal, concluding smooth transition between a certain decayed part of the test signal to the continuing signal.

The signal excerpt containing one click from orca pod and porpoise evaluated with the algorithm depicted in Fig. 4.11. Passing the input signal through Hilbert transform operator provides the imaginary part of the analytic signal which then passed through Cartesian to polar ( $C \rightarrow P$ ) transform block along with the input signal to obtain ( $r = |x_A(t)|$ ). The echogram obtained as per Fig. 4.11 is the plot of the logarithm of the magnitude of the analytic signal concerning time.

In Fig. 4.13 and Fig. 4.14, the given echograms, found are certain discontinuities along the time. These discontinuities form sharp momentary decay along the decaying click as shown in the above-said figures. Porpoise of class Dall are found to have two echo decay times in a single

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#### Algorithm 5: Simple Hilbert transform

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**Result:** Natural logarithm of the analytic signal

**Initialization:**

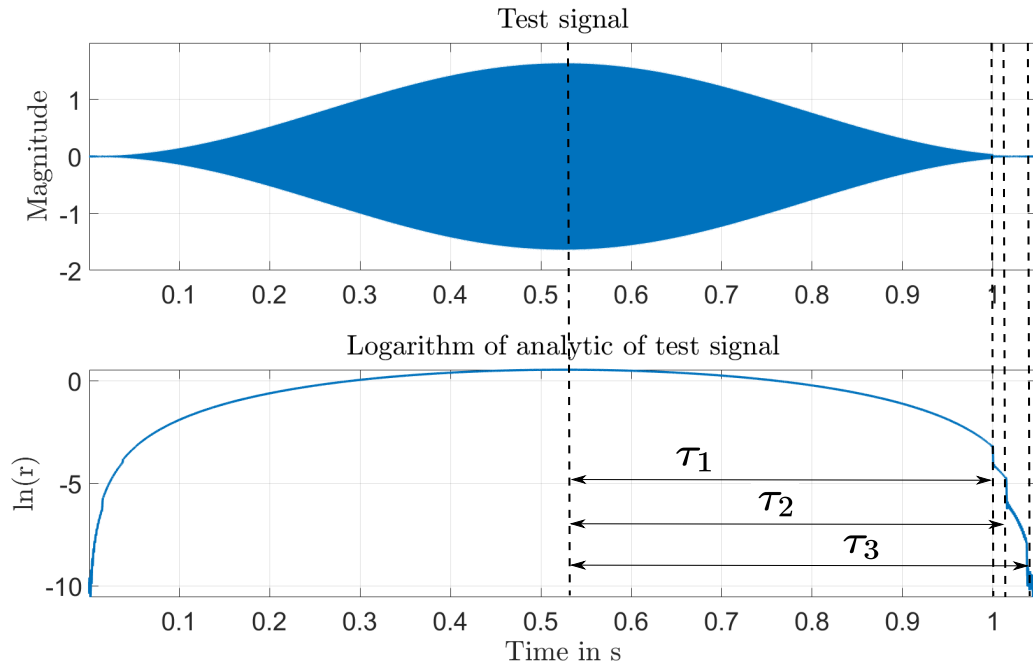
- $x(t)$

**Intermediate step:**

- Defining the vector  $v(i)$  as in Eq. 4.26

**Final step:**

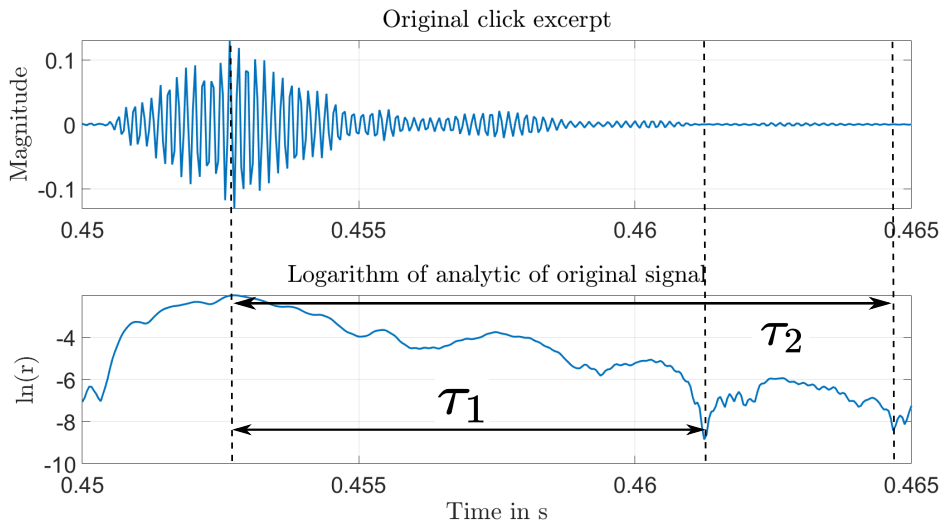
- Implementation of steps presented in Sec. 4.4.1.2 to compute  $x_{A_i}(t)$
  - $x(t) = x_{A_i}(t)$
  - Calculate  $r$  using Eq. 4.22
  - Calculate the natural logarithm using Eq. 4.25
-



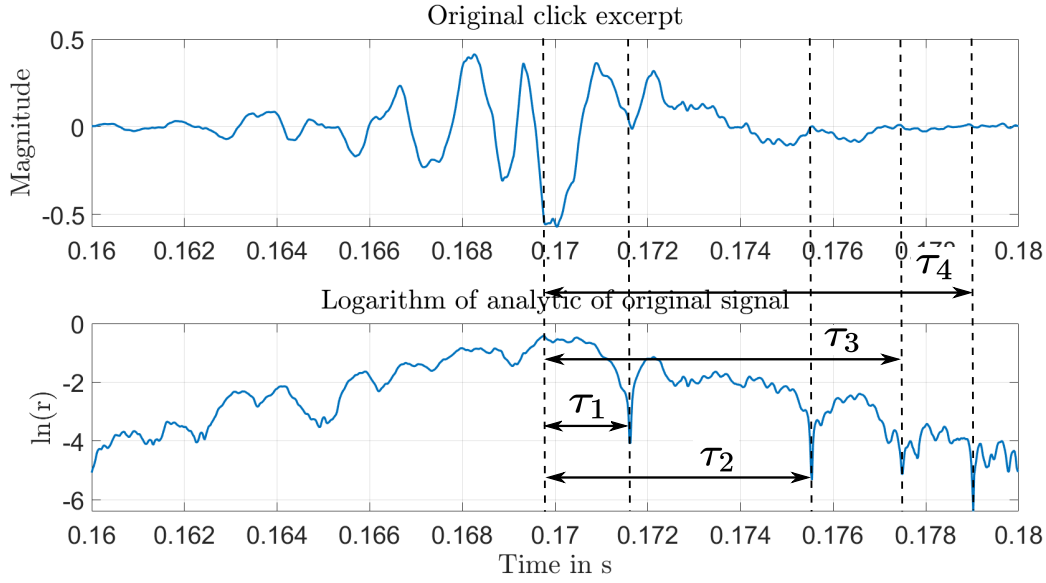
**Figure 4.12:** *Top:* The original overlapped decaying random sine signals. *Bottom:* The natural logarithm of the magnitude of analytic of original signal

click structure and on the contrary when considering killer whales or orca; four such echo decay times recorded. Extracting the maxima of the concerned standalone click and calculating the distance to each of the sharp dips are calculated defining them as the echo decay time and representing the entities as  $\tau_{d=1,...,D}$ .

One firm conclusion can be made based on the echogram result is that a single click pulse contains superimposed pulses after being emanated by whales as the picture in the top of Fig. 4.14. On the contrary  $\tau_{d=2,...,D}$  can be seen as the reflected sounds of the target of other



**Figure 4.13:** *Top:* The original click excerpt from porpoise audio. *Bottom:* The natural logarithm of the magnitude of the analytic of original click excerpt



**Figure 4.14:** *Top:* The original click excerpt from killer whale audio. *Bottom:* The natural logarithm of the magnitude of the analytic of original click excerpt

mammal involved in hunting.

Hence, to make a comparative study of echo decay time, the penultimate decay time or  $\tau_D$  is tabulated. Beaked whale, three classes of porpoise, false killer whale, pilot whale, narwhal, sperm whale and two classes of dolphins are studied to give detail on the claims that clicks are unique for every species and listed in Table.4.4. The echo decay time also varies with the activity; while searching click of a killer whale taken as a standard. The echo decay time increases at glacier bay can be accounted for lower sound speed at a low temperature; demonstrated in Fig.4.1(a) and Fig.4.1(b). The hunting click's decay time is the most and can be argued so because of their pod-like hunting practices which substantially follows the general communication principle - listen before speaking.

The shortcomings of this method are that it does not give a higher degree of insight about the echoes in general or rather the visualisation of the individual clicks present within a superimposed click. The superimposition of numerous pulses in clicks leads to the next topic of continuous wavelet transform which helps in localising the frequencies over time thus provides visualisation of the energy concentration at a particular time denoting the number of individuals present in a specific scenario.

#### 4.4.2 Continuous wavelet transform

Correlation processing is a general signal processing method in case of sonar technology where the number of the signal path, structure of the environment and location of transmitter and receiver are the important pieces of information required for detecting echo content of the signal. An optimisation is tried to achieve with a trial and error method in Sec.4.3 regarding developing a synthetic impulse response considering all the environmental factors with limited success. Emphasizing the ever-changing aquatic environment; scatter phenomenon formed by multipath scattering deteriorates the interpretation of echoes. Split-spectrum processing [72]



| Name of the species | Echo decay time (ms)   |                      |                      |
|---------------------|------------------------|----------------------|----------------------|
| Beaked whale        | 11.8                   |                      |                      |
| Porpoise            | Dall - 8.5             | Finless - 9.5        | Harbor - 13.1        |
| False killer whale  | 36.2                   |                      |                      |
| Killer whale        | Searching click - 15.5 | Glacier bay - 20.2   | Hunting click - 29.3 |
| Pilot whale         | 27.4                   |                      |                      |
| Narwhal             | 8.9                    |                      |                      |
| Sperm whale         | 2.1                    |                      |                      |
| Dolphin             | Spinner - 14.8         | Rough-toothed - 14.3 |                      |

**Table 4.4:** *Difference between the peak of a single click to the last dent ( $\tau_D$ ) due to the decay of echo produced by the click*

technique is a method in which the received signal split over a set of selected frequencies by digital filtering, and some minimisation algorithm is then applied to the resulting narrowband signals to extract the echo from the signal. The band-pass filters used in the decomposition have a constant bandwidth which is a disadvantage. The received signal, in that case, maybe a combination of low and high-frequency contributions which characterises the need for an adaptive method that efficiently localises both short and long variations.

Wavelet transform refers to a collection of windowed Fourier like adapted transform algorithms. Given any signal, these methods provide a particular family of elementary functions, representing any signal relative to it. Regarding split-spectrum processing, the wavelets are filters with constant relative bandwidth and optimal time and frequency concentration, making the splitting process of wideband signals more efficient with wavelets than with standard filters. The wavelet transforms associates to each signal a 2D representation in the time-frequency plane. This representation localises the frequency content of the signal in time, and for the diagnostic purpose, this is an excellent advantage over time-only or frequency-only representations of the signal.

#### 4.4.2.1 Mathematical background: Choice of a particular wavelet family

The process involves three phase:

1. Defining a family of "sample wavelets" by  $\sqrt{2\pi}\psi_{a,b}(t) = \chi(t - 2b\pi)e^{iat}$ ,  $a, b \in \mathbb{Z}$ , where  $\chi(t)$  being the rectangular window function. This form helps to inbound the wavelet within a specified bound and then shifting it over time.
2. Forming the sample wavelets into an orthonormal basis of the Hilbert space of square integrable (i.e. finite energy) functions
3. Using this as a basis for expanding the signal in consideration

Describing Eq.4.5, in detail, the choice of an analysing wavelet  $\psi$ , the other wavelet of the family is defined by two parameters: position and size. Rewriting Eq.4.5 with  $a > 0$  as the size parameter and  $b \in \mathbb{R}$  as position parameter:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) \quad (4.27)$$

The wavelet coefficients  $W_x(a,b)$  of the signal  $x(t)$  are given by:

$$W_x(a,b) = \int_{-\infty}^{\infty} x(t) \psi_{a,b}^*(t) dt, \quad (4.28)$$

where  $\psi_{a,b}^*(t)$  represents the orthonormal basis of the wavelet helping in expansion of  $x(t)$ . As the sources are assumed to be Gaussian distributed during source separation, Gaussian analyzing wavelet is preferred and Morlet's wavelet is considered which may be formulated as:

$$\psi(t) \approx \pi^{-\frac{1}{4}} e^{j\omega_0 t} e^{-\frac{t^2}{2}} \quad (4.29)$$

From Fig. 4.15, a few conclusions may be drawn to justify the choice of the wavelet to locate the echo of a primary signal. Foremost, wavelet offers a *compact support* which essentially means that it is valid for a certain period. The other characteristics are that the area under the curve is zero; distributing the energy equally in positive and negative directions. The translation and scale parameters are the deciding factors for characterising a wavelet. Translation is used to move the wavelet back and forth in time while correlating the wavelet with the signal in consideration to get the coefficients. The scale and frequency in case of the wavelet is related inversely with a proportionality constant which is termed *center frequency* ( $F_{eq}$ ) of the wavelet and is due to wavelet's bandpass characteristics. So to define the equivalent frequency to interpret the graph, following mathematical equation is adopted:

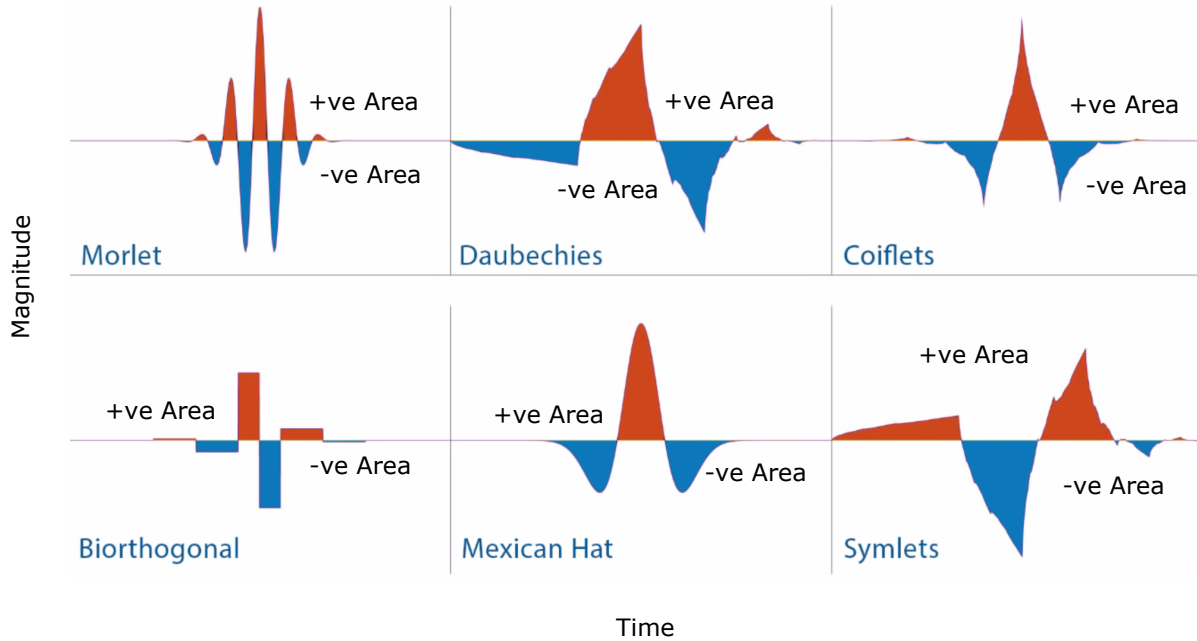


Figure 4.15: Different wavelets

| Wavelet scale                | 2          | 4          | 8          | 16          |
|------------------------------|------------|------------|------------|-------------|
| Equivalent freq ( $F_{eq}$ ) | $F_{eq}/2$ | $F_{eq}/4$ | $F_{eq}/8$ | $F_{eq}/16$ |

**Table 4.5:** Wavelet scale vs equivalent frequency

$$F_{eq} = \frac{C_f}{aT_s}, \quad (4.30)$$

Where  $C_f$  is the centre frequency of the chosen wavelet,  $a$  is the scale factor and  $T_s$  is the sampling time of the signal to be analysed. Hence a higher scale factor  $a > 1$  gives a low frequency, or a stretched wavelet is tracking the slowly varying changes whereas a lower scale factor  $0 < a < 1$  provides higher frequency which shrinks a wavelet helping in tracking abrupt changes in a signal.

Vanishing moments are the next characterising element in choosing the right wavelet for analysing a waveform and hence defining  $k^{th}$  moment ( $m_k$ ) of function  $f(x)$  as:

$$m_k = \int_{-\infty}^{\infty} f(x)x^k dx \quad (4.31)$$

, where  $k^{th}$  moment vanishes when the integral is zero. The implication suggests that a wavelet with a higher number of vanishing moment more accurately represents a signal than a wavelet with lower vanishing moment. A related aspect is the regularity of a wavelet. A wavelet of higher regularity creates a smoother representation of the signal than a signal with lower regularity which concludes that higher the vanishing moment, better is its regularity and thus generates an accurate representation of a signal.

#### 4.4.2.2 Thresholding Criterion

Thresholding is applied based on the intuition that signal bounced off from potential target is at most 50% of the maximum click power recorded by the PAM instrument with preceding and trailing echoes alongside overlapped with the estimated clicks in each pulse. The thresholding applied to each scale of a particular wavelet is as follows:

$$P_x = W_x^2 \quad (4.32)$$

$$P_x = \frac{100 \cdot P_x}{\int P_x} \quad (4.33)$$

$$threshold = \frac{\max(P_x(a, :)) - \min(P_x(a, :))}{2} + \min(P_x(a, :)), \quad (4.34)$$

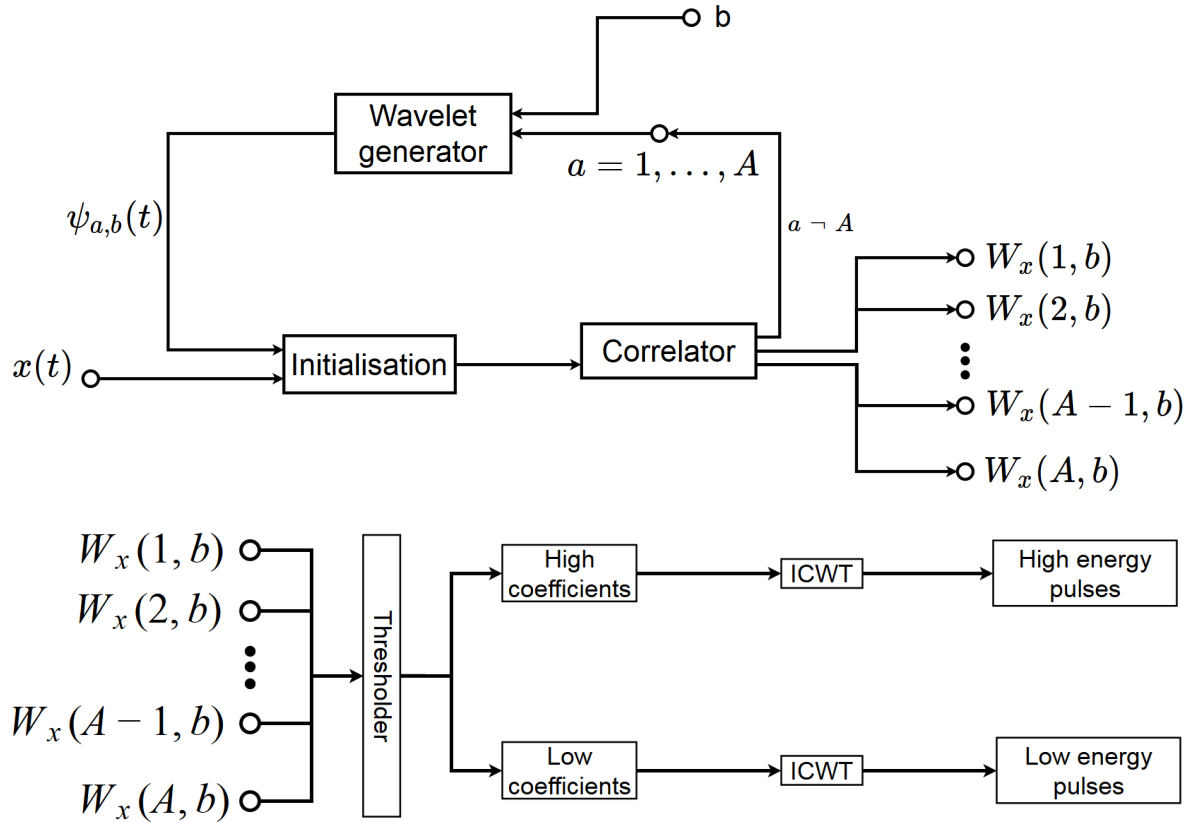
Where  $W_x$  are the wavelet coefficients and  $P_x$  represents the power of coefficients derived from CWT process. The implemented threshold is a hard threshold. The hard thresholding

subsequently translates into 0 for any value lower than the threshold, whereas the higher values are left as they are when clicks are concerned. To extract the high energy content from the echo; each value of coefficients higher than the threshold are pulled down to 0, as they considered to be the least scattered reception and keeping the values lower than the threshold as they are.

#### 4.4.2.3 Implementation

Continuous wavelet transform are implemented according to Fig. 4.16. The initialisation step involves the input signal, generating the wavelet of choice and ultimately the scaling factor. Afterwards, correlating the input signal with the wavelet, modified with each scale and for a certain value of position parameter  $b$ . The correlation of each wavelet  $\psi_{a,b}(t)$  gives rise to a specific set of coefficients which accumulated together into a matrix of wavelet coefficients. The coefficients are plotted using scalogram function which provides complex information about the time localised frequencies. The y-axis represents equivalent frequency which ensues from Eq. 4.30 and x-axis denotes the time which is equivalent to the duration of the analysed signal.

The translation is taken to be single sample shift to get every coefficient to get every possible information. The process of constructing wavelet scales proffered detailed attention due to its inverse relation to the signal frequency shown in Table. 4.5. Continuous wavelet transform helps in analysing a signal at an intermediary octave which allows in restricting the frequency of



**Figure 4.16:** Implementation method for continuous wavelet transform to localize the energies in frequencies and time

| Name of the parameter            | Value of the Parameter |
|----------------------------------|------------------------|
| $a$ (scale parameter)            | 5, ..., 16             |
| Scale interval                   | 0.02                   |
| Mother wavelet                   | Morlet                 |
| $C_f$                            | 0.8125 Hz              |
| $b$ (position / shift parameter) | 1                      |

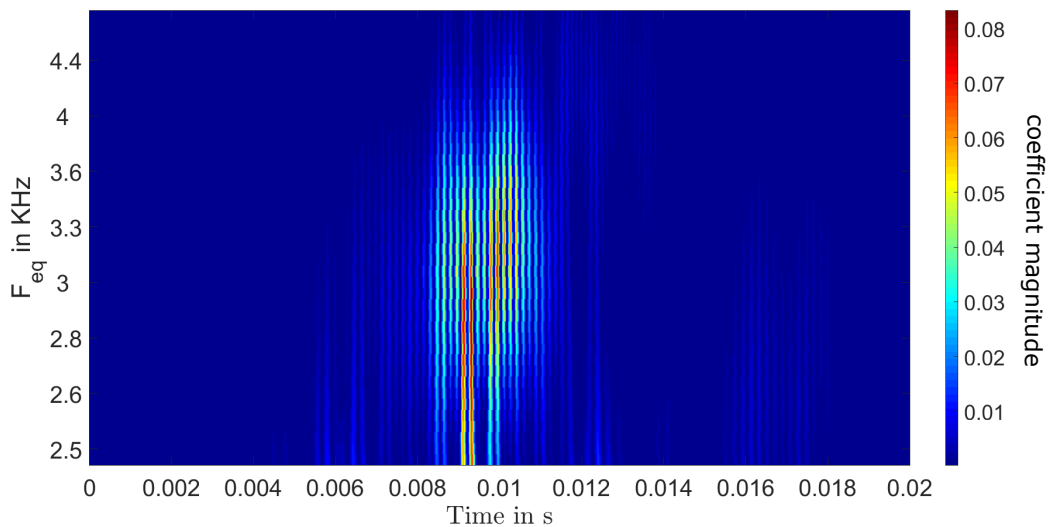
**Table 4.6:** Parameters used in CWT computation

operation to  $f_s/2$  from a desirable low frequency or in other words keeping the upper boundary of the scale factor to a minimum value. The choice of Morlet wavelet typified its higher regularity and consequently subsequently higher order of vanishing moments thereby representing the complex click signal better than that of a wavelet with lower vanishing moments like a Mexican hat, Debauchies wavelet class and others.

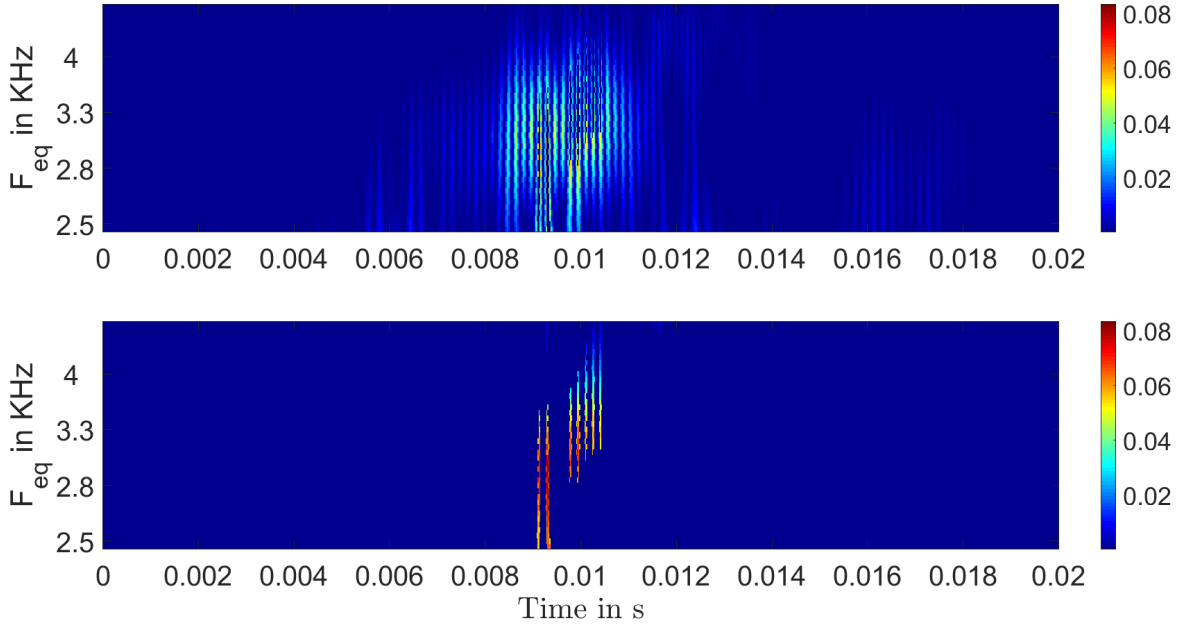
#### 4.4.2.4 Results

The output of continuous wavelet transformation are the coefficients, the depiction of it is in Fig. 4.17. The Eq. 4.34 is used to separate the echo content in each pulse. The argument cited in [67] is counter validated that each click pulse contains superimposed clicks, which can be emanated either by a single species or they determine the number of species talking at the same time. Echoes are in the top figure of Fig. 4.18 determines the number of the object reflecting the clicks. Energies are found concentrated within  $2.5\text{ kHz} - 4.3\text{ kHz}$  and are possible to analyse due to a fine-scale analysis employing continuous wavelet transform. The hard thresholding applied to the coefficients and afterwards through inverse continuous wavelet transform; Fig. 4.19 presents a visualisation of the click pulses.

The peaks in inlying clicks of a pulse are correctly visible and the echoes during that particular

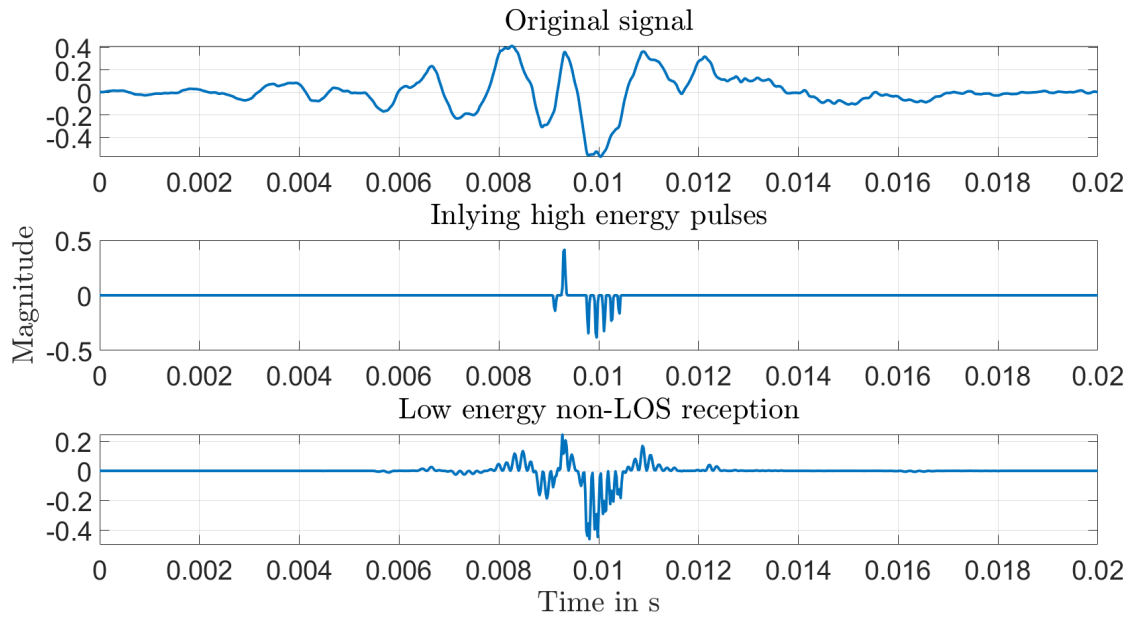


**Figure 4.17:** CWT of a single pulse from killer whale or orca



**Figure 4.18:** *Top:* Estimation of the echo content, *Bottom:* Estimation of the clicks in one pulse based on hard thresholding described in Sec. 4.4.2.2

time seen in the bottom graph of Fig. 4.19. Arguing on the basis of structure of clicks shown in [71] due to the position of hydrophone recording the sound, attributing the positive peak to the fact that killer whale producing the click was in front of the device (hydrophone) and the decaying clicks saw in the middle graph of Fig. 4.19, are recorded when the whale went past the device. Bottom graph of Fig. 4.19 shows the different multi-path propagated sounds captured because the whale is in constant motion during hunting.



**Figure 4.19:** *Top:* Original denoised signal, *Middle:* Estimated high energy content in the signal, *Bottom:* Estimated echo of the signal

## 4.5 Energy based correlation

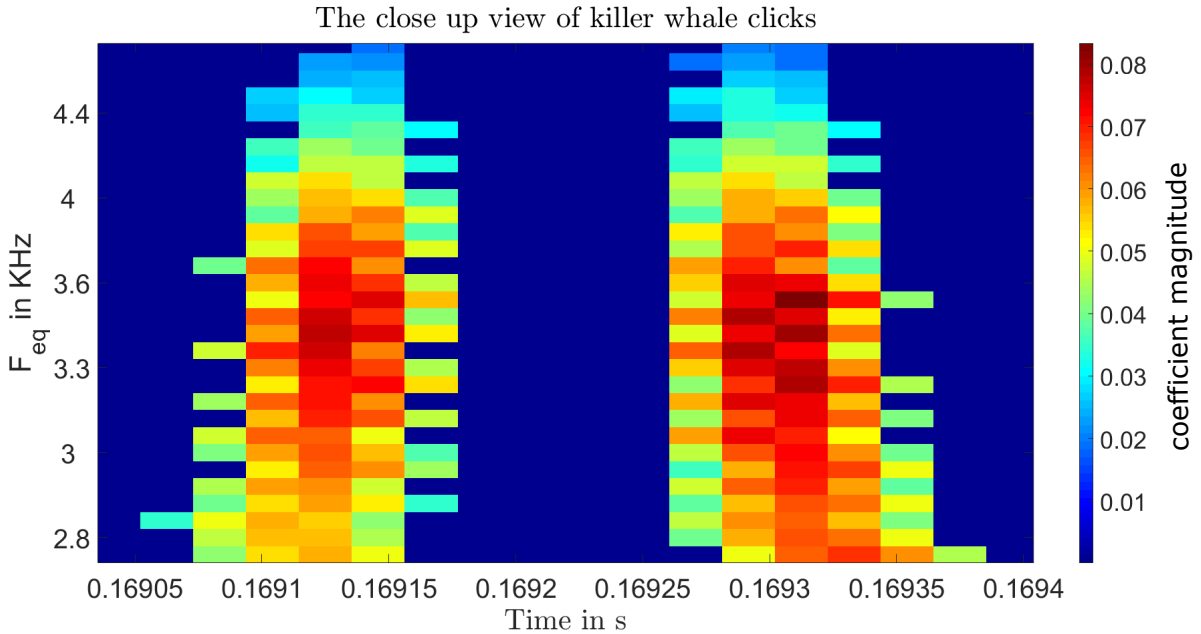
### 4.5.1 Motivation

Chirps are ubiquitous [73] in nature and are used in communication by different classes of animals. Chirp characterises the echolocation clicks used by toothed whales and is analogous to those of bats. The time localised frequencies found through continuous wavelet transform showed chirp-like characteristics. The variation in frequency over time is linear in some cases (for Narwhals) and exponential in others (Killer whales). Fig. 4.20 finds the two most robust clicks of killer whale hunting recordings after continuous wavelet transform motivating the method of chirp correlating with the recording used in the thesis to find the click length of any toothed whale species.

Chirp or compressed high-intensity radar pulse was first formulated by Sidney Darlington [74] for long range and high-resolution performance of radar system. The most important characteristic of the chirp is that on reception, correlating the received signal with the transmitted reference results in an absolute peak which can provide the distance between source and receiver.

### 4.5.2 Mathematical background

Chirp is characterised by varying frequency over time, either linear or exponential. The following equation gives the mathematical definition of chirp defined by  $z_c(t)$ .



**Figure 4.20:** Close up view of the strongest clicks in killer whales from one single pulse.  
Derived from bottom of Fig. 4.18

$$z_c(t) = \sin(\theta_c(t)) \quad (4.35)$$

The instantaneous frequency  $f_c(t)$  is defined as the derivative of phase of the signal which is given by:

$$f_c(t) = \frac{1}{2\pi} \frac{d\theta_c(t)}{dt} \quad (4.36)$$

and the rate of change of frequency,  $c(t)$ , is given by the derivative of instantaneous frequency:

$$c(t) = \frac{1}{2\pi} \frac{d^2\theta_c(t)}{dt^2} \quad (4.37)$$

$$\Leftrightarrow c(t) = \frac{df_c(t)}{dt} \quad (4.38)$$

Chirp is classified into linear and exponential based on frequency rate. The frequency rate changes linearly over time for linear chirps and exponentially with time for exponential chirp. The frequency again can either increase or decrease with time depending on it, each class of chirp is further divided into up-chirp and down-chirp. If the starting frequency,  $f_0$ , and the end frequency,  $f_t$  is defined, the rate of change of frequency is defined as:

$$c_l(t) = \frac{f_t - f_0}{T} \quad (4.39)$$

$$c_e(t) = \left( \frac{f_t}{f_0} \right)^{\frac{1}{T}} \quad (4.40)$$

, where  $T$  is the time taken to sweep from  $f_0$  to  $f_t$ ,  $c_l(t)$  denotes linear frequency change and  $c_e(t)$  is the exponential frequency change. Rearranging Eq. 4.39 and Eq. 4.40, thereby integrating over time,  $\theta_c(t)$  is obtained for linear chirp,  $z_l(t)$ , and exponential chirp,  $z_e(t)$ , which are as follows:

$$z_l(t) = \sin \left[ \theta_0 + 2\pi \left( f_0(t) + \frac{c_l(t)}{2} t^2 \right) \right] \quad (4.41)$$

$$z_e(t) = \sin \left[ \theta_0 + 2\pi \left( f_0(t) + \frac{c_l(t)}{2} t^2 \right) \right] \quad (4.42)$$

$$(4.43)$$

Chirps are varied over time in frequencies, and the idea of correlating them with the click signals helps in knowing the length of a single echolocation click. Hence, clicks are singled out for this analysis as they are common characteristics across toothed whale species used in hunting process and are pointed out as a distinguishable feature of this species. In next sections, the correlation technique developed to extract features of any click and followed by the results derived, reviewing the findings as a conclusion to this section.



| Parameter name     | Value                               |
|--------------------|-------------------------------------|
| Amplitude of chirp | 1                                   |
| Duration of chirp  | 0.1 ms...20 ms                      |
| $f_0$              | 0 kHz...(f <sub>s</sub> /2 - 3) kHz |
| $f_s$              | 3 kHz...f <sub>s</sub> /2 kHz       |

**Table 4.7:** Parameters used to create the reference chirps

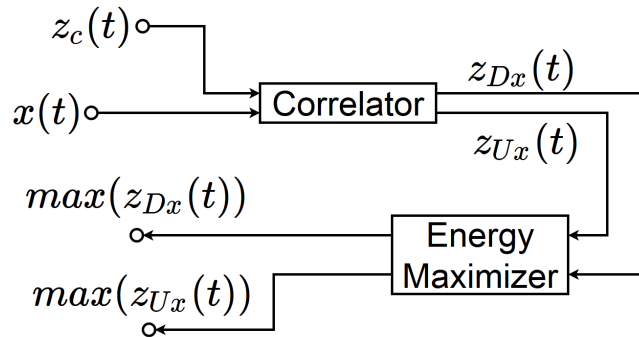
### 4.5.3 Proposed correlation technique: Implementation

In an attempt to abstract the process of finding the click length of any toothed whale developing an energy-based correlation process involves the creation of a synthetic chirp as shown in Fig. 4.22. The reference chirp generator creates chirp with unit amplitude, the start and end frequency. These created chirps repeated and performing correlation with the input signal, thereby storing the results obtained in the process.

The stored results further iterated to compute the ratio of the energy of the original signal to the area under correlation curve, maximising which provides the best fitted correlated chirp with a particular set of parameter. The chirp length is varied from 1 s to 20 ms in steps of 1 ms. In order to mitigate the effect of phase shifts, up and down chirps are created using Eq. 4.42 and Eq. 4.43 along with cosine counterparts. The correlation results are squared, added and after that taken the square root of the result to plot the magnitude and thereby to neglect the influence of phase.

The correlation is done in a windowed approach. The resultant correlation with up-chirp is found in the following way:

$$z_{Ux}(t) = \sqrt{\left(\sum_t x(t) \cdot z_U(t)\right)^2 + \left(\sum_t x(t) \cdot z_U(t + \frac{\pi}{2})\right)^2} \quad (4.44)$$



**Figure 4.21:** Energy based correlation workflow

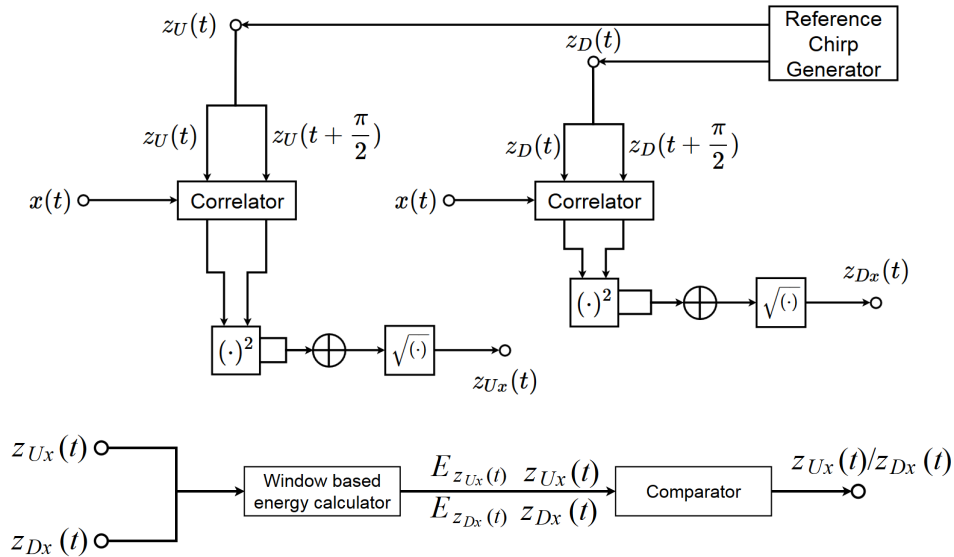
and correlation with down chirp is derived as:

$$z_{Dx}(t) = \sqrt{\left(\sum_t x(t) \cdot z_D(t)\right)^2 + \left(\sum_t x(t) \cdot z_D(t + \frac{\pi}{2})\right)^2} \quad (4.45)$$

The flow diagram explained in Fig. 4.22 is the correlator method as shown in Fig. 4.21 utilized for estimating click length and centre frequency. Table. 4.7 shows the parameters aided to create the synthetic chirps for correlation of the input signal with the input signal. After calculating the correlation coefficient, the energies calculated at each step is further analysed to see at which exact frequency the synthetic chirps correlates most with the input signal and thereby finding the click length of that particular species.

Energy based correlation incorporates the idea of maximising the energy in each correlation window, which is given by the reference chirp length. Chirp length is varied as given in Table. 4.7 and further analysing the energies from each of the correlation results retrieves the maximum energy as elaborated in the previous paragraph, computed as the ratio of the correlated output to the original energy content of the signal and normalized to the length of the original signal. A value close to unity is considered to be the maximum attainable value with unity being the ideal scenario. Correlated result conforming to the above criterion is then selected and plotted against the original signal giving two most essential information:

1. Length of a click
2. Center frequency of a click



**Figure 4.22: Expanded correlator block (Top):** Proposed correlation technique to understand the click length of a toothed whale species. **Bottom :** Expanded energy maximizer block

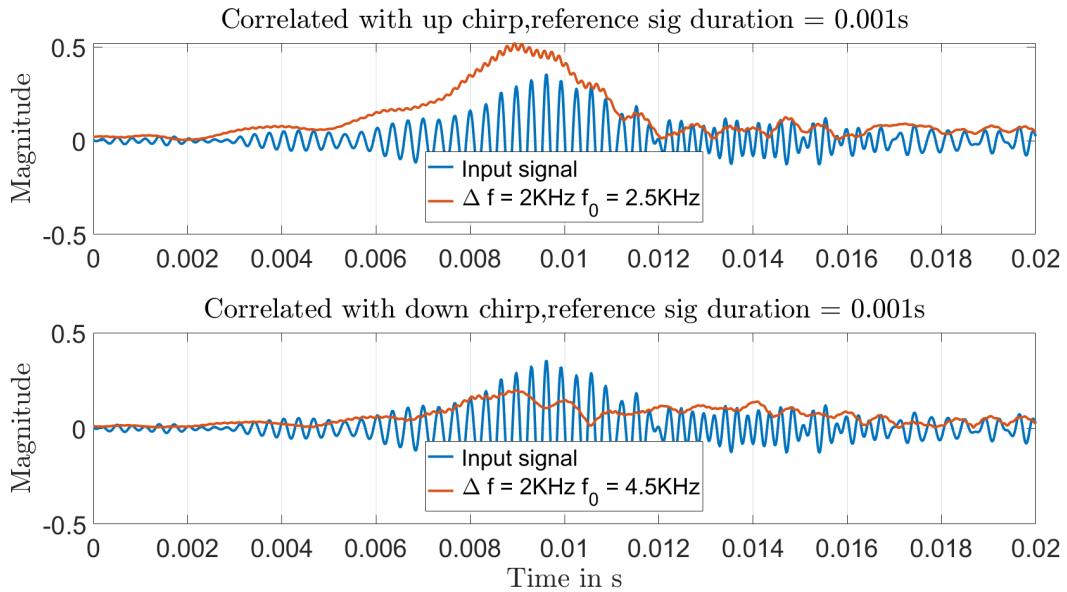
#### 4.5.4 Results

The result section considers two parts; firstly, it visualises the inner content of a random extracted estimated click obtained after the application of the separation algorithm to extract them and finally presents a table tabulating the centre frequency and the click length obtained by an energy-based correlation method. The result of correlation are pictured in Fig.4.23 and 4.24 obtained by implementing proposed method.

The creation of bandpass signal of the estimated click signal from each toothed whale species performed with the frequency band results obtained from the continuous wavelet transform. The coefficient range from the resultant CWT plot aided in extracting the frequency range localising the energy of the signal content, and after that, the extracted click is band-limited to the extracted values which and thought of as the input signal. This bandpass signal is then passed through the energy based correlator block as shown in Fig. 4.22

Fig.4.23 shows the correlation of a random extracted click estimated by source separation algorithm from killer whale audio. Results shown are the correlation with up-chirp in the top of the figure and down-chirp at the bottom of it, which again intuitively gave the chirp content in killer whale echolocation clicks. Killer whale clicks correlated with up-chirp of length  $1\text{ ms}$  and with a centre frequency of  $3.5\text{ kHz}$ , which is determined by energy maximisation technique. Chirp content in the click signal better visualised with the provided down-chirp correlation at the bottom of Fig. 4.23 which is minimum for the same set of parameters.

The proper reading of the graph is as per following rule. Input signal signifies the bandpass click signal, in which the high energy contents are visualised and designated with a blue colour. The orange coloured line denotes the correlation with the highest ratio. Bandwidth or the difference between the start and end frequency of the chirp is found to be  $2\text{ kHz}$  and starting frequency



**Figure 4.23:** *Top:* up-chirp in killer whale click is found to be of  $1\text{ ms}$  with a bandwidth of  $2\text{ kHz}$ , *Bottom:* correlation with down-chirp for killer whale click is maximum with a chirp length of  $1\text{ ms}$  with a bandwidth of  $2\text{ kHz}$  and suggests that here is no down-chirp component

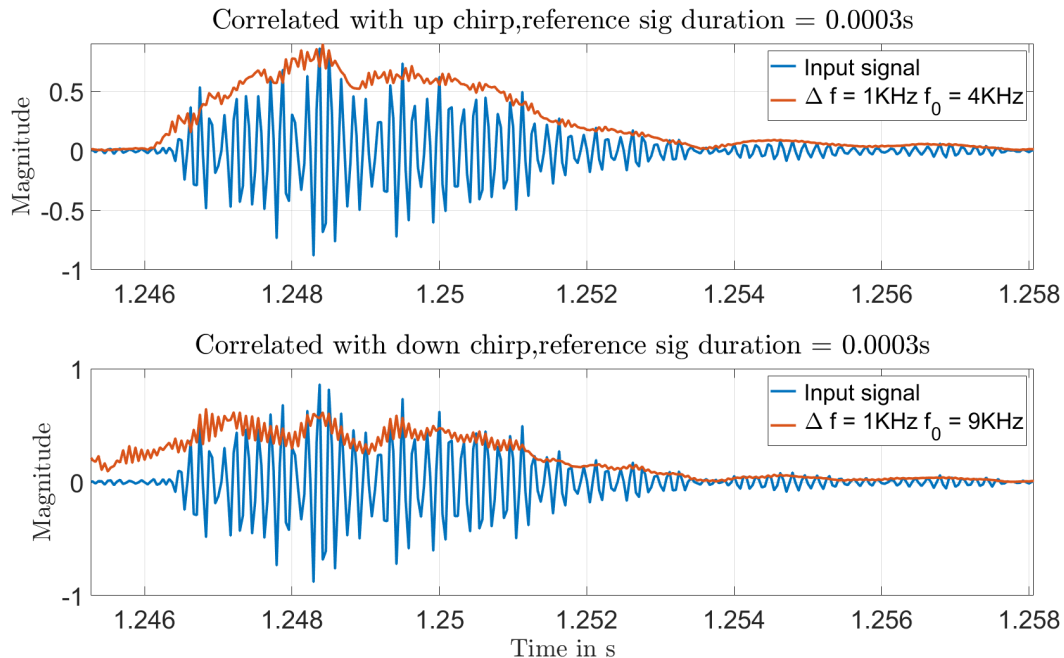
| Name of the species | Center frequency of click (kHz) | Click length (ms) |
|---------------------|---------------------------------|-------------------|
| Beaked whale        | 16.0                            | 11.8              |
| Dall Porpoise       | 4.5                             | 0.3               |
| False killer whale  | 18.25                           | 1.1               |
| Killer whale        | 4.5                             | 1                 |
| Pilot whale         | 7.5                             | 0.1               |
| Sperm whale         | 5.5                             | 5                 |

**Table 4.8:** Click length and centre frequency of some toothed whale species.

in case of an up-chirp is  $2.5\text{ kHz}$  which intuitively derives the fact that the starting frequency for down chirp is  $4.5\text{ kHz}$ .

Fig. 4.24 represents the chirp type present in the porpoise click, and findings conclude that the high energy click is a variation of up-chirp with a chirp length of  $0.3\text{ ms}$  and a centre frequency of  $4.5\text{ kHz}$ . Further experiments resulted in the finding that high-frequency elements are down chirp with a centre frequency of  $8.5\text{ kHz}$ .

The second fraction of the result is more oriented towards tabulating the click length and the centre frequency of the click from different toothed whales obtained by this implementation and are tabulated in Table. 4.8.



**Figure 4.24:** **Top:** up-chirp in porpoise click is found to be of  $0.3\text{ ms}$  with a bandwidth of  $3\text{ kHz}$ , **Bottom:** down-chirp in porpoise click is found to be of  $1\text{ ms}$  with a bandwidth of  $3\text{ kHz}$

## 4.6 Further improvements

The most important work can be done with improving the generation of synthetic impulse response which may help in better click visualisation and helping in estimating the whale concentration and their positioning with respect to the hydrophone. The second improvement can be done with respect to the chirp correlator technique. Fig. 4.19 helped in visualising the inlying clicks as time-varying amplitude and gradually diminishing characteristics. The chirps as implemented in Sec. 4.5, can be modified with varying amplitude by multiplying with a Gaussian function and thereby creating a chirplet. With proper scale and translation factor, correlating the chirplet will help in determining even accurate chirp content and the information about chirp length and centre frequency. The third improvement can be extrapolated on basis of deconvolved signal, which can help in setting a much more informed threshold to separate the high energy echo to low energy echo of any singular echolocation click. The basic step to start improving the analysis could start with a definite knowledge of the hardware recording these animals.



## Chapter 5

# Conclusion

Marine mammal being the indicator of ocean health, conservation of the same became an utmost priority as schools of these mammals ascertain prosperity in an oceanic ecosystem which is where this work plays an important role. Toothed whales, being highly active socialised species, having a distinct vocalisation feature called click helped in estimating different species. The sounds collected through *Discoveries* and *NatGeo* had vocalisations containing clicks in most cases.

The structure of recordings was visualised implementing short time Fourier transformed and revealed that the clicks have drum beat like structure. This finding encouraged to apply the convolutive mixing class of blind source separation techniques using generalised Wiener filtering approach. Non-negative matrix factorisation, media filtering and kernel additive modelling are the separation techniques used for separating clicks from harmonic parts of the recordings which also include creaks. The evaluation of separation algorithm with optimised parameters well documented in Chapter 3. The evaluation process includes signal-to-distortion ratio, peak-to-noise ratio and different simple errors which includes Pearson correlation coefficient; magnitude squared coherence and root mean square error. The evaluation prompted to apply non-negative matrix factorisation with median filtering to separate the entire recorded signal into a whistle and click datasets for further analysis as this particular combination helps in reducing the computational time.

Extraction process being successful in estimating the clicks from a recording, visualising the echo and extracting them became the centre-point of Chapter 4. The click signal is further denoised using discrete wavelet transform attaining a maximum possible signal-to-noise ratio. The analysis of the denoised signal with a stripped Hilbert-Huang transform; proposed as simple-Hilbert transform is calculating the echo decay time of an estimated click signal obtained via separation. While estimating echo decay time from the click, notches observed either indicated the number of objects reflecting the clicks or superimposed clicks. The study on superimposition of click pulses performed via continuous wavelet transform. The transformation gave a clear visualisation of the number of high energy contents from a single estimate click (also termed the pulse in work). Thresholding the energy to a limit, the echoes are extracted and form the signal made of multi-path propagation.

Chapter 4 concluded with an exciting result which could pave the way for further improvements in this field of conservation of marine mammals. Exploiting the fact that vocalisation from the animal kingdom is chirp-like, an energy-based correlation is performed to estimate the click

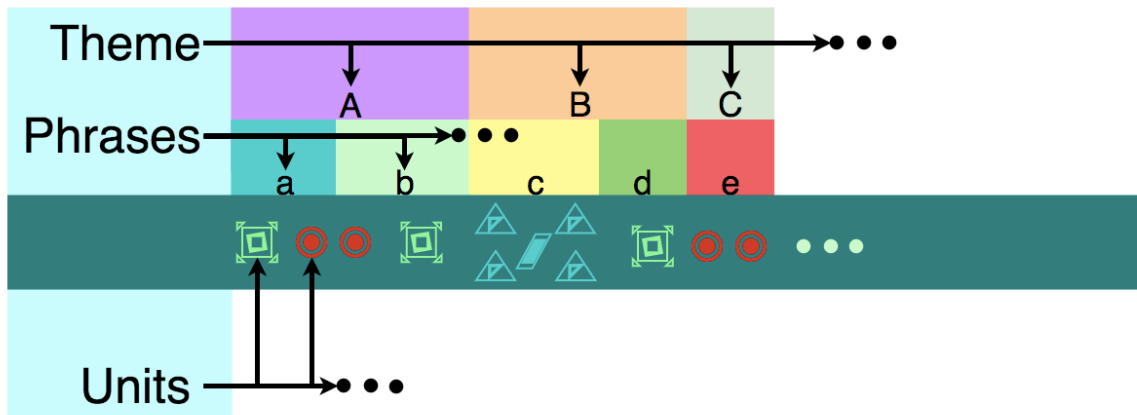
length from marine mammals; helping in estimating the species blindly. Sec. 5.1 introduces to the limitation of the work followed by areas of further improvements.

## 5.1 Limitation

The explanation of a certain set of limitations of the experiments are as follows:

1. The separation techniques aided in separating the clicks from harmonics separates the whistle and clicks when the species are in hunting orientation with optimised parameter values. However complex social interaction with the same species creates a problem. Separation of clicks and harmonics performed with the same parameter values and harmonic part of the signal contains whistles, moans, groans and other vocal structures; they are all classified as whistles only.
2. The learning process implemented in non-negative matrix factorisation and kernel additive modelling to learn the smooth structures over frequency and time to classify the whistle and click contents. However, when learning all the signals in an additive way and results applied to separate a blind recorded mammal vocal source, the process failed to separate the source into its whistle and click entity.
3. Synthetic impulse response is created in work for an underwater channel in an attempt to deconvolve the recorded signal for better localisation of click energy. However, even with all gathered parameter from different research works like, reflection coefficients, signal attenuation factor and others, the deconvolved signal degraded from the estimated separated signal and hence not researched further. A specific reason is that the ocean changes constantly and so knowledge of the environment through hardware, recording the species, is important

The reason mentioned in Item. 1 in Sec. 5.1, is the other reason why the work did not consider baleen whales in general and a brief idea of their vocalisation structure portrayed in Sec. 5.1.1.



**Figure 5.1:** Probable structure of whale song



### 5.1.1 Whale Songs

Whale songs are complex which ensembles units like moans, cries, groans, trills, chirps into phrases and assembling of these phrases form themes which form a predictable pattern creating the well-known songs of whales. This complicated structure requires separation differently by modelling each characteristic meticulously. Fig. 5.1 gives an idea of how a song is created. The other characteristics of the sound production are that the song contains certain reverberations which are produced in the air sac within its body to produce trailing phenomenon.

## 5.2 Areas of further improvement

The thesis work is dependent on various second-hand information about underwater surroundings which primarily is the reason for failing to create a correct estimation of an impulse response. The attenuation is taken to be  $20\text{ dB}$  and the reflection coefficients are taken based on a real-time test series with the surface reflection coefficient taken at 0.88, whose error is negligible, but reflection coefficient due to another mammal is taken arbitrarily and thus producing the erroneous results. Moreover, echoes are also thought to have at most 50% of the peak to peak value of the original power which might contradict if the objects are near to each other. Hence, extensive first-hand information makes the system more robust in determining any toothed whale blindly and help in contributing to the conservation process.



# List of Abbreviations

## A

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|     |                                                                     |
|-----|---------------------------------------------------------------------|
| ANT | Allgemeine Nachrichtentechnik (Signal Processing and Communication) |
| AAR | Autonomous Acoustic Recorder                                        |

## B

---

|       |                                |
|-------|--------------------------------|
| BSS   | Blind Source Separation        |
| bior# | biorthogonal class of wavelets |

## C

---

|       |                              |
|-------|------------------------------|
| CWT   | Continuous Wavelet Transform |
| coif# | Coiflet class of wavelets    |

## D

---

|     |                             |
|-----|-----------------------------|
| DWT | Discrete Wavelet Transform  |
| db# | Debauchies class of wavelet |

## F

---

|     |                                  |
|-----|----------------------------------|
| FFT | Fast Fourier Transform           |
| fk# | Fejer-Korovkin class of wavelets |

## H

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|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| HSU-HH | Helmut-Schmidt-University/University of the Federal Armed Forces Hamburg |
| HT     | Hilbert Transform                                                        |

## I

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|      |                                                |
|------|------------------------------------------------|
| ICA  | Independent Component Analysis                 |
| IUCN | International Union for Conservation of Nature |

## K

---

|     |                           |
|-----|---------------------------|
| KAM | Kernel Additive Modelling |
|-----|---------------------------|

## M

---

|            |                                          |
|------------|------------------------------------------|
| MF         | Median Filtering                         |
| $\mu$ MARS | micro Marine Autonomous Recording System |
| MARU       | Marine Autonomous Recording Unit         |
| MSC        | Magnitude Squared Coherence              |

## N

---

|     |                                   |
|-----|-----------------------------------|
| NMF | Non-negative Matrix Factorization |
|-----|-----------------------------------|

|      |                                                 |
|------|-------------------------------------------------|
| NOAA | NAtional Oceanic and Atmospheric Administration |
| NDT  | Non-Destructive Testing                         |

**P**

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|     |                                 |
|-----|---------------------------------|
| PAM | Passive Acoustic Monitoring     |
| PNR | Peak-to-Noise Ratio             |
| PCC | Pearson Correlation Coefficient |

**R**

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|       |                                        |
|-------|----------------------------------------|
| RMS   | Root-Mean-Square                       |
| rbio# | reverse biorthogonal class of wavelets |

**S**

---

|      |                              |
|------|------------------------------|
| sym# | Symlet class of wavelet      |
| SDR  | Signal-to-Distortion Ratio   |
| SAR  | Signal-to-Artifact Ratio     |
| STFT | Short-Time Fourier Transform |
| SSM  | State-Space Modelling        |

**T**

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|      |                                  |
|------|----------------------------------|
| TF   | Time-Frequency                   |
| TUHH | Hamburg University of Technology |

**U**

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|        |                                                                      |
|--------|----------------------------------------------------------------------|
| UNESCO | The United Nations Educational, Scientific and Cultural Organisation |
|--------|----------------------------------------------------------------------|

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